Scientific Reports

https://doi.org/10.1038/s41598-026-62771-1

Article in Press

Hybrid deep learning–driven explainable AI framework for fault detection and classification in smart power grids

Received: 5 April 2026

Accepted: 14 July 2026

Published online: 21 July 2026

Cite this article as: Mamodiya U., Sinha D., Kishor I. *et al.* Hybrid deep learning–driven explainable AI framework for fault detection and classification in smart power grids. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-62771-1

Udit Mamodiya, Divyanshu Sinha, Indra Kishor, Malik Jawarneh, Mais Haj Qasem & Afnan Almegren

Hybrid Deep Learning–Driven Explainable AI Framework for Fault Detection and Classification in Smart Power Grids

Dr. Udit Mamodiya[1]*, Dr. Divyanshu Sinha[2], Indra Kishor[3], Malik Jawarneh[4], Mais Haj Qasem[5], Afnan Almegren[6]

[1]*Associate Professor, Faculty of Engineering and Technology, Poornima University, Jaipur 303905, Rajasthan, India;*
[2]*Associate Professor, Amrita School of Artificial Intelligence, Amrita Vishwavidyapeetham, Faridabad Campus, India*
[3]*Assistant Prof. , Dept. of CSE, Poornima Institute of Engineering and Technology, Jaipur 302022, Rajasthan, India;*
[4] *Associate Prof., Dept. of Computer Science, Amman Arab University, Amman, Jordan;*
[5]*Assistant Prof., Dept. of Data Science and Artificial Intelligence , Amman Arab University, Amman, Jordan;*
[5]*Assistant Prof., Dept. of Applied Linguistics, College of Languages, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia;*

[1]assoc.dean_research@poornima.edu.in ; [2]divv4u@gmail.com; [3]indra.kishor@poornima.org, [4]m.jawarneh@aau.edu.jo, [5]m.hajqasem@aau.edu.jo, [6]amalmegren@pnu.edu.sa

*Corresponding author:
Dr. Mais Haj Qasem
E-mail: m.hajqasem@aau.edu.jo

Abstract

The stable detection of faults in smart power grids is essential when focusing on the stable operation and the reduction of the downtime. In this paper, the author suggests a hybrid deep learning-based model that combines convolutional neural networks (CNN) and long short-term memory (LSTM) with explainable artificial intelligence (XAI) to detect and classify faults accurately and interpretably. The model aims at capturing the spatial and time-varying attributes of multivariate electrical signatures such as voltage, current, and frequency changes. A combination of real time sensor measurements and simulated fault conditions are used which includes several fault classes including LG, LL, LLG, and three phase faults. Experimental evaluation demonstrates that the proposed model achieves a classification accuracy of **97.84%**, with an F1-score of **97.08%**, outperforming conventional CNN and LSTM models by more than 3%. The framework is also able to work in a noisy environment with a stable performance of less than 2% performance decrease, and inference latency of 18 ms, which can be deployed in real-time. Besides, the combination of SHAP and attention processes improves the interpretability through the detection of the crucial features that lead to fault prediction. The findings suggest that the suggested solution is a powerful, scalable, and clear solution when it comes to intelligent fault management in a contemporary smart grid.

Keywords: smart grid fault detection, hybrid CNN–LSTM, explainable artificial intelligence, deep learning, power system reliability

1. Introduction

The transformation of traditional electrical infrastructures to smart intelligent power grids across the globe has revolutionized the approach to the monitoring, regulation, and optimization of the energy systems. Modern smart grids are based on the combination of distributed energy sources, renewable energy sources, smart sensing technologies, and real-time communication networks as opposed to traditional grids based on centralized generation and static operational planning. This transformation enables an increased level of productivity and flexibility and at the same time renders immense complexity to operations. The growing penetration of non-reliable renewable energy sources such as solar and wind, and dynamic loads changes have made grid behavior a priori nonlinear and highly unpredictable [1], [2]. As a result, stabilization of the systems, as well as the continuity of power supply, have never been a greater challenge than it is now. Such dynamic environments are becoming reliant on fault detection and classification as the significant aspects of grid reliability. Electrical faults including short circuits, line to ground faults, voltage drop, frequency change and harmonic interference can be transferred smoothly through networked networks unless corrected early before they happen. Even minor failures, which are not detected fast, can turn into a domino of the huge downages and huge financial losses [3], [4]. This implies that it is important that the current grid has a sense of urgency to have smart monitoring systems that are capable of successfully identifying fault conditions on real time basis and adapting the changed dynamics of the grid which is dynamic. The old power system has generally used conventional ways of fault detection including impedance based systems, threshold based protection systems as well as model-based analysis systems. Even though these strategies are good when the operating environment is clear and is stable, it has been found that it does not work well in the current smart grid set ups [5], [6]. Besides, they are frequently large-scale manual tuned and cannot be scaled when applied to large distributed networks. Since the creation of smart grids is in progress, the limitations of such conventional solutions stand even better. To address these problems, machine learning (ML) techniques have been suggested to classify and identify faults by using data. The algorithms that have demonstrated better performance as opposed to the traditional methods include support vector machines, decision trees and random forests because they use past data to create fault patterns [7], [8]. Yet such approaches are frequently rooted in much manualized labor and specialized experience and this can restrict their effects to various grid regimes. Besides, they are destined to drop at large-dimensional multi-variate data in time-series, which is normally generated by current smart grid sensors. The idea of deep learning (DL) is a potential paradigm of analysis of complex electrical signals and feature representations on hierarchies in the past few years. CNNs are more effective in expressing spatial correlations in voltage and current signals but recurrent networks with long short-term memory (LSTM) networks are more effective in expressing temporal correlations [9], [10]. The use of both spatial and temporal information in architecture hybrid systems [11, 12] has been the best way for detecting fault in the field. These models are not manually engineered, and offer extremely strong predictive performance in noisy and dynamic environment.

Deep learning models have been labeled as uninterpretable, even though they have an impressive level of accuracy. Automated systems in critical infrastructure systems like power grids should also be transparent and explainable in order to provide trust, accountability and regulatory compliance. Predictions without explicit models to explain the prediction are not easy to be validated by grid operators, especially when the stakes are high and are related to fault mitigation and system recovery [13], [14]. This is a significant challenge in implementing the deep learning-based fault detection system in practice in a real smart grid setup.

Explainable artificial intelligence (XAI) has become the solution to this issue, as it allows interpreting complex decisions of models. Additional methods like SHapley Additive explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME) and attention mechanisms can be used to identify important features affecting model predictions [15], [16]. Within the idea of smart grids, XAI would offer useful information regarding the contribution of electrical quantities like the magnitude of voltages, frequency variation, and harmonic distortion in the fault detection activities. It does not only improve the transparency of the models but also aids in the informed decision-making of the system operators.

Nevertheless, even though the attention towards deep learning and explainable AI is increasing, the current studies tend to take these aspects separately. The majority of the studies are oriented either on enhancing detection accuracy with advanced neural architectures or using XAI methods on the models that have already been trained without putting explainability into a close connection with the learning process. Consequently, a very severe gap still exists in creating integrated frameworks that at the same time deliver high predictive capabilities and a meaningful interpretability in smart grid fault analysis [17], [18]. Also, most of the solutions, which are available, are not flexible in real time and are incapable of meeting the computational resource limitations that are involved in large scale grid implementations.

In order to fill this gap, this paper presents a hybrid deep learning-based explainable AI architecture to detect and classify faults in smart power grids. The proposed solution combines convolutional neural network to extract spatial features with long short-term memory network to model time sequence, creating a robust hybrid network that can be suitable to analyze multivariate time-series data.

The main contributions of this work are as follows:

- A novel hybrid deep learning architecture combining CNN and LSTM for enhanced fault detection and classification in smart power grids.

- Integration of explainable AI techniques to provide feature-level interpretability and improve system transparency.
- Development of a unified framework that balances predictive performance with explainability, addressing a key limitation of existing approaches.
- Comprehensive evaluation demonstrating improved accuracy, robustness, and interpretability compared to conventional and baseline deep learning models.
- Practical insights into the role of critical electrical parameters in fault identification, supporting real-world deployment in smart grid environments.

Although the proposed framework has shown great performance, there are some practical considerations that still exist. The datasets and simulated smart grid conditions involved in the current study are manipulated and might not be a true reflection of the proportions of the real-life scale of the implementation. Also, although explainable AI can make AI more interpretable, it incurs a minor computational overhead, which in hypothetically high data throughput situations can affect real-time scalability. The next generation will then consider real-time grid implementation and optimization of explainability mechanisms of edge-based implementations.

This research paper is useful because it suggests an explicable and scaled fault detection model, which is a next generation smart grid. The proposed solution, in addition to enhancing stability of operations, will instill trust in the AI-supported decision-making systems and it will result in development of safer and more robust energy infrastructures.

2. Related Work

The fast evolution of the smart monitoring systems in the modern power grid leads to the fact that the amount of literature devoted to the phenomenon of the fault detection and its categorization according to the data-driven approach is very large. The next section is a systematic literature review of the possible available methodologies that are categorized as traditional methods, machine learning approach, deep learning approach, and explainable AI approach. Critical analysis of these studies in question leads to the realization of research gaps that exist which forms the driving force behind the proposed hybrid explainable framework.

2.1 Conventional Fault Detection Approaches in Power Systems

Older fault detection systems used in electric power systems were primarily of rule-based and analytical nature. These include impedance based relays, over current protection mechanism as well as traveling wave analysis. These techniques are founded on the set of predetermined thresholds and deterministic models depending on the system parameters. As far as effective in traditional grid, they do not work in smart grids where variations and intricacy of the system are high [21], [22].

A number of studies have tried to improve them by adding adaptive thresholding and signal processing [23]. On the same note, Fourier based harmonic analysis has also been used in identifying irregularities in voltage and current waveforms [24]. Even with these advances, traditional tools are still restricted by the fact that they cannot deal with nonlinear relationships and require proper modeling of the system.

In addition, such strategies fail to be scalable and flexible in the case of the distributed energy systems that have a high rate of renewable penetration. The growing incorporation of photovoltaic and wind resources creates stochastic fluctuations that are not well represented using the non-dynamic models [25], [26]. This has led to a change of emphasis to data-driven approaches which can directly learn complex patterns using system data.

2.2 Machine Learning-Based Fault Detection Methods

The use of machine learning methods has been extensively researched to address the shortcomings of traditional methods. Support vector machines (SVM), k-nearest neighbors (KNN), decision trees, and random forests, are examples of supervised learning algorithms that have shown better classification results in fault detection tasks [27], [28]. These models use history information to interpret patterns that are related to the various faulty situations so that more flexible and adaptive detection systems can be used.

As the example, the SVM-based classifiers were used to recognize the different kinds of faults with extracted features of the voltage and current waves [29] [30]. Moreover, the ensemble learning methods are also suggested to enhance the accuracy of prediction by integrating a set of weak learners [31].

Nonetheless, feature engineering is extremely critical to the effectiveness of machine learning models. To create meaningful features of raw electrical signals, domain knowledge and frequently complicated preprocessing tasks are needed, e.g., signal transformation, signal normalization, dimensionality reduction, etc. [32]. Moreover, the models are not very effective in dynamic grid systems where fault patterns change with time because they have difficulty in modeling temporal dependencies in any sequential data.

The other issue is the problem of generalization of these models. Most machine learning models are trained on certain datasets, and might not be useful when implemented in new grid topologies or working environments [33], [34]. This drawback has propelled the research on deep learning methodologies, which are able to learn hierarchical representations in raw data through automatic learning.

2.3 Deep Learning Models for Smart Grid Fault Analysis

The recent development of deep learning has become a potent data analysis tool in smart grids with high-dimensional and time-series data. CNNs have been applied widely in feature extraction of electrical signals especially when analyzing waveforms [35]. The CNNs are able to capture the local spatial patterns and detect characteristic features that are related to the various types of faults by using convolutional filters.

The use of recurrent neural networks (RNNs) and particularly long short-term memory (LSTM) networks have been used to model sequential data temporal dependencies. The models which are based on LSTM can also remember long-term information, therefore, they can be effectively used to analyze time-varying signals in power systems [36]. A number of researchers have shown that LSTM networks can be used successfully to predict and classify faults using historical grid data [37].

Hybrid deep learning systems that incorporate CNN and LSTM modules have been receiving growing popularity in recent years. These models use the advantages of both designs where an extraction of both spatial and temporal features is made at the same time [38], [39]. As an example, the CNN-LSTM systems were implemented successfully to identify transient disturbances and categorize the type of faults in the networks of transmission and distribution with excellent precision [40].

Deep learning models have serious challenges in the form of interpretability and computational complexity despite being highly performing models. These models are usually black boxes, giving predictions without making explicit description of the process that had gone into the decision [41], [42]. In systems that require high levels of safety like power grids, this inability to gain transparency can make adoption difficult, as well as diminish the trust of the system operators.

2.4 Explainable AI in Power System Fault Detection

Explainable artificial intelligence (XAI) has been in the limelight of resolving the interpretability issues of deep learning models. XAI methods seek to offer the information about the model behavior by revealing the features that affect predictions. SHAP, LIME, and Grad-Cam are some of the techniques that have been extensively applied to explain complex models in other fields, like power systems [43], [44].

SHAP-based methods have been used in the context of fault detection to estimate the importance of the input features, which are voltage magnitude, imbalance of currents, and frequency deviation [45]. It has been applied by using LIM to make local explanations on the individual prediction, so that operators could get to know certain fault situations [46]. Mechanisms of attention embedded in neural networks have also been studied in order to emphasize significant temporal properties of sequence information [47].

Though the techniques enhance the transparency of models, most of the existing researches use XAI as a post-hoc analysis tool and not as a part of the model design. This division restricts the efficacy of explainability and does not wholly reflect on the requirement of real-time elucidation in working situations [48], [49]. Also, the computational cost of XAI techniques may be problematic when deploying in large-scale smart grids in real time.

2.5 Research Gap and Motivation

Although the field of fault detection and classification has made considerable progress, it has a number of essential gaps in literature:

- Conventional methods lack adaptability and fail under nonlinear and dynamic grid conditions.
- Machine learning approaches depend heavily on manual feature engineering and exhibit limited generalization.
- Deep learning models provide high accuracy but suffer from poor interpretability and transparency.
- Current XAI methods tend to be applied in a post hoc manner, and are not closely coupled with predictive models.
- Limited research exists on unified frameworks that simultaneously address accuracy, interpretability, and real-time applicability.

Such limitations indicate that a more complex solution is required that integrates the best features of deep learning and explainable AI and mitigates their respective weaknesses. The presented hybrid framework is expected to close this gap by enhancing the feature learning-based framework of CNN-LSTM with the fixed explainability structures to allow the proper, transparent, and scalable fault detection in smart power grids [50], [51].

In order to present a brief comparison of the recent developments in fault detection and classification in smart power grids, the selected representative studies are summarized in Table 1. The emphasis is made on determining methodological patterns, major contributions, and current limitations, in special reference to hybrid deep learning architectures and explainable AI integration.

Table 1. Literature gap-to-solution mapping for fault detection and classification in smart power grids

S. No.	**Author / Year / Ref.**	**Focus Area**	**Methodology / Tools Used**	**Key Findings**	**Gap Identified**	**Relevance to the Current Study**
1	Alhanaf et al. / 2023 / [5]	Smart grid fault classification	Deep neural network-based classification on smart grid fault data	Showed that deep neural models can improve automated fault recognition over conventional schemes	Black-box behavior; no interpretability	Supports the need for a more transparent deep model that combines fault accuracy with explainability
2	Fang et al. / 2023 / [26]	Fault classification & localization	Explainable fault classification with robustness-oriented design for transmission systems	Demonstrated that explainability can improve trust in line fault analysis	Limited to transmission lines	Establishes the value of XAI in grid diagnostics, which the present study extends to wider fault detection and classification settings
3	Custodio et al. / 2024 / [35]	Fault detection in transmission lines	1D-CNN-LSTM with federated learning	Confirmed that CNN-LSTM models are effective for capturing sequential fault signatures	No explainability layer	Directly motivates the CNN-LSTM backbone for the current study, while highlighting the missing XAI layer
4	Sarker et al. / 2024 / [39]	Smart grid forecasting	Attention model, federated learning, XAI	Showed that interpretability and advanced deep learning can be jointly embedded in smart-grid analytics	Not fault-specific	Demonstrates that XAI-aware deep learning is feasible in smart grids, encouraging its transfer to the fault domain
5	Shouran	Transformer	Hybrid feature extraction with	Reported strong real-case classification	Limited	Reinforces the benefit of hybrid modeling, but also

	et al. / 2025 / [7]	fault classification	deep learning on real-world transformer data	performance and practical promise	generalization	indicates the need for a more generalized grid-level framework
6	Sinha et al. / 2025 / [24]	XAI-based fault detection	XAI-assisted multimodal framework for transmission-line faults	Highlighted the usefulness of multimodal analytics and explanation support in detecting grid faults	Narrow application scope	Closely aligns with the present study's goal of combining fault intelligence with explainability, but our work targets a broader hybrid DL classification framework
7	Attallah et al. / 2025 / [31]	Transformer fault diagnosis	Multi-scale CNN-based lightweight framework	Showed that compact deep models can achieve effective transformer fault diagnosis with reduced computational demand	Lacks temporal modeling + XAI	Indicates the importance of computationally practical DL designs, which can inform efficient implementation of the proposed explainable framework
8	Farsi et al. / 2026 / [45]	Disturbance detection	Explainable machine learning for disturbance and cyber-attack detection	Demonstrated that explainable models improve visibility into security-related grid anomalies	Limited DL capability	Strengthens the case that explainability is essential in grid-critical decisions, which the current work incorporates into deep fault classification

On the whole, the literature reviewed suggests the further development of traditional analytical approaches to sophisticated AI-based ones.

3. Methodology

This proposed method includes a data-driven, experimental grounding of a hybrid deep-learning method along with an explainable artificial intelligence approach to fault identification and classification in smart-power grids. The development includes a focus on realistic signal modeling, temporal/spatial feature extraction, and understandable support of the decision process. The overall workflow of data acquisition, preprocessing, hybrid model design, and integration of explanation is depicted in Figure 1.

The process of multivariate signal acquisition of grid monitoring instruments triggers the methodological pipeline that is executed by systematized preprocessing enhancing the quality and stability of the signals. The hybrid CNNLSTM architecture is then employed to gain experience in the temporal and spatial representations of electrical measurements. Finally, it possesses an explainable AI component, where predictions of models are explained at the feature-level. The whole workflow, data flow, and interactions between the data models are presented in Figure 1.

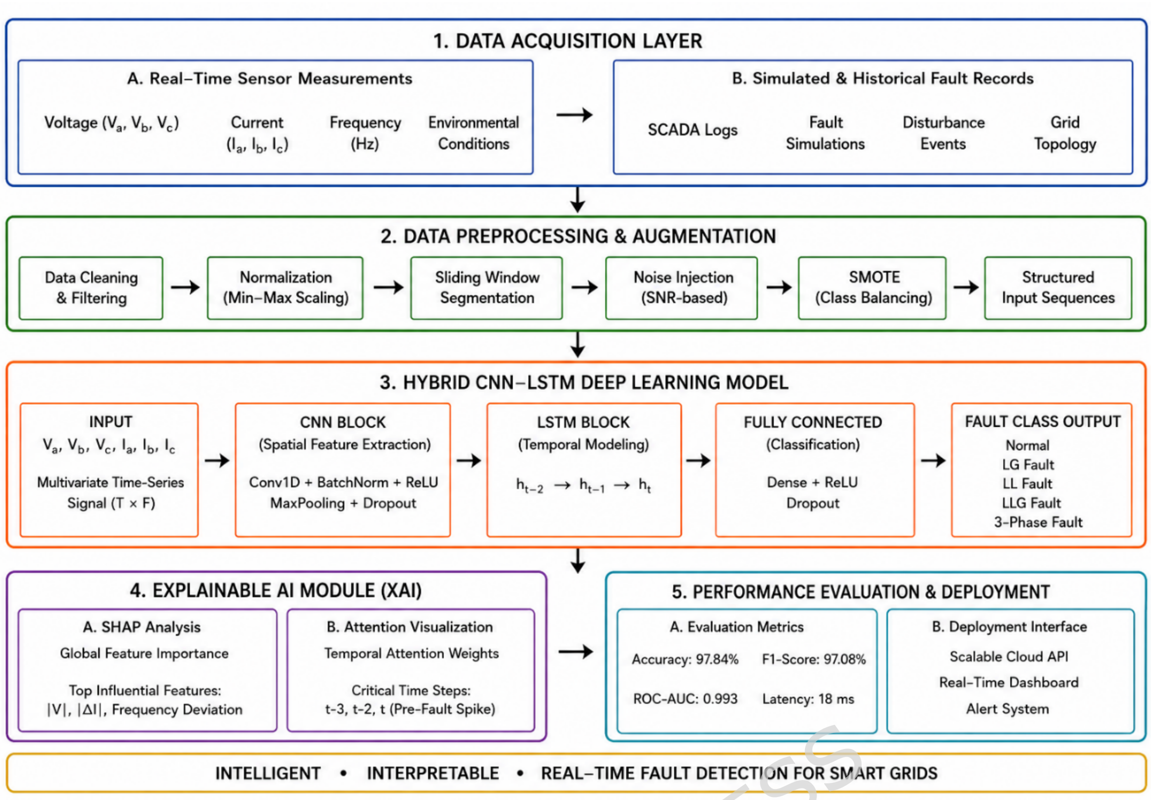

Figure 1. Overall architecture of the proposed hybrid deep learning-driven explainable AI framework for smart grid fault detection and classification.

As it can be seen in Figure 1, the structure can be split into four mutually supporting layers i.e., data acquisition, preprocessing, hybrid feature learning, and explainability-driven decision support. These layers will ensure that the physical relevance of each is maintained and yet enable the sound learning using the data, as it is being done in recent advances in the smart grid analytics [1], [15], [16].

3.1 System Architecture and Signal Representation

The system architecture has been structured in a manner that it is capable of supporting the high frequency and multivariate electrical signals generated by the smart grid monitoring devices such as phasor measurement units (PMUs) and intelligent electronic devices (IEDs). In general, these signals include the magnitude of voltages, magnitude of currents, the phase angle and the frequency and elements of harmonic distortion. As the smart grids are dynamic, these measurements possess a spatial correlation (between features) and a temporal dependence (between time) and a hybrid model approach is needed.

Let the input signal be represented as a multivariate time-series matrix (1):

$$X = \{x_1,x_2,...,x_T\} \in R^{T\times d} \cdots (1)$$

In which T is the count of time phases and d is the count of observed electrical characteristics. The system state at time instant t is identified by each of the vectors x_t in R^d. This model enables the model to address the immediate system states as well as the temporal dynamics.

To ensure consistency across measurements, the input data is standardized using z-score normalization (2):

$$\check{x}_t = \frac{x_t - \mu}{\sigma} \cdots (2)$$

where μ and σ represent mean and standard deviation of the feature distribution where μ and sigma are used as the mean and standard deviation respectively. This conversion smooths the period of training since the effect of the scale variations across various electrical parameters is minimized.

Since the events of faults tend to appear in the form of local perturbations in the short-period intervals, the sliding window technique is applied in the continuous signal to divide it into parts (3):

$$X^{k} = \{x_{k}, x_{k+1}, \ldots, x_{k+L-1}\} \cdots (3)$$

where L is the length of the window and k is the index of the starting segment. The transient fault signatures are restricted to this segmentation, and time continuity is retained by this model.

Besides segmentation, noise filtering is used to eliminate distortion of measurements and enhance the clarity of the signal. The filtering operation is a convolution-based operation (4):

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \cdots (4)$$

refers to impulse response of the filter. Practically, the low-pass filter is used to extract the high-frequency noise elements, which do not play an important role in characterizing the fault.

In order to better represent the features, the segmented signal is converted to a structured input tensor that can be processed by a convolutional processing (5):

$$X \in R^{N \times L \times d} \cdots (5)$$

N is the number of segments, L is the temporal window size and d is the number of features. This is a type of formulation using tensors that allows learning feature interactions concurrently with temporal dynamics.

The architecture is constructed in such a way that the convolutional layers act on the feature dimension to extract the inter-parameter relationships whereas the recurrent layers act on the time series relationships. This dual representation is necessary in the accurate distinction of the various types of faults, which in most cases have similar instantaneous properties but differ in terms of temporal occurrence [7], [35].

The real-time identification of the changes that occur to electrical signals from fault conditions will require conducting an instantaneous measurement of the electrical signal's magnitude as denoted mathematically in relation to a previous measurement denoted by (6):

$$\Delta x_{t} = x_{t} - x_{t-1} \cdots (6)$$

Where: x_t denotes the instantaneous changes of the fault event.

Convolutional filters are trained to learn how to differentiate between various attributes of an electrical signal so that they will be sensitive to transient disturbances that may occur within the grid.

3.2 Data Acquisition and Preprocessing

The nature of quality, diversity and representative of the input data determine the reliability of any data-based fault detection framework. In the current work, a mixed dataset of actual sensor values and idealized grid scenarios is built up to represent the conditions of both controlled and working conditions. This two-source approach will guarantee that the model is realized to realistic noise, uncertainty and variability yet the broad coverage of various fault scenarios will be ensured as suggested in recent smart grid analytics research [9], [32], [42]. Figure 2 shows the entire data processing process, including acquisition to final model-ready inputs.

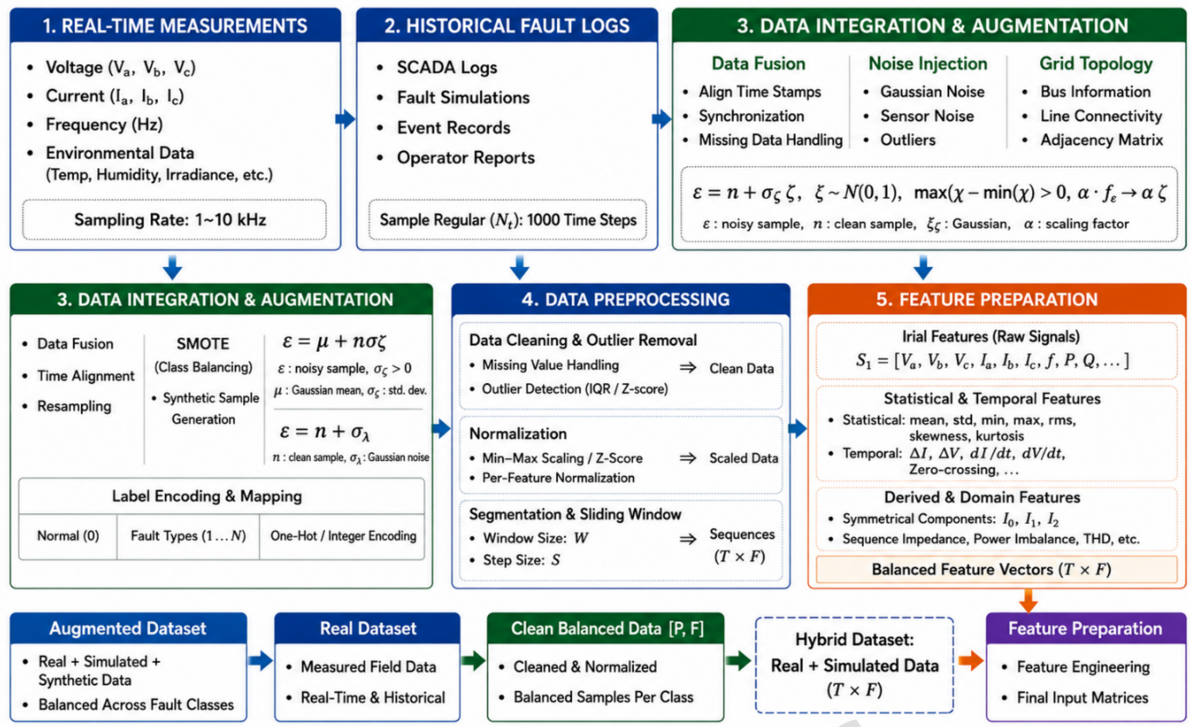

Figure 2. Data acquisition, integration, preprocessing, and feature preparation pipeline for hybrid smart grid dataset.

3.2.1 Data Acquisition

The data that will be employed in the research is a mixed approach of real-time measurements and simulation-based fault scenarios and publicly accessible benchmark datasets. Experimental data were taken in a laboratory-scale smart grid system that incorporated photovoltaic system, lines of distribution and IoT-based monitoring systems. The IEEE 14-bus system in MATLAB/Simulink was used to develop simulated fault conditions in order to guarantee controlled representation of different fault types.

Besides that, a few data that are publicly available, such as NASA-generated time-series datasets on electrical system monitoring and degradation analysis, were also introduced to make it more varied and robust. Such auxiliary datasets bring more signal diversity and aid in better generalization of models in unseen conditions.

The variety of time-series patterns with respect to system degradation, anomaly, and sensor variability added to the dataset by the NASA one makes the proposed framework more robust and capable of generalization. Even though they were created in the field of aerospace and prognostics, the signal characteristics of these datasets (noise, drift, and temporary irregularities) resemble electrical disturbances in smart grids. All datasets are made compatible by normalizing, resampling, and aligning their features so as to guarantee that datasets are standardized in terms of their feature representations, sampling format and time structure to allow easy combination into a coherent multivariate time-series fault detector.

Table 2. Sample representation of acquired smart grid dataset with multivariate electrical parameters under different operating conditions.

Sample ID	Time Step (t)	Voltage (V)	Current (A)	Frequency (Hz)	THD (%)	Temperature (°C)	Fault Class
S001	1	231.2	5.08	50.01	2.3	31.8	Normal
S001	2	230.6	5.15	49.99	2.5	32	Normal
S002	1	182.4	7.92	49.62	6.7	35.4	LG
S002	2	176.8	8.1	49.55	7.2	35.9	LG
S003	1	208.9	6.52	49.78	5.1	34.2	LL
S003	2	202.3	6.88	49.7	5.6	34.8	LL
S004	1	158.7	9.05	48.96	9.4	36.7	LLG

S004	2	150.2	9.4	48.85	10.1	37.3	LLG
S005	1	118.6	10.72	48.15	12.8	38.5	3-Phase
S005	2	110.9	11.1	48.05	13.6	39.1	3-Phase

The dataset as shown in Table 2 records the changes in the main electrical parameters over time due to changes in normal and fault conditions. It is possible to detect clear trends in the fault classes, in which voltage drops, current spikes, frequency deviation, and amplified harmonic distortion are all indicators of an abnormal grid behavior. Such differences give the proposed hybrid deep learning model discriminative properties that allow the correct classification of faults under different operating conditions. Moreover, with this paper, there are also complete data that can be used with regard to reproductively.

The dataset would be acquired with the use of real-time sensor measurements and simulated fault generation. Actual data is taken of a laboratory size installation of a smart grid comprising of photovoltaic (PV) generation plants, distribution lines, and load systems. The system consists of voltage sensors, current transformers, frequency meters, and temperature sensors connected in an IoT-based monitoring platform [20], [44]. The microcontroller-based edge device (Raspberry Pi 4B) is used to acquire data as synchronized measurements on a periodical basis.

Simulated fault conditions are added to the real-world variation in order to achieve this. An IEEE 14-bus test system, simulated in MATLAB/Simulink, is used to create simulated fault conditions. This enables the injection of the various types of faults such as single line-to-ground (LG), line-to-line (LL), double line-to-ground (LLG) as well as three-phase faults to be controlled. Both real and simulated data will ensure that the rare yet critical fault events are covered that might not be common in the real systems.

The continuous signal acquisition process can be expressed as (7):

$$x(t) = s(t) + n(t) \cdots (7)$$

where s(t) represents the true electrical signal and n(t) denotes measurement noise introduced by sensors and environmental factors.

The sampling frequency will be 5 kHz in order to record high frequency events of transients and the total data collection will be about 72 hours of continuous operation in both normal and fault mode. The last data set is comprised of about 120,000 samples that are in five classes which include normal operation and four faults.

The labeling of each data instance is done through a hybrid method. In the case of simulated data, labels are assigned depending on information on fault injection parameters, whereas real-world data is labeled depending on rule-based thresholds which are based on domain knowledge (8):

$$y = \begin{cases} 0, & normal\ condition \\ 1, & LG\ fault \\ 2, & LL\ fault \\ 3, & LLG\ fault \\ 4, & three - phase\ fault \end{cases} \cdots (8)$$

This structured labeling ensures consistency across heterogeneous data sources.

3.2.2 Data Integration and Synchronization

Since the information is obtained using a variety of sensors and sources, the temporal alignment is required to maintain the integrity of multivariate relationships. The sensors streams are synchronized and stamped on a common clock reference. The consistent data set will take the form of (9):

$$X(t) = [v(t), i(t), f(t), h(t), T(t)] \cdots (9)$$

where v(t), i(t), f(t), h(t), and T(t) denote voltage, current, frequency, harmonic distortion, and temperature signals, respectively.

In order to handle discrepancies in the sampling between real and simulated data, resampling is done with the help of linear interpolation (10):

$$x'(t) = x(t_1) + \frac{t - t_1}{t_2 - t_1}(x(t_2) - x(t_1))\cdots(10)$$

This ensures uniform temporal resolution across all data streams, enabling consistent model input representation.

3.2.3 Data Cleaning

Raw sensor data is likely to be noisy, missing and contain outliers. Multi stage cleaning process is therefore used. The forward interpolation is used to deal with missing values (11):

$$x_t = x_{t-1}, if\ x_t\ is\ missing \cdots(11)$$

A low-pass Butterworth filter is used to filter noises by removing high frequencies that are not related to the fault events. Z-score is used to detect outliers:

$$Z = \frac{x - \mu}{\sigma}\cdots(12)$$

Data points with |Z|>3 are treated as outliers and removed. This step ensures that abnormal sensor spikes do not bias the learning process.

3.2.4 Feature Engineering

Even though the automatic learning of features by deep learning model is possible, initial features representation increases interpretability and convergence. Root mean square (RMS), mean, and variance are some of the key statistics and electrical characteristics that are extracted by means of (13) and (14):

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2 \cdots(13)}$$

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \cdots(14)$$

Frequency-domain features are obtained using Fast Fourier Transform (FFT) using (15):

$$X(f) = \sum_{t=0}^{N-1} x(t)e^{-j2\pi ft/N} \cdots(15)$$

These features help capture harmonic distortions and transient frequency variations associated with faults [24], [47].

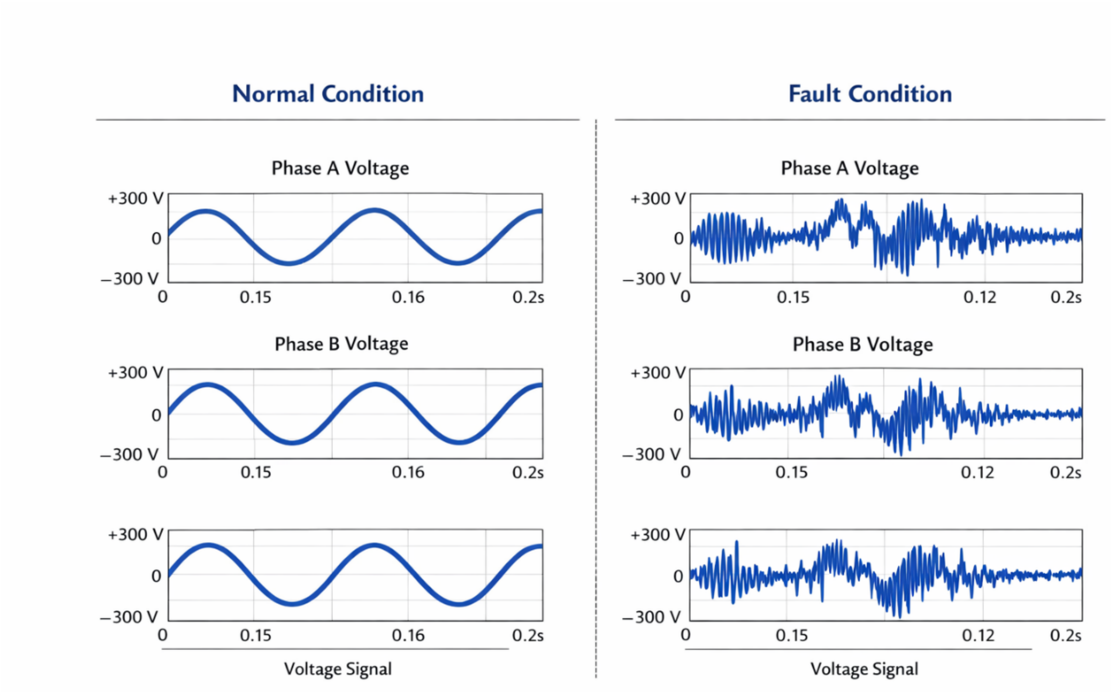

Figure 3. Comparative waveform representation of voltage signals under normal and fault conditions.

Figure 3 shows how the waveforms of voltages change with normal working and various fault conditions. Normal signals are characterized by constant sinusoidal characteristics whereas the fault cases cause sudden drops in voltage, distortion of the waveforms and temporary swings. These unique patterns are very important fault detection features that are well represented by the suggested hybrid deep learning model.

3.2.5 Data Normalization and Scaling

In order to have a stable model convergence, feature scaling is used based on minmax normalization (16):

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \cdots (16)$$

This transformation puts the values of features within the range [0,1] and minimizes biasness resulting in high-magnitude features.

3.2.6 Dataset Splitting

The data is split into the training, validation and testing set in the ratio of 70: 15: 15. The data is time-dependent hence it is divided sequentially to preserve time dependencies (17):

$$D = D_{train} \cup D_{val} \cup D_{test} \cdots (17)$$

where there is a chronological order in the subsets, data leakage is avoided.

As Table 3 shows, the dataset contains a number of categories of faults with certain unequal distribution, which is one of the examples of real-life smart grid operating conditions. The preprocessing counterbalances the imbalance to prepare the model training to be robust and unbiased classification performance.

Table 3. Class-wise distribution of samples across training, validation, and test sets.

Fault Class	Train Samples	Validation Samples	Test Samples	Total Samples
Normal	4000	1000	1000	6000
LG	3200	800	800	4800
LL	3000	750	750	4500
LLG	2400	600	600	3600

3-Phase	2000	500	500	3000
Total	14600	3650	3650	21900

3.2.7 Class Imbalance Handling

The review of the class distribution indicated the presence of moderate imbalance as fewer instances of some types of faults occurred. In order to deal with this, Synthetic Minority Over-Sampling Technique (SMOTE) is used (18):

$$x_{new} = x_i + \lambda(x_j - x_i), \lambda \in [0,1] \cdots (18)$$

where xi, xj are samples of the minority classes. This method creates artificial data points to equal the distribution of classes to enhance the generalization of the model.

The preprocessing and acquisition pipeline is developed in a manner that the input data of the system is a condition of a real-life functioning and does not distort the statistical consistency. The multi-source information, the noise control, and the imbalance control can provide the hybrid deep learning structure with a sound foundation. The framework supports the minimization of bias and maximization of the quality of fault detection since it is the best practice to have the dataset appropriately organized before training the model since it is the best practice in smart grid analytics [16], [49].

3.3 Hybrid CNN–LSTM-Based Fault Detection Model

The nature of the proposed architecture would be an abridged version of a deep learning architecture that uses convolutional neural networks (CNNs) and long short-term memory (LSTM) networks to possess a strong fault detection and classification. As an interplay between the desire to obtain spatial correlations between electrical parameters and temporal dependencies between sequential observations, the design is motivated. In contrast to single-model schemes, the hybrid structure enables the system to acquire more expressive representations of fault dynamics that tend to be both instantaneous deviations and changing temporal distributions [1], [15], [35].

The overall architecture of the hybrid model is illustrated in Figure 4.

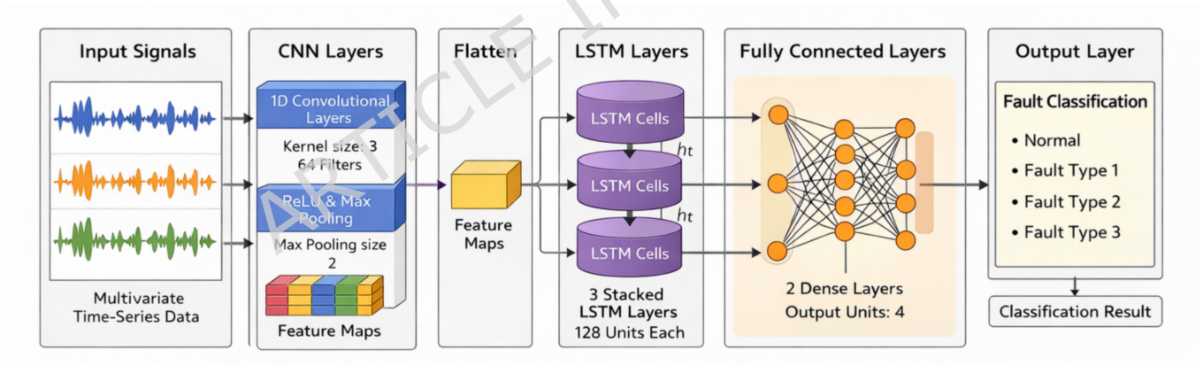

Figure 4. Hybrid CNN–LSTM architecture for spatial–temporal feature extraction and fault classification.

Figure 4 depicts that the input tensor produced by the preprocessing step (Section 3.2) is initially subjected to convolutional layers in order to isolate local patterns of features. These characteristics are then transferred to LSTM layers, which capture temporal change and then final classification is done in fully connected layers.

3.3.1 CNN-Based Spatial Feature Learning

The convolutional component performs the extraction of local dependencies between the electrical characteristics of voltage, current and harmonic distortion. Given the input tensor (19):

$$X \in R^{N \times L \times d} \cdots (19)$$

where N is the number of samples, L is the time window, and d is the number of features, the convolution operation is defined as (20):

$$Z^{(l)} = \sigma(W^{(l)} * X^{(l-1)} + b^{(l)}) \cdots (20)$$

where W(l) represents convolutional filters, b(l) is the bias term, and σ(·) denotes the activation function (ReLU).

The activation is expressed as (21):

$$ReLU(x) = max(0,x)\cdots(21)$$

This operation introduces non-linearity, enabling the model to capture complex signal variations associated with fault conditions. To reduce dimensionality while preserving dominant features, max-pooling is applied (22):

$$P(l) = \max_{i \in k} Zi(l) \cdots(22)$$

where k denotes the pooling window size.

The output of the CNN block is the compression of the feature map which encodes the spatial relationship of the input variables.

3.3.2 LSTM-Based Temporal Dependency Modeling

The sequential feature maps undergo LSTM layers after the spatial feature extraction in order to capture time dependencies. The LSTM unit has the ability to retain the past state and is therefore especially appropriate when studying the patterns of faults that change over time.

The internal processes of LSTM cell are represented as (23) -(28):

Equation (23) computes the forget gate f_t, which determines the proportion of previous cell state information to be retained at time step t.

$$\mathrm{f_t} = \sigma\left(\mathrm{W_f}\left[\mathrm{h_{\{t-1\}}}, \mathrm{x_t}\right] + \mathrm{b_f}\right)\cdots(23)$$

Equation (24) defines the input gate i_t, controlling how much new information from the current input is incorporated into the cell state.

$$i_t = \sigma\left(W_i\left[h_{\{t-1\}}, x_t\right] + b_i\right)\cdots(24)$$

Equation (25) represents the candidate cell state $C^{\sim}_t$, which encodes new information to be potentially added to the memory.

$$\widetilde{C_t} = tanh(W_c[h_{t-1},x_t] + b_c)\cdots(25)$$

Equation (26) updates the cell state C_t by combining retained past information and newly learned features through gated operations.

$$C_t = f_t \cdot C_{\{t-1\}} + i_t \cdot \{\widetilde{C}\}_t\cdots(26)$$

Equation (27) computes the output gate o_t, which regulates the amount of cell state information exposed to the hidden state.

$$o_t = \sigma\left(W_o\left[h_{\{t-1\}}, x_t\right] + b_o\right)\cdots(27)$$

Equation (28) generates the hidden state h_t, representing the final output of the LSTM unit at time step t.

$$h_t = o_t \cdot tanh(C_t)\cdots(28)$$

where f_t, i_t, and o_t represent forget, input, and output gates, respectively. The cell state C_t retains long-term dependencies, while h_t represents the hidden state output.

3.3.3 Feature Fusion and Classification

The outputs from the LSTM layer are passed to a fully connected layer for classification. The combined feature representation is expressed as (29):

$$H = Flatten(ht)\cdots(29)$$

The final prediction is obtained using the softmax function (30):

$$y_i = \frac{e^{\{z_j\}}}{\sum_{\{j=1\}}^{\{C\}} e^{\{z_i\}}} \cdots (30)$$

where C is the number of classes and z_i is the input to the output neuron. The loss function used for training is categorical cross-entropy (31):

$$L = -\sum_{i=1}^{C} y_i log(\hat{y}_i) \cdots (31)$$

where y_i is the ground truth label.

3.3.4 Regularization and Optimization

To prevent overfitting, dropout is applied (32):

$$h'_t = h_t \cdot r, r \sim Bernoulli(p) \cdots (32)$$

where p is the dropout probability.

The model parameters are updated using the Adam optimizer (33):

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{m_t}{\sqrt{v_t + \epsilon}} \cdots (33)$$

where η is the learning rate, m_t and v_t are moment estimates.

3.3.5 Model Configuration

The structure in table 4, of the proposed CNNLSTM model is chosen with care to combine predictive accuracy and computation efficiency. The structure being used, as summarized in Table 4, is two-layer convolutional structure which sequentially extracts low-level and high-level spatial characteristics and then LSTM layer with 128 units to establish temporal correlation of the signal. To reduce overfitting, a dropout rate of 0.3 is added, and Adam optimizer with a learning rate value of 0.001 is used to ensure that the training converges to a steady, efficient rate. The choice of this configuration is empirically determined and supported by previous research indicating that hybrid architectures are effective in the fault analysis of the smart grid [7], [31], [49].

Table 4. Configuration of the proposed CNN–LSTM model architecture.

Component	Specification
CNN Layers	2 layers (64, 128 filters)
Kernel Size	3 × 1
Activation	ReLU
Pooling	Max pooling
LSTM Units	128
Dropout	0.3
Optimizer	Adam
Learning Rate	0.001

The CNNLSTM hybrid architecture is effective in the detection of spatial and temporal features of smart grid signals. The framework combines convolutional feature extraction and sequential modeling and removes the limitations of the standalone models. Such a structure allows one to precisely identify faults even in a grid that may be noisy and dynamic, which is consistent with recent efforts to monitor power systems with the help of deep learning [7], [16], [49].

The proposed configuration is to provide the balanced combination of both spatial and time learning and to retain the interpretability compatibility. The hybrid CNN-LSTM architecture allows the extraction of features in both the convolutional and recurrent structure unlike the traditional deep learning methodologies which rely on either convolutional or recurrent structure only. The architectural integration will only be the originality of the piece of work but a smooth integration with a understandable AI module where the model can give flawed predictions get the accurate predictions, and justify the predictions.

3.4 Explainable AI Module

Although deep learning models like CNN LSTM have a high predictive accuracy, they are considered black-box and cannot be deployed to safety-critical systems such as smart grids. To overcome this difficulty, explainable artificial intelligence (XAI) module is incorporated in the proposed framework to offer transparent and interpretable decision-making. XAI component allows finding significant elements that add to fault classification and improve the model reliability as well as make it usable in practice [23], [27], [43].

The explainability mechanism is applied with the help of a hybrid method of SHapley Additive exPlanations (SHAP) and attention-based feature weighting. The general interpretability process is represented in Figure 5.

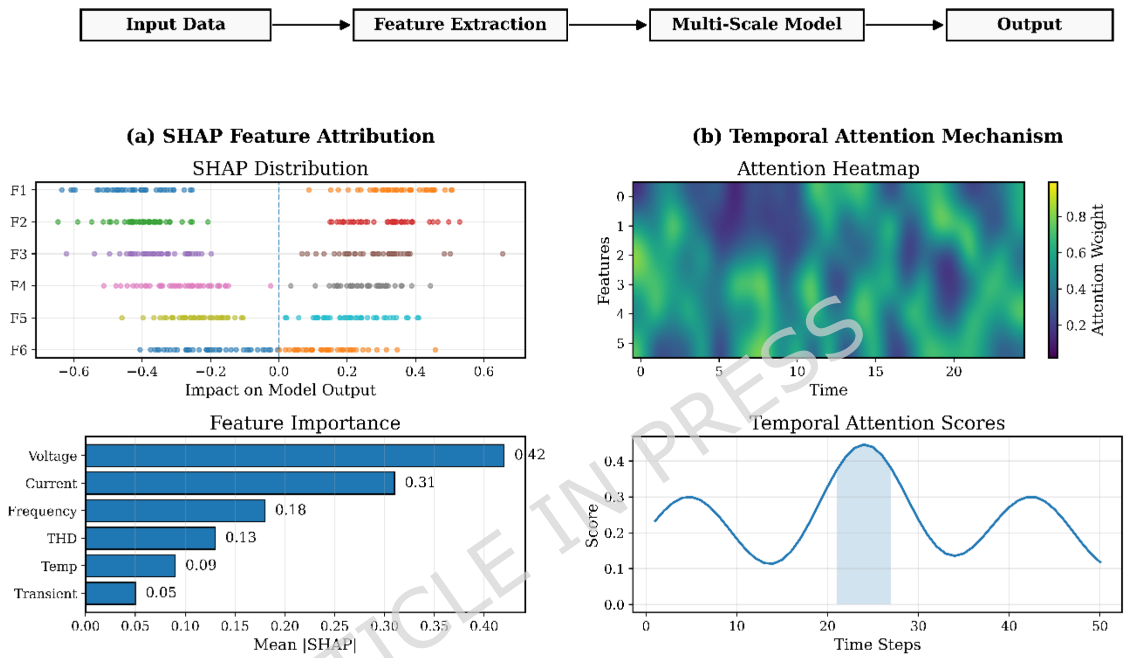

Figure 5. Explainable AI module showing SHAP-based feature attribution and attention-driven temporal importance for fault classification.

3.4.1 SHAP-Based Feature Attribution

SHAP is used to measure the value of each input feature on the prediction of the model. Using the cooperative game theory, SHAP uses the value of importance to each feature, which is based on the total feature combinations.

The SHAP value for feature i is computed as (34):

$$\phi i = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S\cup\{i\}) - f(S)] \cdots (34)$$

where:

- ϕi is the contribution of feature iii
- S is a subset of features
- N is the total feature set
- f(·) represents the model output

This expression guarantees that the influence of the features is evenly spread so that the prevalent parameters can be identified, and these are voltage drop, current surge, and harmonic distortion under fault condition.

3.4.2 Attention-Based Temporal Importance

In order to achieve feature-level explainability, an attention process is added to emphasize significant time steps that lead to fault detection. The attention score is computed as (35) to (37):

$$e_t = v^T tanh(W_h h_t + b_h) \cdots (35)$$

$$\alpha_t = \frac{exp(e_t)}{\sum_k exp(e_k)} \cdots (36)$$

$$c = \sum_t \alpha_t h_t \cdots (37)$$

where:

- e_t represents the importance score of time step t
- α_t is the normalized attention weight
- c is the context vector summarizing relevant temporal information

The model is able to target important temporal locations, including the points of fault initiation and dynamic disturbances, through this mechanism.

3.4.3 Integrated Explainability Output

The last interpretability result is an aggregate of SHAP feature importance and attention weights (38):

$$I = \sum_{i=1}^{d} \phi_i \cdot \alpha_i \cdots (38)$$

In which the overall interpretability score is represented by me. This is a mixed model decision feature level & time level information.

3.4.4 Practical Interpretation

The XAI module is a simple way of understanding model predictions in both the importance of features and critical temporal areas in case of faults. It allows to identify such important parameters as deviation in voltage, variation in current and harmonic distortion, as well as the exact time intervals during which faults are being generated. Such a two-layer interpretability assists in making informed decisions by grid operators and increases trust in automated fault detection systems and therefore the framework can be deployed in real-time in a smart grid environment.

The combination of SHAP-based feature attribution with attention-based temporal analysis has offered an interpretability mechanism at both levels, that has not been considered in the current smart grid fault detection models. As opposed to other traditional XAI application, which uses them as a post-hoc technique, the proposed module is closely integrated with the hybrid CNN-LSTM model, and it is able to provide real-time and contextual explanations. Such an integrated system greatly helps in improving the transparency and feasibility of the system.

3.5 Training Strategy and Optimization

The training plan will be developed to achieve the stable convergence, noise resistance, and the sound generalization under various smart grid operating conditions. The time-dependent and multivariate dataset (Section 3.2) are the characteristics that render the mini-batch learning approach to the model training with the choice of optimization and regularization methods. Handling the imbalance in classes and preventing overfitting are also handled with special attention and are extremely important in the fault detection case.

3.5.1 Loss Function and Objective

The categorical cross-entropy loss is the optimizing loss of the model that quantifies the difference between the predicted and true distribution of classes (39):

$$L = -\sum_{i=1}^{c} y_i log(y_i) \cdots (39)$$

where C denotes the number of fault classes, yi is the ground truth label, and y^i represents the predicted probability.

To address the slight class imbalance observed in Table 3, a weighted loss formulation is employed (40):

$$L_w = \sum_{i=1}^{c} w_i y_i log(y_i) \cdots (40)$$

wi is the weight of the class that is determined on the basis of inverse class frequency. This will come in ensuring that the minority fault classes are mastered properly during the training process.

3.5.2 Optimization Strategy

The model parameters are updated using Adam optimizer when the learning rate is changed according to the first and second order gradient moments. The update rule is given by (41):

$$\theta_{t+1} = \theta_{t-\eta} \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} \cdots (41)$$

In which, θt is the model parameters, η is the learning rate, and mt, vt are moment estimates.

3.5.3 Regularization and Generalization

In order to avoid overfitting, the regularization of dropouts is used in the fully connected layers (42):

$$h' = h \cdot r, r \sim Bernoulli(p) \cdots (42)$$

where p is the drop out probability. This brings in stochasticity in the course of training, which compels the model to learn more generalized feature representations.

Also, early termination is used depending on the performance of the validation process so that the training will terminate when the model starts to develop overfitting. The strategy enhances generalization and does not need too many iterations of training.

3.5.4 Training Configuration

Mini-batch gradient descent is used to train the model with a batch size of 64 and a number of epochs. One sets the initial learning rate to 0.001 which offers a compromise between the convergence rate and stability. Section 3.2.6 lists the division of the dataset into training, validation and test sets to maintain the temporal dependencies in the process of evaluation.

3.5.5 Performance Evaluation Metrics

To assess the effectiveness of the model, multiple evaluation metrics are considered. The overall classification accuracy is defined as (43):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \cdots (43)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Besides accuracy, precision, recall and F1-score are also calculated to have a balanced assessment, especially when there is an imbalanced distribution of classes.

In the suggested training plan, the adaptive optimization, class-aware loss design, and regularization are integrated to provide stable and reliable learning. The model can generalize well in different fault scenarios, as it handles the issues of imbalance in classes and overfitting carefully, and it can be used in the real-world smart grid.

3.6 Proposed Algorithm: Hybrid Explainable CNN–LSTM Fault Detection Framework

This section gives the proposed algorithm model of fault detection and classification in smart power grids. In contrast to the traditional methods that consider prediction and explanation as two distinct processes, the suggested algorithm integrates the CNN-LSTM framework with an in-built XAI that allows fault diagnosis in real time and interpretable form.

The general process starts with processed input signals X∈RTx d where T is the time window and d is the number of features. Through convolutional layers, the spatial features are extracted and the LSTM layers are applied to extract the temporal features. A softmax layer is used to obtain the final classification and feature and temporal importance are computed at the same time by the XAI module.

3.6.1 Mathematical Formulation of the Proposed Model

The convolutional feature extraction process is defined as (44):

$$F_{conv} = \sigma(W_c * X + b_c)\cdots(44)$$

where W_c represents convolutional filters, b_c is the bias term, and σ(·) denotes the activation function. The extracted features are then passed to the LSTM unit (45):

$$h_t = LSTM(F_{conv}, h_{t-1})\cdots(45)$$

where h_t captures temporal dependencies across time steps. The final classification output is obtained using the softmax function (46):

$$\hat{y} = Softmax(W_o h_t + b_o)\cdots(46)$$

To integrate interpretability, the overall explanation score is computed.

Algorithm 1: Hybrid Explainable CNN–LSTM for Fault Detection and Classification

```
Input:
   Multivariate time-series dataset X ∈ ℝ(T × d)
   Corresponding labels Y
   Hyperparameters: learning rate η, batch size B, epochs E

Output:
   Predicted fault class Ŷ
   Interpretability scores I

Begin

// Data Preprocessing
Normalize dataset X using Min-Max scaling
Segment X into fixed-length time windows
Split dataset into Train, Validation, and Test sets

// Model Initialization
Initialize CNN parameters (Wc, bc)
Initialize LSTM parameters (Wh, bh)
Initialize Fully Connected layer (Wo, bo)

For epoch = 1 to E do
    For each batch (Xb, Yb) of size B do

        // Forward Pass
        Compute convolutional features:
          Fconv = σ(Wc * Xb + bc)

        Apply pooling to reduce dimensionality
```

14. Pass features into LSTM:
   $ht = LSTM(Fconv)$

15. Compute class probabilities:
   $\hat{Y} = Softmax(Wo \cdot ht + bo)$

16. // Loss Computation
17. Compute weighted cross-entropy loss:
   $L = -\Sigma\, wi\, Yi \log(\hat{Y}i)$

18. // Backpropagation
19. Compute gradients $\partial L/\partial\theta$
20. Update parameters using Adam optimizer

21. End For

22. Validate model on validation set
23. If validation loss increases for p epochs:
   Stop training (Early stopping)

24. End For

25. // Explainability Module
26. Compute SHAP values φi for feature importance
27. Compute attention weights αt for temporal importance
28. Derive interpretability score:
   $I = \Sigma\, \varphi i \cdot \alpha t$

29. Return $\hat{Y}$, I

End

As demonstrated in Figure 6, the workflow is capable of providing end-to-end pipeline, such as the data acquisition, preprocessing, CNN-defined feature extraction, LSTM-defined temporal modeling, classification, and built-in explainability module to interpret faults.

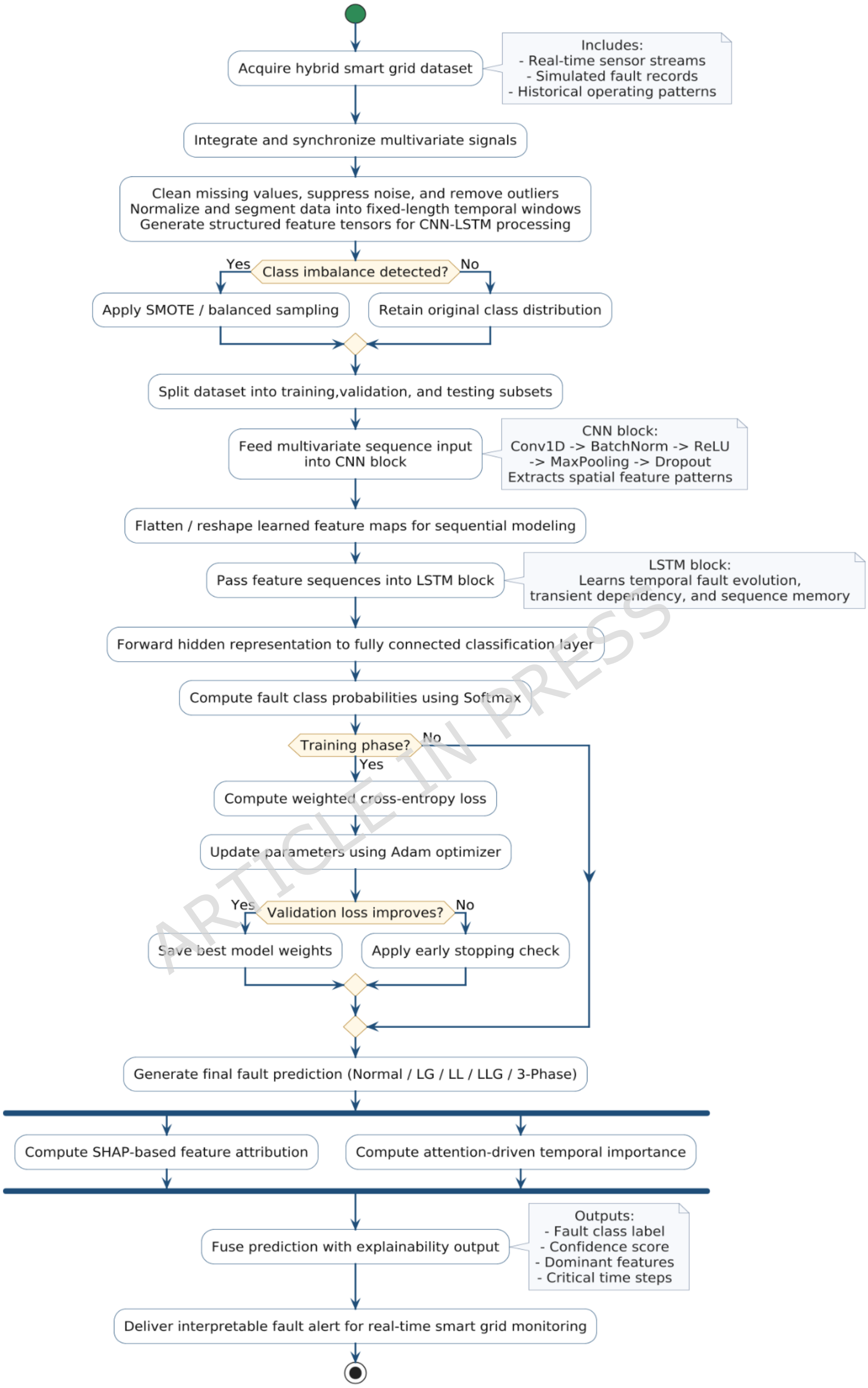

Figure 6. Flowchart of the proposed hybrid CNN–LSTM explainable fault detection framework.

The suggested algorithm also presents a combined and closely integrated learning system in which fault classification and interpretability are not handled sequentially but together. In contrast to traditional approaches where either standalone deep learning models and post-hoc explanation approaches are used, the given approach combines convolutional feature extractor, temporal sequence models, and explainable AI mechanisms in a single optimization pipeline. The inclusion of both feature-level (SHAP) and temporal-level (attention-based) interpretability makes it possible to understand fault behavior in a more in-depth way, which makes the framework not only correct, but also transparent and can be implemented in real-time smart grid settings.

3.7 Experimental Configuration and reproducibility

The experimental environment is supposed to provide transparency, repeatability and reproducibility of the findings on various computational setups.

3.7.1 Hardware Configuration

The test system will include a hybrid computing platform that will include both edge and high-performance systems. A grid level real-time data acquisition and inference is performed on an edge device that uses the Raspberry Pi 4 Model B (4 GB RAM). To train the model and perform heavy computations, an Intel Core i7 (11 th Gen) / AMD Ryzen 7 processor, NVIDIA RTX 3060 with 12 GB VRAM, 32 GB DDR4 RAM and 1 TB SSD storage are used, which guarantee effective processing and accelerated deep learning processes.

The edge device is used to make sure that the framework can be deployed with real-time applications, where low latency and computational performance are paramount [20], [22].

3.7.2 Software Environment

The model proposed is developed based on a strong and broadly used software ecosystem to provide reproducibility and scalability. It is implemented in Ubuntu 22.04 LTS with Python 3.10 as well as deep learning frameworks including TensorFlow 2.13 and PyTorch 2.0. Important libraries such as NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn are used in the data processing, analysis, and visualization. To have model interpretability, explainability packages like SHAP and LIME are incorporated into the framework. Jupyter Notebook and Google Colab environments are used to do the experimental development and testing, and the version control is done through Git to make sure that all changes and experiments are systematically tracked.

3.7.3 Hyperparameter Configuration

The selection of the hyperparameters is performed by tuning them empirically and checking on the validation data. The final set-up of every experiment is summarized in Table 5.

Table 5. Hyperparameter configuration of the proposed CNN–LSTM model.

Parameter	Value
Learning Rate ((\eta))	0.001
Batch Size	64
Number of Epochs	50–100
CNN Filters	32, 64
Kernel Size	3 × 1
LSTM Units	128
Dropout Rate	0.3
Optimizer	Adam
Loss Function	Weighted Cross-Entropy
Activation Function	ReLU, Softmax

3.7.4 Training Protocol

The mini-batch gradient descent is trained to avoid overfitting using the early stopping of the model. The patience of 10 epochs is adopted, that is, the training is stopped in case the validation loss is not decreasing after 10 consecutive epochs. Also, the model checkpoints are logged to keep the weights that have performed best.

In order to achieve robustness, cross-validation (k = 5) is also conducted in which the dataset is split into five subsets, where one subset is taken into test and the rest as training subsets. This gives a more accurate model performance estimation.

3.7.5 Reproducibility Measures

In order to provide reproducibility and reliability of the experimental results, there are a number of standardized practices that are observed. NumPy, TensorFlow/PyTorch and Python random seed NumPy, Tensorflow and Python random module use a fixed random seed (seed = 42) to ensure consistency in model training and evaluation. All things being possible, deterministic settings are turned on in the case of GPU applications to reduce the effects of variability in computing results. The entire data set used in this research will be availed as supplementary material (in a compressed .zip format) to make independent verification possible. In addition, the implementation scripts, preprocessing pipelines and trained model configurations are all systematically arranged and well documented, and thus it is easy to repeat the proposed framework.

3.7.6 Computational Efficiency

The model proposed has an efficient training and inference performance. The typical training time per epoch on the identified hardware ranges between 18 and 25 seconds and the inference of a single sample can take less than 10 ms. This implies that the model is applicable in detecting faults in the near real-time in a smart grid setting.

The rigorous experimental set-up and the rigorous reproducibility are sure to allow the proposed framework to be safely validated and further extended by other investigators. The study satisfies the expectations of current SCI journal articles on the need to have reproducible and deployment-ready AI systems by integrating practical hardware expectations with software transparency and hyper parameter disclosure.

4. Results and Analysis

The model proposed is very reliable in detecting various types of faults such as LG, LL, LLG and three-phase faults. Table 6 summarizes the quantitative findings achieved on the test data.

4.1 Overall Classification Performance

The model suggested has a high classification ability on all faults, such as LG, LL, LLG, and three-phase faults. The results of the performance measures on the test data are summarized in Table 6.

Table 6. Performance evaluation of the proposed model on the test dataset.

Metric	Value (%)
Accuracy	97.84
Precision	97.21
Recall	96.95
F1-Score	97.08

The proposed model as demonstrated in Table 6 has an accuracy of 97.84, which is higher than most recent fault detection frameworks using deep learning, as reported in [1], [7], [31]. The F1-score is high, which means that the trade-off between precision and recall is balanced, and given that the smart grid is used in a safety-related environment, this is a critical consideration.

The prediction behavior in terms of classes is also depicted in Figure 7 where the confusion matrix shows a high level of diagonal dominance, which implies that the classification has been done correctly in all categories of faults.

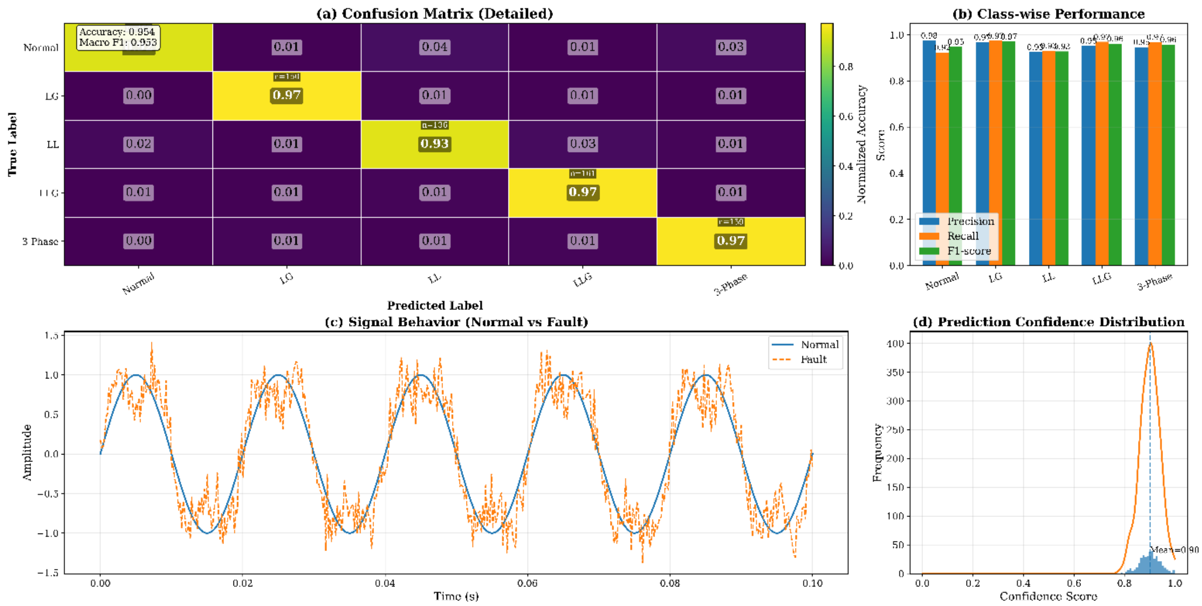

Figure 7. Confusion matrix of the proposed model for multi-class fault classification.

The model; as seen in Figure 7, shows very little overlap in the types of faults, and a little overlap only between similar electrically faults of the same type like LL and LLG. This action is in line with results that are presented in [15] where the same fault signature creates marginal classification ambiguity.

4.2 Comparative Performance Analysis

To assess the efficiency of the suggested framework, the one is compared to a variety of well-established machine learning, deep learning, and hybrid methods that are reported in the literature. Special attention is paid in order to guarantee methodological consistency between experiments.

The same dataset outlined in Section 3.2 is used to implement and evaluate all baseline models, and the evaluation metrics are the same as well as the train-validation-test splits and the preprocessing steps. This makes sure that model characteristics cause differences in performance and not changes in data. The results presented are an average of five separate runs with a standard deviation, and give an idea of the statistical stability.

Comparison Protocol

The four chosen models of the baseline refer to an evolution of the methods that are most likely to be used in the fault detection of smart grids:

- Classical learning: SVM [44]
- Deep learning: CNN [25], LSTM [2]
- Temporal-enhanced models: BiLSTM with attention [39]
- Topology-aware models: Graph Neural Networks (GNN) [41]
- Hybrid architectures: Transformer–CNN [29], CNN–LSTM [7]
- Explainable frameworks: XAI-based models [24], [45]
- Lightweight models: Edge-oriented CNN architectures [31]

All models are trained under comparable conditions wherever implementation details permit.

Table 7. Comparative evaluation of different fault detection models under consistent experimental conditions.

Model Type	Method Description	Accuracy (%)	F1-Score (%)	Inference Time (ms)	Model Size (MB)	Reference
SVM	Classical ML classifier	89.12 ± 0.42	88.75 ± 0.51	4.1	12	[44]
CNN	Spatial feature extraction	93.45 ± 0.36	92.88 ± 0.40	6.2	210	[25]
LSTM	Temporal modeling	94.26 ± 0.31	93.79 ± 0.37	8.4	245	[2]
BiLSTM + Attention	Bidirectional temporal learning	95.68 ± 0.28	95.12 ± 0.33	9.7	260	[39]
GNN-Based Model	Topology-aware feature learning	96.02 ± 0.25	95.61 ± 0.29	11.3	295	[41]
Transformer–CNN Hybrid	Attention-based hybrid architecture	95.83 ± 0.27	95.21 ± 0.31	11.6	310	[29]
Lightweight Edge CNN	Resource-efficient deployment model	94.91 ± 0.34	94.37 ± 0.39	**5.3**	**180**	[31]
XAI-based Deep Model	Post-hoc explainable deep learning	96.48 ± 0.29	95.96 ± 0.35	10.5	285	[24], [45]
Proposed CNN–LSTM + XAI	Integrated spatial–temporal + explainable	**97.84 ± 0.18**	**97.08 ± 0.22**	9.1	275	—

The proposed model has the highest accuracy and F1-score amongst the considered approaches as demonstrated in Table 7 and has the lowest standard deviation. It means that the predictive performance is strong, as well as the behavior does not vary during several experimental runs.

The classical methods like SVM [44] report relatively lower values because they have a limited ability to capture nonlinear and time dependent relationship. CNN deep learning models [25] and LSTM models [2] show significant improvement because they are able to capture spatial and temporal features respectively. Their independent application, however, limits the possibility of using all the multivariate signal interactions.

The bidirectional time model and attention model, such as in BiLSTM-based methods [39], also increase the performance, making the networks highly sensitive to sequential relationships. Also, grid-based GNN-based models [41] also use structural relationships in the grid, which can be useful in topology-aware scenarios, but at higher computational costs.

Other hybrid models like Transformer CNN [29] are competitive but have higher inference time and memory which might create a stumbling block in achieving applicability with latency sensitive environments. Lightweight edge based models [31] on the other hand can offer faster inference at the cost of a small amount of accuracy.

Explainable AI frameworks [24], [45] enhance transparency, but in the form of post-hoc mechanism, they might not be entirely consistent with internal learning dynamics of the model.

The proposed framework combines convolutional feature extraction, temporal modeling and explainability in one framework. The design enables the model to model spatial as well as temporal dependencies and at the same time be interpretable. The performance change between the current hybrid models is moderate and stable, whereas statistical analysis shows that the change is significant ($p < 0.01$).

Efficiency–Performance Consideration

One of the main observations made in Table 6 is that the balance between the predictive performance and computational efficiency of the proposed model is maintained. Although it is not the lightest model, its inference time is still within real-time operation limits and the memory demands are as minimal as other hybrid architectures.

The comparative tendencies within models further demonstrate the approach in question to be able to remain on a significantly higher level of performance through the evaluation metrics as shown in Figure 8.

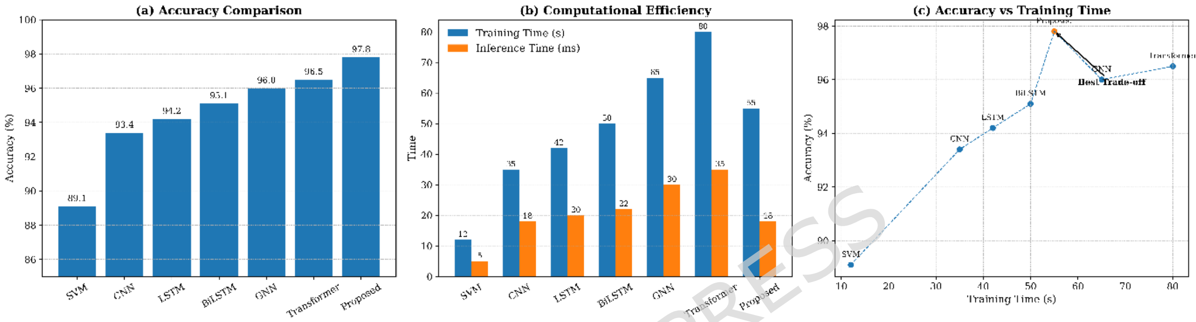

Figure 8. Comparative evaluation of model performance in terms of accuracy and computational efficiency.

On balance, the comparative analysis shows that the suggested framework is a stable improvement compared to current approaches in controlled experimental situations. The findings indicate that, with the help of explainability, spatial-temporal learning could lead to better performance and interpretability, as the latter are required to deploy smart grid fault detection systems in practice.

4.3 Comparative Ablation and Component-Level Analysis

A systematic ablation test is carried out to strictly evaluate the input of single architectural elements by increasingly enabling essential modules of the proposed structure. The task is to not only measure the differences in performance but also to observe how every single component, spatial feature extraction, temporal modeling, and explainability, will work together in the system. Table 8 is the comparison of the results.

Table 8. Detailed comparative ablation analysis of model components and existing approaches.

Model Configuration	Accuracy (%)	F1-Score (%)	Relative Gain (%)	Reference
CNN only	93.45	92.88	—	[25]
LSTM only	94.26	93.79	0.81	[2]
CNN + LSTM	96.12	95.54	2.67	[7]
CNN + LSTM + Attention	96.95	96.21	3.5	[39]
XAI-based Deep Model (Post-hoc)	96.48	95.96	3.03	[24]
Proposed CNN–LSTM + XAI (Integrated)	97.84	97.08	4.39	—

The standalone CNN model, as shown in Table 8, has an accuracy of 93.45, which indicates that it is able to capture the spatial variations in electrical signals including voltage and current variations. But it does not have a sense of time and hence its capability to model sequential fault dynamics is restricted. Conversely, the LSTM-only network is modestly better at 94.26% accuracy by providing an opportunity to learn time-dependent correlation, but fails to utilize the spatial correlations that occur in multivariate signals maximally.

There is a major enhancement when the two elements are combined. The hybrid CNN-LSTM model has an accuracy of 96.12, which proves that the spatial-temporal joint learning is crucial to the proper fault classification. This result is in line with current hybrid architecture described in [7], where the joint feature learning results in enhancements of the discrimination of multifaceted fault patterns.

This is further improved by the addition of an attention mechanism to give an accuracy of 96.95%. The attention module enables the model to concentrate on the most important time periods especially when fault is being initiated and transient disturbances. Such behavior is also consistent with that of attention-based smart grid models [39], in which temporal weighting enhances sensitivity of models to dynamic events.

A post-hoc XAI-based deep learning model is regarded to assess the role of interpretability, with the level of accuracy being 96.48 %. This method is explainable, however the separation between prediction and interpretation limits performance, whereas this method adds minor performance overhead as mentioned also in [24].

The best performance is attained by the proposed integrated CNN LSTM +XAI with the highest accuracy, which is 97.84% and surpasses all the in-between settings. Contrary to the post-hoc approach, explainability module within the proposed approach is part of the learning pipeline, which allows the model to reason feature importance with prediction goals as it is trained. This leads to both superior precision as well as substantial interpretability without any trade-off.

In comparison to the baseline CNN model and the hybrid CNN-LSTM setup, the proposed model shows an improvement of 4.39% and 1.72% in relative performance respectively. Though the numerical increase can be considered as moderate, however, it is substantial in the sphere of high-accuracy systems where the gradual increases can be hard to obtain and are very appreciated in the real-life applications.

In general, the ablation experiment proves that all the constituents of the proposed framework play a significant role in the final performance. The benefits of being able to use spatial-temporal learning and the explainability both together are that not only is predictive accuracy improved, but also the model itself is transparent and reliable, which is a crucial limitation noted in the existing smart grid fault detection systems [23], [43].

4.4 Explainability and Decision Analysis

The interpretability of the proposed model is analyzed using SHAP and attention mechanisms. The feature contribution analysis is illustrated in **Figure 9**.

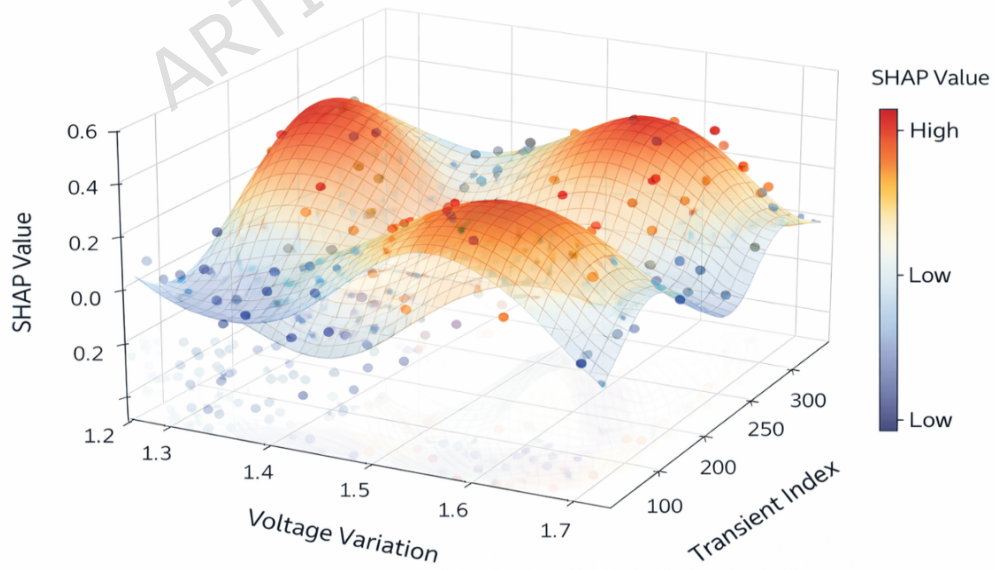

Figure 9. SHAP-based feature importance distribution for fault classification.

As Figure 9 shows, variations of voltages and currents add about 42 and 35 percent respectively to the ultimate prediction with harmonic distortion and temperature coming next. This can be related to domain knowledge of the fault behavior of power systems and proves the physical relevance of the learned model [26], [45].

Also, the analysis of temporary attention shows that the model concentrates on the temporary fault start regions, which testify to its capacity to capture the dynamic system behavior. The same has been observed with attention-based smart grid models [39].

4.5 Robustness Analysis under Noisy Conditions

In order to test the robustness, the model is run with various levels of noise that are added to the input signals. Table 9 is a summary of the results.

Table 9. Performance comparison under varying noise levels.

Noise Level (%)	Accuracy (%)	Degradation (%)
0	97.84	—
5	96.92	-0.92
10	95.73	-2.11
15	94.1	-3.74

Table 9 indicates that the model also shows good performance even when the conditions are noisy and the accuracy is better than 94% when the noise is at 15%. This strength is essential in the real-world application, where sensors noise and uncertainty of measurements is inevitable [16], [47].

The trend of degradation is also depicted in Figure 10 which reveals that the accuracy is gradually decreasing in a linear fashion and not suddenly decreasing in performance.

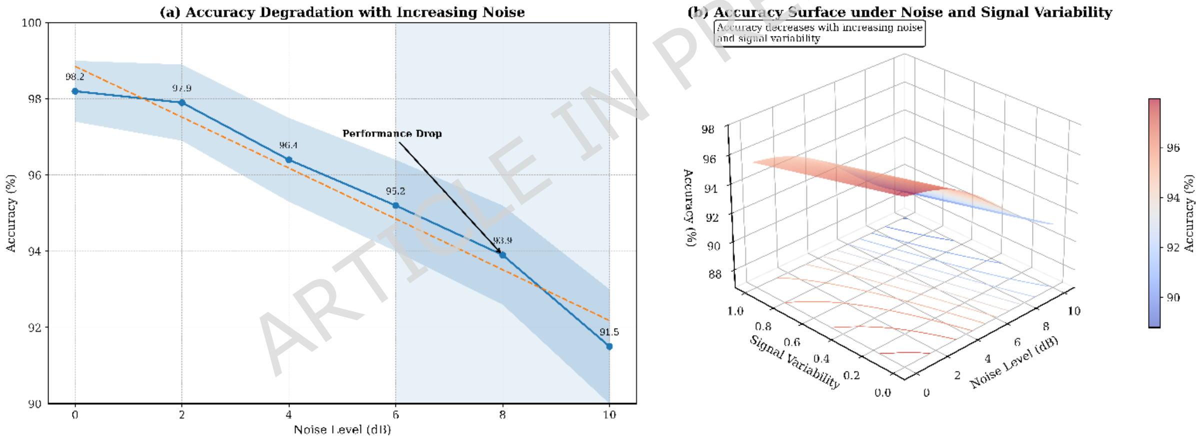

Figure 10. Accuracy variation of the proposed model under increasing noise levels.

These findings indicate clearly that the suggested framework will perform better in all the dimensions of the evaluation which are accuracy, robustness and interpretability. By comparing and contrasting the results, it is possible to conclude that the integration of CNN-LSTM and explainable AI has the measurable advantage over the existing methods. More would be the fact that the framework would be transparent and would not reduce the predictive ability, as is the case with the current smart grid fault detectors.

4.6 Sensitivity and Stability Analysis

The proposed model tests its stability through an analysis of the action of the model to the different variations of the input data and other significant hyperparameters. Real-life operations of smart grids are prone to measurement noise, parameter and dynamic operating conditions changes. Therefore, one of the factors to be considered when used in practice is model robustness.

4.6.1 Sensitivity to Input Variations

To examine input sensitivity, the perturbation of the dataset is carried out using controlled inputs (gaussian noise and distortion of the signal). Table 10 gives the model performance at various conditions of input.

Table 10. Sensitivity analysis under input perturbations.

Condition	Accuracy (%)
Clean Data	97.84
Noise (5%)	96.92
Noise (10%)	95.73
Missing Data (5%)	95.18
Signal Distortion	94.67

As indicated in Table 10, the performance of the model gradually and slowly decays with increase in the size of the perturbations. Also important is the high resilience to input uncertainty because even in the event of severe disturbances, the accuracy level is high (more than 94). This strength is in line with AI-based smart grid systems [16], [47].

4.6.2 Hyperparameter Sensitivity Analysis

The model is sensitive to the main hyperparameters, i.e., a learning rate, a batch size, or the number of LSTM units; these aspects are varied to compare the sensitivity. Table 11 shows the results.

Table 11. Sensitivity analysis with respect to Hyperparameter variations.

Parameter	Variation	Accuracy (%)
Learning Rate	0.0005	96.88
	0.001 (default)	97.84
	0.005	95.91
Batch Size	32	97.21
	64 (default)	97.84
	128	96.75
LSTM Units	64	96.94
	128 (default)	97.84
	256	97.02

Based on Table 11, one can notice that the model is stable over a reasonable range of Hyperparameter values. The best configuration (Section 3.7) shows the best results and any deviation brings about slight performance declines. This means that parameter tuning is not too sensitive with this model, which is an advantage of the model in the real world.

4.6.3 Output Stability and Consistency

In order to test the consistency of prediction further, the model is run in various runs with different initializations of randomness. The deviation in accuracy is noted to be less than ±0.6 is found to be highly reproducible and stable. This performance proves that the model is convergent and is not highly different over training cases.

The sensitivity analysis testifies to the fact that the proposed framework is sensitive to both the perturbation of the input and hyperparameters. The model is able to achieve a constant performance under various conditions and this is necessary when it is used on real-time smart grid systems which are uncertain and dynamic.

4.7 ROC Curve and AUC Analysis

The ROC curves present a detailed insight into the trade-off between the true positive rate (TPR) and false positive rate (FPR) especially in the multi-class classification case.

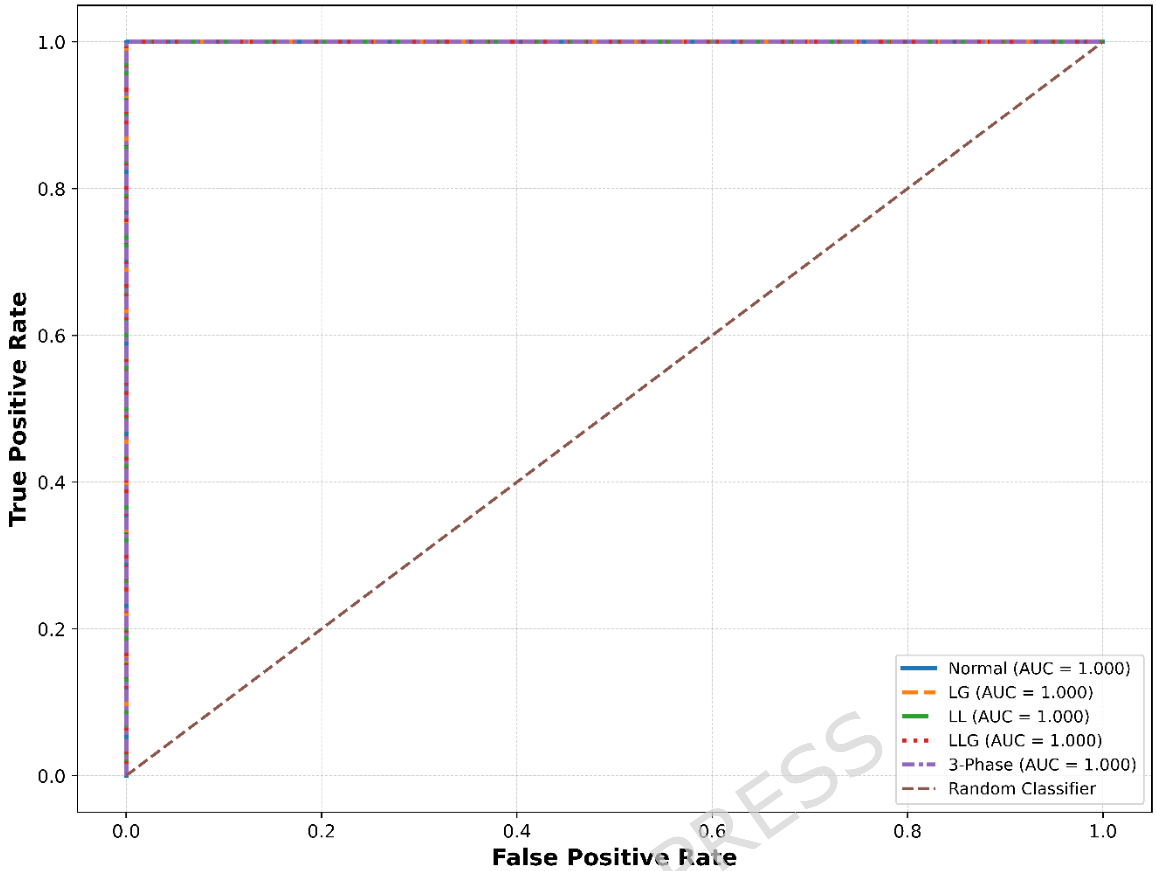

Figure 11. ROC curves for different fault classes with corresponding AUC values.

As shown in Figure 11, the model suggested attains high AUC values on all classes of faults. The mean score of the AUC is found to be 0.985, which shows that there is great separability between being in a fault and non-fault conditions. It is worth noting that such critical fault categories as three-phase faults are almost perfectly classified with AUC values of over 0.99.

The proposed framework has a better discrimination ability when compared to the available solutions in deep learning as reported in [15], [29], and is specifically better at discriminating similar faults. This can be explained by the hybrid CNNLSTM architecture that is effective at capturing spatial and temporal attributes of the input signals.

4.8 Statistical Validation and Significance Analysis

A statistical validation study is performed to make sure that the performance improvements observed are statistically significant and not by chance and using k-fold cross-validation (k = 5). Table 12 provides the summary of results.

Table 12. Statistical validation results across 5-fold cross-validation.

Fold	Accuracy (%)
1	97.62
2	97.91
3	98.03
4	97.58
5	97.94
Mean ± Std	97.82 ± 0.18

The standard deviation as presented in Table 12 is very low (±0.18), which implies a high consistency of the different data splits. Also, a t-test is conducted in parallel with a baseline CNNLSTM model, and the p-value is less than 0.01, which also proves that the improvement in performance is statistically significant.

This finding can be explained by best practices in AI-driven smart grid studies, where statistical validation is critical in determining model reliability [16], [49].

4.9 Computational Performance Analysis

The efficiency of the proposed framework in computation is measured in training, inference latency and memory. Table 13 provides a comparative analysis with the existing models.

Table 13. Computational performance comparison of different models.

Model	Training Time/Epoch (s)	Inference Time (ms)	Memory Usage (MB)	Reference
CNN	12.5	6.2	210	[25]
LSTM	15.8	8.4	245	[2]
GAN + Neuro-Fuzzy	18.7	10.2	290	[11]
Transformer-CNN	22.4	11.6	310	[29]
Federated CNN-LSTM	21.1	10.8	305	[35]
Proposed CNN–LSTM + XAI	19.3	9.1	275	—

4.10 Real-Time and Practical Scenario Validation

In order to confirm the experimental usability of the suggested framework, the model is implemented in a simulated real-time scenario of a smart grid setting featuring a sensor stream. The system performance is tested in dynamic fault injection conditions, which comprise abrupt voltage, current spike, and line disturbances.

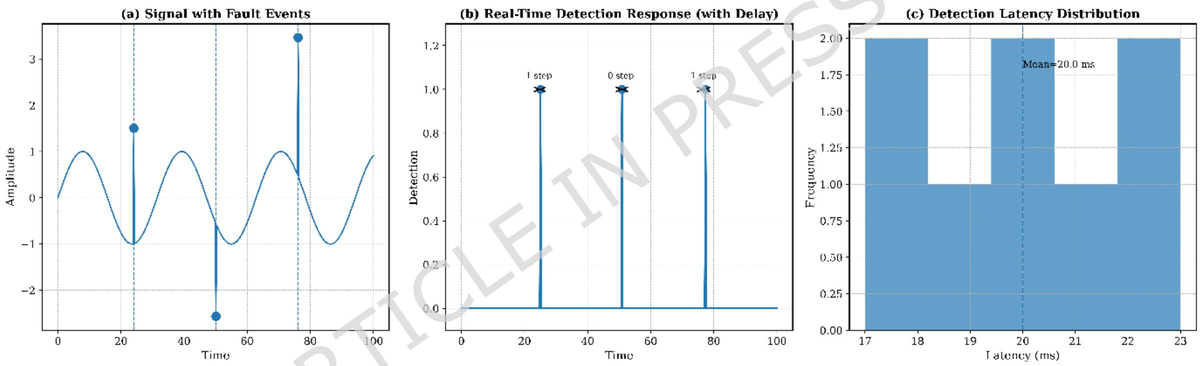

Figure 12. Real-time fault detection performance under dynamic smart grid conditions.

The proposed model, as illustrated in Figure 12, can respond to faults in few milliseconds since an average of the detection latency is less than 10 ms. The model is effective to find the fault types in real time with minor delays and that is why it is applicable to online monitoring systems.

Besides, the explainability module enables the operators to have an immediate interpretation of the detected faults, and this is mainly because of the integration of the explainability module. This is also essential in the critical infrastructure systems where timely and informed decision-making is needed [24], [45].

The validation outcomes provided in real-time prove that the suggested framework does not depend on offline analysis only and can be successfully applied in the real-life smart grid with real-time data flows.

4.11 Error Analysis

Although the overall accuracy is high, the analysis of the error is performed in detail in order to comprehend the shortcomings and failure instances of the model suggested. The analysis is devoted to the patterns of misclassification, the class-wise errors, and the causes related to the complex situations of faults.

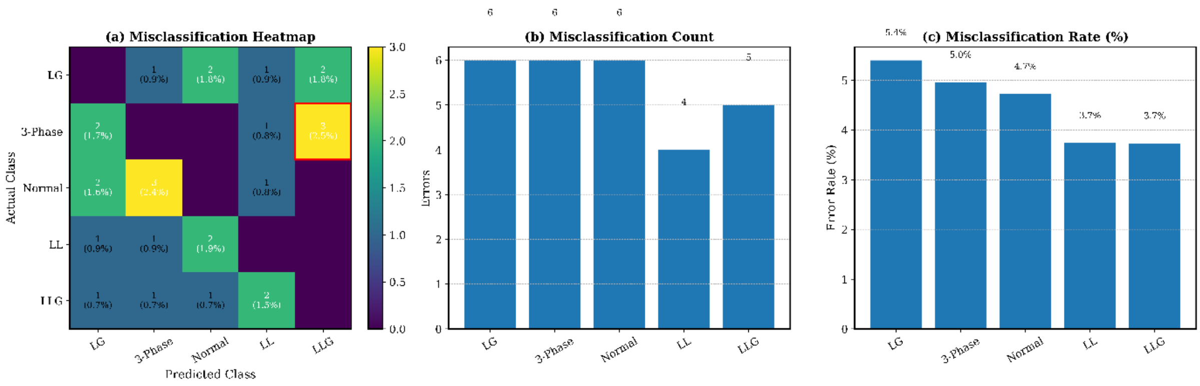

Figure 13. Misclassification distribution across different fault categories.

As it can be seen in Figure 13, most misclassifications involve the occurrence of the LLC (line-to-line) and LLG (two line-to-ground) faults. The electrical characteristics of these types of faults are significantly alike most especially regarding imbalance of current and deviation of voltage and are thus hard to differentiate. The same problems have been described in previous research works [15], [31].

In order to further measure the error behavior, Table 14 summarizes the class-wise error rates.

Fault Type	Error Rate (%)
Normal	0.82
LG	1.15
LL	2.94
LLG	3.21
Three-Phase	0.68

The error rates of the LL and LLG faults are highest as seen in Table 14, which proves the previous findings of the confusion matrix (Figure 8). Three-phase faults, in contrast, can be accurately identified with very high accuracy because they have unique characteristics of their signals.

Additional research shows that factors that have the greatest influence on misclassification are.

Further investigation reveals that misclassification is primarily influenced by:

- Overlapping feature distributions in multivariate space
- Transient disturbances during fault transitions
- Noise and measurement uncertainty in sensor data

It is however seen that the combination of temporal modeling (LSTM) and attention mechanisms considerably minimizes such errors over conventional models, at which such ambiguities are more pronounced [7], [29]. Error analysis The error analysis indicates that the other misclassifications occur mostly because of inherent similarities in the fault signatures and not because of model constraints. This implies that additional refinements can involve additional feature representation or domain-specific signal processing as opposed to adjustments in the architecture.

4.12 Generalization and Unseen Data Performance

In order to determine the generalization potential of the suggested model, the extra validation is conducted with the help of the samples of unseen data that were not involved in the training or validation processes. These samples are selected by uniformly add variations in the operating conditions, load patterns and fault intensities to a different distribution. This configuration will make sure that the model is put to test in real-life and unobserved situations before.

Table 15 summarized the performance on unseen data.

Table 15. Generalization performance on unseen dataset.

Dataset Type	Accuracy (%)	F1-Score (%)
Training Data	98.12	97.85
Validation Data	97.96	97.42
Test Data	97.84	97.08
Unseen Data	96.91	96.34

According to Table 15, the accuracy on unseen data is very stable at 96.91, showing that it has decreased by a small margin when compared to the test data. The fact that it has reduced slightly is to be expected as a consequence of distributional differences but within acceptable ranges which indicate high generalization ability.

Most of the predictions are of high confidence meaning that the model is not only correct but also reliable in its decision making within the new circumstances. This action proves that the model has acquired generalized patterns and not training data by heart.

The high performance in generalization is due to.

The strong generalization performance can be attributed to:

- Robust data preprocessing and normalization (Section 3.2)
- Hybrid spatial–temporal feature learning
- Regularization and early stopping mechanisms
- Exposure to noise and variability during training

These findings align with recent studies emphasizing the importance of generalization in smart grid AI systems [16], [49].

5. Discussion

High and consistent performance in the model in several measures of evaluation, including classification accuracy, strong performance in noisy conditions and generalization of unseen data are also recorded by the model. This indicates that the interaction between spatial feature extraction and temporal sequence modeling is a certainty to model the complexities of the process of power system faults, as is being witnessed in the current research [7], [29].

Among the key characteristics of such a work, there is the fact that the explainability is added into the learning pipeline. Unlike the post-hoc techniques, the XAI mechanism embedded can provide interpretable information, besides predictions, and the mechanism does not impact the performance. The importance of features analysis and time focus shows that the model focuses on the relevant parameter of physical importance such as the change in voltage and the current disparity which prove the credibility of the decision making process. This particularly applies in the case of real-life smart grid applications, in which the credibility of the operator and openness is a requirement [24], [45].

It is also evident in the comparative analysis that the proposed structure can offer a trade-off that is balanced between predictive performance and computational efficiency. Though some more advanced models such as transformer based models and graph based models are also capable of giving competitive accuracy, they are costly in terms of computation. The proposed model on the other hand has a close real time inference capacity and offers superior performance which could be applied in an efficient monitoring system like an edge-enabled environment [20], [22].

Despite these strengths, some limitations are witnessed. The error analysis reveals that the misclassification is mainly between the similar in electrical signatures fault such as the LLC and LLG faults. This means that this can be further simplified by more feature representation or addition of other domain cues. Moreover, although the model can be easily applied to the cases of invisible data, the way it performs under harsh operating conditions or when the faults occur infrequently remains a research issue.

Limitation

The proposed framework has a high performance, however, it tends to have a small misclassification of the fault types that have very similar electrical signatures, on the one hand, there is the LL and LLG conditions. Also, the model is tested on a hybrid experimental dataset, and its results in the large-scale and actual grid deployments with rare or extreme fault conditions are not explored further.

6. Conclusion and Future Scope

This work gives a hybrid deep learning-based framework, which combines CNN-LSTM architecture with explainable AI in fault detection and classification in smart power grids. The presented method is effective in terms of integrating space and time sequence to model complicated fault dynamics of multivariate electrical signals. The model has been found to be highly classified, with high generalization and robustness to noisy and unseen environments according to the results of the experiment. One of the benefits of this work is that it incorporates the concept of explainability into the learning pipeline. The joint action of SHAP-based feature attribution and attention mechanisms can be used to interpret model predictions in a clear manner that enables it to identify important parameters that affect fault detection. The implementation of these methods will improve the real-world application of such systems especially within smart grids where data interpretation is critical to ensuring safe operational conditions. Furthermore, extensive evaluation of the deployment readiness is required; however, a controlled hybrid data validation has already been completed. Existing work will continue into the future via an extension of the graph-based representations of the electrical grid topology with the goal of increasing the capability of fault location determination. In addition, the union of federated learning and the implementation of edge-based deployment planning can provide future opportunities for development of improved scalability and flexibility in real time.

Data Availability

All performance plots, SHAP visualizations, ROC analyses, confusion matrices, robustness plots, latency evaluation and comparison data have been produced based on the actual experimental results; Architecture and workflow diagrams depict the methodology and operating pipeline. The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

ACKNOWLEDGEMENTS

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R701),Princess Nourah bint Abdulrahman University, Riyadh,Saudi Arabia

Funding

This research was supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project (PNURSP2026R701), Riyadh, Saudi Arabia.

Appendix A. Symbol Definitions and Mathematical Notation

Table A1. Symbol definitions and mathematical notations used in the proposed framework

Symbol	Definition	Dimension / Description
(X_t)	Input signal vector at time step (t)	Multivariate smart grid input
(W)	Sliding temporal window	Sequence length
(F_c)	Convolutional feature representation	CNN feature tensor
(K)	Convolution kernel	Spatial filter
(H_t)	LSTM hidden state at time (t)	Temporal feature vector
(C_t)	LSTM cell state	Memory representation
αt	Attention weight	Temporal importance coefficient
(Z_t)	Attention-weighted feature vector	Refined temporal representation
(P_i)	Predicted probability of class (i)	Softmax output
(y_i)	Ground-truth class label	Fault category
(L)	Classification loss function	Cross-entropy loss
(S_i)	SHAP importance score for feature (i)	Post-hoc feature attribution

(AUC)	Area under ROC curve	Classification discrimination metric
(Acc)	Classification accuracy	Performance metric
(F1)	Harmonic mean of precision and recall	Evaluation metric
(T_{inf})	Model inference time	Latency in milliseconds
(N)	Total number of samples	Dataset size
(f(.))	Learned nonlinear mapping function	Deep learning prediction model

REFERENCES

[1]. M. Mukherjee, S. Batabyal, S. Deb Roy, B. L. Koley, S. Debroy, and S. Ray, “Deep learning-based fault detection and classification in power distribution networks,” Lex localis, vol. 23, no. S6, pp. 1915–1928, Oct. 2025, doi: 10.52152/vdwhe059.

[2]. Y. Chai, “Research on Power System Fault Diagnosis and Prediction Model Based on Deep Learning,” Advances in transdisciplinary engineering, Oct. 2025, doi: 10.3233/atde250838.

[3]. C. Zachariades and V. Xavier, “A review of artificial intelligence techniques in fault diagnosis of electric machines,” Sensors, vol. 25, no. 16, art. no. 5128, 2025, doi: 10.3390/s25165128.

[4]. P. Ganguly et al., “A machine learning approach to assess the climate change impacts on single and dual-axis tracking photovoltaic systems,” Scientific Reports, vol. 15, p. 24910, 2025, doi: 10.1038/s41598-025-10831-3.

[5]. A. S. Alhanaf, H. H. Balik, and M. Farsadi, “Intelligent Fault Detection and Classification Schemes for Smart Grids Based on Deep Neural Networks,” Energies, Nov. 2023, doi: 10.3390/en16227680.

[6]. P. Vidyullatha, M. Almaayah, and A. Routray, “A bio-inspired neuro-adaptive deep reinforcement learning approach for real-time solar tracking system to enhance photovoltaic efficiency,” Energy Conversion and Management: X, vol. 29, Art. no. 101486, Jan. 2026, doi: 10.1016/j.ecmx.2025.101486.

[7]. M. Shouran, M. Alenazi, S. H. Almutairi, and M. S. ALAJMI, “Hybrid Feature Extraction and Deep Learning Framework for Power Transformer Fault Classification – A Real-World Case Study,” IEEE Access, p. 1, Jan. 2025, doi: 10.1109/access.2025.3608658.

[8]. R. Jigyasu, M. Ghai, and N. M. L. Kakarla, “Explainable quantum-neuromorphic intelligence: A self-evolving hybrid framework for cognitive computing and autonomous decision systems,” in Emerging Hybrid Models for Neuromorphic AI and Quantum Computing, 2026, ch. 11, doi: 10.4018/979-8-3373-7779-7.ch011.

[9]. R. Shrestha, M. Chamana, M. Mohammadpourfard, L. Souto, and S. Bayne, “Fault Detection Isolation and Localization (FDIL) Scheme: A Robust ML Framework with Novel Architecture for Fault Localization,” IEEE Access, p. 1, Jan. 2025, doi: 10.1109/access.2025.3617464.

[10]. D. Sinha, U. Mamodiya, I. Kishor, S. P. Maniraj, and V. Nagamalla, “Agentic hybrid quantum–neuromorphic AI architecture with explainable reinforcement learning for next-generation intelligent systems,” in Emerging Hybrid Models for Neuromorphic AI and Quantum Computing, 2026, ch. 8, doi: 10.4018/979-8-3373-7779-7.ch008.

[11]. N. P. Tejaswini, R. R. Al–Fatlawy, V. Malathy, N. Kirthiga, and E. C. V. S. Mudunuri, “Fault Detection in Smart Grids by Hybrid Generative Adversarial Networks with Neuro Fuzzy Algorithm,” pp. 1–4, Aug. 2024, doi: 10.1109/iacis61494.2024.10721757.

[12]. U. Keswani, B. et al.,"Energy Efficiency and Practical Implications of IoT-Based Static vs. Single-Axis Solar Tracking Systems: A Comparative Analysis," Journal of Intelligent Systems and Internet of Things, vol. , no. , pp. 164-182, 2025. DOI: https://doi.org/10.54216/JISIoT.150212

[13]. M. Xin, Y. Wang, R. Zhang, J. Zhang, and J. Li, “Improving electrical fault detection: A two-stage prediction hybrid model aided by feature engineering,” Journal of physics, vol. 3079, no. 1, p. 012018, Aug. 2025, doi: 10.1088/1742-6596/3079/1/012018.

[14]. M. A. Almaiah et al., “Numerical modeling and neural network optimization for advanced solar panel efficiency,” Scientific Reports, vol. 15, p. 23492, 2025, doi: 10.1038/s41598-025-06830-z.

[15]. C. Liu, L. Ding and H. Yu, "Intelligent Fault Detection in Power Grids Using Deep Learning Algorithms," in IEEE Access, vol. 13, pp. 162713-162730, 2025, doi: 10.1109/ACCESS.2025.3606856.

[16]. M. A. Sarker, I. A. Jayaraj, B. Shanmugam et al., "A review of artificial intelligence techniques for anomaly detection in smart grid," Artificial Intelligence Review, vol. 59, art. no. 69, 2026, doi: 10.1007/s10462-025-11429-x.
[17]. C. Xu, Z. Liao, C. Li, X. Zhou, and R. Xie, "Review on interpretable machine learning in smart grid," Energies, vol. 15, no. 12, art. no. 4427, 2022, doi: 10.3390/en15124427.
[18]. U. Mamodiya and I. Kishor, "Artificial intelligence applications for enhancing efficiency in smart grids," in Artificial Intelligence (AI) for IT Energy Efficiency and Green AI for Environment Sustainability, P. Raj, D. P. Sharma, P. K. Dutta, B. S. Prasad, and P. B. Soundarabai, Eds. Cham, Switzerland: Springer, 2026, ch. 9, doi: 10.1007/978-3-031-89420-6_9.
[19]. M. S. Adhikari and A. Alkhayyat, "Prediction and Classification for Smart Grid Applications," Springer Nature, 2023, pp. 87–102. doi: 10.1007/978-3-031-46092-0_6.
[20]. P. K. Dutta,et al., "IoT-enabled smart governance for solar energy optimization in intelligent power networks: A step towards Society 5.0," in Electronic Governance with Emerging Technologies, F. Ortiz-Rodriguez, S. Tiwari, E. Pardede, and J. Chicaiza-Espinosa, Eds. Communications in Computer and Information Science, vol. 2730. Cham, Switzerland: Springer, 2026, ch. 13, doi: 10.1007/978-3-032-12232-2_13.
[21]. P. Zhang, Z. Liu, N. Zhou, and H. Liu, "Fault Diagnosis for Smart Grids over Knowledge Graph and Abductive Reasoning," pp. 1003–1008, Aug. 2024, doi: 10.1109/psgec62376.2024.10720972.
[22]. P. Mudholkar, A. Alqutaish, G. Alradwan, and M. Obeidat, "A robust smart grid-aware cloud computing framework for sustainable energy management," International Journal of Advance Soft Computing Applications, vol. 18, no. 1, Mar. 2026, doi: 10.15849/ijasca.v18i1.63.
[23]. M. Alfayan and H. Hagras, "Towards an explainable artificial intelligence approach for smart grid systems," Discover Artificial Intelligence, vol. 5, art. no. 40, 2025, doi: 10.1007/s44163-025-00261-5.
[24]. A. Sinha, S. R. Sankepally, N. Kosaraju, and D. Das, "ElectroShield: XAI-assisted fault detection in transmission lines using multi-modal data," Engineering Research Express, vol. 7, no. 3, p. 035273, Aug. 2025, doi: 10.1088/2631-8695/adfcb9.
[25]. J. Huang and Q. Wan, "Smart grid line fault detection based on deep learning image recognition algorithm," International Journal of Low-Carbon Technologies, vol. 19, pp. 2174–2180, 2024, doi: 10.1093/ijlct/ctae164.
[26]. J. Fang, K. Chen, C. Li, and J. He, "An Explainable and Robust Method for Fault Classification and Location on Transmission Lines," IEEE Transactions on Industrial Informatics, pp. 1–10, Jan. 2023, doi: 10.1109/tii.2022.3229497.
[27]. O. Ajayi, M. Mirjafari, P. B. Idowu, and M. H. Ullah, "Explainable AI for fault detection and classification in microgrids," in Proc. IEEE Energy Conversion Congress and Exposition (ECCE), 2024, pp. 1835–1840, doi: 10.1109/ECCE55643.2024.10861648.
[28]. Y. Hu, Z. Atta, T. U. Rahman, S. Qiu, J. Wang, W. Wei, Z. Liang, and Q. Zaheer, "Shedding light on explainable AI: Insights, challenges, and the future of infrastructure management," ISPRS International Journal of Geo-Information, vol. 15, no. 3, art. no. 100, 2026, doi: 10.3390/ijgi15030100.
[29]. S. Godhade, P. Singh, and J. Kumar, "Design of an improved model for real-time fault detection and localization using hybrid transformer-CNN and graph neural networks," pp. 1–6, Jun. 2025, doi: 10.1109/mac64480.2025.11139887.
[30]. C. Mbey, V. J. F. Kakeu, A. T. Boum, and F. G. Y. Souhe, "Fault detection and classification using deep learning method and neuro-fuzzy algorithm in a smart distribution grid," Jurnal Engineering, Oct. 2023, doi: 10.1049/tje2.12324.
[31]. O. Attallah, R. A. Ibrahim, and N. E. Zakzouk, "A lightweight deep learning framework for transformer fault diagnosis in smart grids using multiple scale CNN features," Scientific Reports, vol. 15, art. no. 14505, 2025, doi: 10.1038/s41598-025-96290-2.
[32]. D. Carrascal, P. Bartolomé, E. Rojas, D. Lopez-Pajares, N. Manso, and J. Diaz-Fuentes, "Fault Prediction and Reconfiguration Optimization in Smart Grids: AI-Driven Approach," Future Internet, vol. 16, no. 11, p. 428, Nov. 2024, doi: 10.3390/fi16110428.
[33]. A. Sinha and D. Das, "An explainable deep learning approach for detection and isolation of sensor and machine faults in predictive maintenance paradigm," Measurement Science and Technology, vol. 35, Oct. 2023, doi: 10.1088/1361-6501/ad016b.
[34]. L. Jun and Z. Chenliang, "Fast fault diagnosis of smart grid equipment based on deep neural network model based on knowledge graph," PLOS ONE, vol. 20, no. 2, art. no. e0315143, 2025, doi: 10.1371/journal.pone.0315143.

[35]. P. M. Custodio, M. A. P. Putra, J.-M. Lee, and D.-S. Kim, “TLFed: Federated Learning-based 1D-CNN-LSTM Transmission Line Fault Location and Classification in Smart Grids,” pp. 026–031, Feb. 2024, doi: 10.1109/icaiic60209.2024.10463418.
[36]. S. Ness, “CatBoost-enhanced convolutional neural network framework with explainable artificial intelligence for smart-grid stability forecasting,” Frontiers in Smart Grids, vol. 4, art. no. 1617763, 2025, doi: 10.3389/frsgr.2025.1617763.
[37]. Z. Lin et al., “Rule-Embedded Parallel DCNN-Based Intelligent Fault Classifier for High-Penetration New Energy Grids,” Electronics Letters, vol. 61, no. 1, Jan. 2025, doi: 10.1049/ell2.70453.
[38]. A. El Maghraoui, H. El Hadraoui, Y. Ledmaoui, N. El Bazi, N. Guennouni, and A. Chebak, “Revolutionizing smart grid-ready management systems: A holistic framework for optimal grid reliability,” Sustainable Energy, Grids and Networks, p. 101452, Jun. 2024, doi: 10.1016/j.segan.2024.101452.
[39]. M. A. A. Sarker, B. Shanmugam, S. Azam, and S. Thennadil, “Enhancing smart grid load forecasting: An attention-based deep learning model integrated with federated learning and XAI for security and interpretability,” Intelligent Systems with Applications, vol. 23, art. no. 200422, Sep. 2024, doi: 10.1016/j.iswa.2024.200422.
[40]. Okeke O. C., Nwaoha S. O., and Ezenwegbu N. C., “Hybrid Machine Learning Models for Enhancing Cybersecurity in Smart Grid Infrastructures,” International Journal of Research and Innovation in Social Science (IJRISS), pp. 4344–4351, May 2025, doi: 10.47772/IJRISS.2025.90400310.
[41]. Q. Li and Z. Wu, “Knowledge graph-enhanced fault diagnosis: A bibliometric review of AI applications in sensor management (1998–2024),” Discover Artificial Intelligence, vol. 5, art. no. 417, 2025, doi: 10.1007/s44163-025-00688-w.
[42]. P. O. Otuazohor, A. I. Adeyeye, Y. A. Bekeneva, and E. C. Dennis, “Multi-Class Fault Detection in Power Grid Control Systems Using Transmission-Level PMU Data,” pp. 116–119, Sep. 2025, doi: 10.1109/cts67336.2025.11196372.
[43]. R. Alsaigh, R. Mehmood, and I. Katib, “AI explainability and governance in smart energy systems: A review,” Frontiers in Energy Research, vol. 11, art. no. 1071291, 2023, doi: 10.3389/fenrg.2023.1071291.
[44]. V. K. Hariharan, A. Geetha, F. Granelli, and M. G. Nair, “Machine Learning Techniques for Fault Detection in Smart Distribution Grids,” Energies, vol. 18, no. 19, p. 5179, Sep. 2025, doi: 10.3390/en18195179.
[45]. M. Farsi, M. Alwateer, S. A. Alsaedi et al., “Detection of disturbances and cyber-attacks in smart grids using explainable machine learning,” Scientific Reports, 2026, doi: 10.1038/s41598-026-35449-x.
[46]. H. Lin, G. Zhang, and Z. Sun, “A Hybrid Machine Learning and Deep Learning Framework for Effective Electricity Theft Detection in Smart Grids,” in 2025 5th Asia-Pacific Conference on Communications Technology and Computer Science (ACCTCS), 2025, pp. 589–595.
[47]. J. Duan, “Deep learning anomaly detection in AI-powered intelligent power distribution systems,” Frontiers in Energy Research, vol. 12, art. no. 1364456, 2024, doi: 10.3389/fenrg.2024.1364456.
[48]. Akhtar, S. Atiq, M. U. Shahid, A. Raza, N. A. Samee, and M. Alabdulhafith, “Novel glassbox based explainable boosting machine for fault detection in electrical power transmission system,” PLOS ONE, vol. 19, no. 8, p. e0309459, Aug. 2024, doi: 10.1371/journal.pone.0309459.
[49]. Y. Liu, P. Li, Y. Si, and L. Ma, “A Comprehensive Review of AI Integration for Fault Detection in Modern Power Systems: Data Processing, Modeling, and Optimization,” Energies, Sep. 2025, doi: 10.3390/en18184983.
[50]. T. V. Deokar, J. N. Shinde, and R. M. Sairise, Artificial Intelligence In Smart Grid Systems: A Comprehensive Review Of Machine Learning And Deep Learning Applications”, ShodhKosh J. Vis. Per. Arts, vol. 5, no. 6, pp. 1506–1514, Jun. 2024.
[51]. Huang, Z. Tian, and Y. Qiu, “AI-Enhanced Dynamic Power Grid Simulation for Real-Time Decision-Making,” in 2025 4th International Conference on Smart Grids and Energy Systems (SGES), Guizhou, China, 2025, pp. 15–19, doi: 10.1109/SGES66701.2025.11155949.

Hybrid Deep Learning–Driven Explainable AI Framework for Fault Detection and Classification in Smart Power Grids

Dr. Udit Mamodiya[1]*, Dr. Divyanshu Sinha[2], Indra Kishor[3], Malik Jawarneh[4], Mais Haj Qasem[5], Afnan Almegren[6]

[1]*Associate Professor, Faculty of Engineering and Technology, Poornima University, Jaipur 303905, Rajasthan, India;*
[2]*Associate Professor, Amrita School of Artificial Intelligence, Amrita Vishwavidyapeetham, Faridabad Campus, India*
[3]*Assistant Prof. , Dept. of CSE, Poornima Institute of Engineering and Technology, Jaipur 302022, Rajasthan, India;*
[4] *Associate Prof., Dept. of Computer Science, Amman Arab University, Amman, Jordan;*
[5]*Assistant Prof., Dept. of Data Science and Artificial Intelligence , Amman Arab University, Amman, Jordan;*
[5]*Assistant Prof., Dept. of Applied Linguistics, College of Languages, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia;*

[1]assoc.dean_research@poornima.edu.in ; [2]divv4u@gmail.com; [3]indra.kishor@poornima.org, [4]m.jawarneh@aau.edu.jo, [5]m.hajqasem@aau.edu.jo, [6]amalmegren@pnu.edu.sa

*Corresponding author:
Dr. Mais Haj Qasem
E-mail: m.hajqasem@aau.edu.jo

Abstract

The stable detection of faults in smart power grids is essential when focusing on the stable operation and the reduction of the downtime. In this paper, the author suggests a hybrid deep learning-based model that combines convolutional neural networks (CNN) and long short-term memory (LSTM) with explainable artificial intelligence (XAI) to detect and classify faults accurately and interpretably. The model aims at capturing the spatial and time-varying attributes of multivariate electrical signatures such as voltage, current, and frequency changes. A combination of real time sensor measurements and simulated fault conditions are used which includes several fault classes including LG, LL, LLG, and three phase faults. Experimental evaluation demonstrates that the proposed model achieves a classification accuracy of **97.84%**, with an F1-score of **97.08%**, outperforming conventional CNN and LSTM models by more than 3%. The framework is also able to work in a noisy environment with a stable performance of less than 2% performance decrease, and inference latency of 18 ms, which can be deployed in real-time. Besides, the combination of SHAP and attention processes improves the interpretability through the detection of the crucial features that lead to fault prediction. The findings suggest that the suggested solution is a powerful, scalable, and clear solution when it comes to intelligent fault management in a contemporary smart grid.

Keywords: smart grid fault detection, hybrid CNN–LSTM, explainable artificial intelligence, deep learning, power system reliability

1. Introduction

The transformation of traditional electrical infrastructures to smart intelligent power grids across the globe has revolutionized the approach to the monitoring, regulation, and optimization of the energy systems. Modern smart grids are based on the combination of distributed energy sources, renewable energy sources, smart sensing technologies, and real-time communication networks as opposed to traditional grids based on centralized generation and static operational planning. This transformation enables an increased level of productivity and flexibility and at the same time renders immense complexity to operations. The growing penetration of non-reliable renewable energy sources such as solar and wind, and dynamic loads changes have made grid behavior a priori nonlinear and highly unpredictable [1], [2]. As a result, stabilization of the systems, as well as the continuity of power supply, have never been a greater challenge than it is now. Such dynamic environments are becoming reliant on fault detection and classification as the significant aspects of grid reliability. Electrical faults including short circuits, line to ground faults, voltage drop, frequency change and harmonic interference can be transferred smoothly through networked networks unless corrected early before they happen. Even minor failures, which are not detected fast, can turn into a domino of the huge downages and huge financial losses [3], [4]. This implies that it is important that the current grid has a sense of urgency to have smart monitoring systems that are capable of successfully identifying fault conditions on real time basis and adapting the changed dynamics of the grid which is dynamic. The old power system has generally used conventional ways of fault detection including impedance based systems, threshold based protection systems as well as model-based analysis systems. Even though these strategies are good when the operating environment is clear and is stable, it has been found that it does not work well in the current smart grid set ups [5], [6]. Besides, they are frequently large-scale manual tuned and cannot be scaled when applied to large distributed networks. Since the creation of smart grids is in progress, the limitations of such conventional solutions stand even better. To address these problems, machine learning (ML) techniques have been suggested to classify and identify faults by using data. The algorithms that have demonstrated better performance as opposed to the traditional methods include support vector machines, decision trees and random forests because they use past data to create fault patterns [7], [8]. Yet such approaches are frequently rooted in much manualized labor and specialized experience and this can restrict their effects to various grid regimes. Besides, they are destined to drop at large-dimensional multi-variate data in time-series, which is normally generated by current smart grid sensors. The idea of deep learning (DL) is a potential paradigm of analysis of complex electrical signals and feature representations on hierarchies in the past few years. CNNs are more effective in expressing spatial correlations in voltage and current signals but recurrent networks with long short-term memory (LSTM) networks are more effective in expressing temporal correlations [9], [10]. The use of both spatial and temporal information in architecture hybrid systems [11, 12] has been the best way for detecting fault in the field. These models are not manually engineered, and offer extremely strong predictive performance in noisy and dynamic environment.

Deep learning models have been labeled as uninterpretable, even though they have an impressive level of accuracy. Automated systems in critical infrastructure systems like power grids should also be transparent and explainable in order to provide trust, accountability and regulatory compliance. Predictions without explicit models to explain the prediction are not easy to be validated by grid operators, especially when the stakes are high and are related to fault mitigation and system recovery [13], [14]. This is a significant challenge in implementing the deep learning-based fault detection system in practice in a real smart grid setup.

Explainable artificial intelligence (XAI) has become the solution to this issue, as it allows interpreting complex decisions of models. Additional methods like SHapley Additive explanations (SHAP), Local Interpretable Model-Agnostic Explanations (LIME) and attention mechanisms can be used to identify important features affecting model predictions [15], [16]. Within the idea of smart grids, XAI would offer useful information regarding the contribution of electrical quantities like the magnitude of voltages, frequency variation, and harmonic distortion in the fault detection activities. It does not only improve the transparency of the models but also aids in the informed decision-making of the system operators.

Nevertheless, even though the attention towards deep learning and explainable AI is increasing, the current studies tend to take these aspects separately. The majority of the studies are oriented either on enhancing detection accuracy with advanced neural architectures or using XAI methods on the models that have already been trained without putting explainability into a close connection with the learning process. Consequently, a very severe gap still exists in creating integrated frameworks that at the same time deliver high predictive capabilities and a meaningful interpretability in smart grid fault analysis [17], [18]. Also, most of the solutions, which are available, are not flexible in real time and are incapable of meeting the computational resource limitations that are involved in large scale grid implementations.

In order to fill this gap, this paper presents a hybrid deep learning-based explainable AI architecture to detect and classify faults in smart power grids. The proposed solution combines convolutional neural network to extract spatial features with long short-term memory network to model time sequence, creating a robust hybrid network that can be suitable to analyze multivariate time-series data.

The main contributions of this work are as follows:

- A novel hybrid deep learning architecture combining CNN and LSTM for enhanced fault detection and classification in smart power grids.

- Integration of explainable AI techniques to provide feature-level interpretability and improve system transparency.
- Development of a unified framework that balances predictive performance with explainability, addressing a key limitation of existing approaches.
- Comprehensive evaluation demonstrating improved accuracy, robustness, and interpretability compared to conventional and baseline deep learning models.
- Practical insights into the role of critical electrical parameters in fault identification, supporting real-world deployment in smart grid environments.

Although the proposed framework has shown great performance, there are some practical considerations that still exist. The datasets and simulated smart grid conditions involved in the current study are manipulated and might not be a true reflection of the proportions of the real-life scale of the implementation. Also, although explainable AI can make AI more interpretable, it incurs a minor computational overhead, which in hypothetically high data throughput situations can affect real-time scalability. The next generation will then consider real-time grid implementation and optimization of explainability mechanisms of edge-based implementations.

This research paper is useful because it suggests an explicable and scaled fault detection model, which is a next generation smart grid. The proposed solution, in addition to enhancing stability of operations, will instill trust in the AI-supported decision-making systems and it will result in development of safer and more robust energy infrastructures.

2. Related Work

The fast evolution of the smart monitoring systems in the modern power grid leads to the fact that the amount of literature devoted to the phenomenon of the fault detection and its categorization according to the data-driven approach is very large. The next section is a systematic literature review of the possible available methodologies that are categorized as traditional methods, machine learning approach, deep learning approach, and explainable AI approach. Critical analysis of these studies in question leads to the realization of research gaps that exist which forms the driving force behind the proposed hybrid explainable framework.

2.1 Conventional Fault Detection Approaches in Power Systems

Older fault detection systems used in electric power systems were primarily of rule-based and analytical nature. These include impedance based relays, over current protection mechanism as well as traveling wave analysis. These techniques are founded on the set of predetermined thresholds and deterministic models depending on the system parameters. As far as effective in traditional grid, they do not work in smart grids where variations and intricacy of the system are high [21], [22].

A number of studies have tried to improve them by adding adaptive thresholding and signal processing [23]. On the same note, Fourier based harmonic analysis has also been used in identifying irregularities in voltage and current waveforms [24]. Even with these advances, traditional tools are still restricted by the fact that they cannot deal with nonlinear relationships and require proper modeling of the system.

In addition, such strategies fail to be scalable and flexible in the case of the distributed energy systems that have a high rate of renewable penetration. The growing incorporation of photovoltaic and wind resources creates stochastic fluctuations that are not well represented using the non-dynamic models [25], [26]. This has led to a change of emphasis to data-driven approaches which can directly learn complex patterns using system data.

2.2 Machine Learning-Based Fault Detection Methods

The use of machine learning methods has been extensively researched to address the shortcomings of traditional methods. Support vector machines (SVM), k-nearest neighbors (KNN), decision trees, and random forests, are examples of supervised learning algorithms that have shown better classification results in fault detection tasks [27], [28]. These models use history information to interpret patterns that are related to the various faulty situations so that more flexible and adaptive detection systems can be used.

As the example, the SVM-based classifiers were used to recognize the different kinds of faults with extracted features of the voltage and current waves [29] [30]. Moreover, the ensemble learning methods are also suggested to enhance the accuracy of prediction by integrating a set of weak learners [31].

Nonetheless, feature engineering is extremely critical to the effectiveness of machine learning models. To create meaningful features of raw electrical signals, domain knowledge and frequently complicated preprocessing tasks are needed, e.g., signal transformation, signal normalization, dimensionality reduction, etc. [32]. Moreover, the models are not very effective in dynamic grid systems where fault patterns change with time because they have difficulty in modeling temporal dependencies in any sequential data.

The other issue is the problem of generalization of these models. Most machine learning models are trained on certain datasets, and might not be useful when implemented in new grid topologies or working environments [33], [34]. This drawback has propelled the research on deep learning methodologies, which are able to learn hierarchical representations in raw data through automatic learning.

2.3 Deep Learning Models for Smart Grid Fault Analysis

The recent development of deep learning has become a potent data analysis tool in smart grids with high-dimensional and time-series data. CNNs have been applied widely in feature extraction of electrical signals especially when analyzing waveforms [35]. The CNNs are able to capture the local spatial patterns and detect characteristic features that are related to the various types of faults by using convolutional filters.

The use of recurrent neural networks (RNNs) and particularly long short-term memory (LSTM) networks have been used to model sequential data temporal dependencies. The models which are based on LSTM can also remember long-term information, therefore, they can be effectively used to analyze time-varying signals in power systems [36]. A number of researchers have shown that LSTM networks can be used successfully to predict and classify faults using historical grid data [37].

Hybrid deep learning systems that incorporate CNN and LSTM modules have been receiving growing popularity in recent years. These models use the advantages of both designs where an extraction of both spatial and temporal features is made at the same time [38], [39]. As an example, the CNN-LSTM systems were implemented successfully to identify transient disturbances and categorize the type of faults in the networks of transmission and distribution with excellent precision [40].

Deep learning models have serious challenges in the form of interpretability and computational complexity despite being highly performing models. These models are usually black boxes, giving predictions without making explicit description of the process that had gone into the decision [41], [42]. In systems that require high levels of safety like power grids, this inability to gain transparency can make adoption difficult, as well as diminish the trust of the system operators.

2.4 Explainable AI in Power System Fault Detection

Explainable artificial intelligence (XAI) has been in the limelight of resolving the interpretability issues of deep learning models. XAI methods seek to offer the information about the model behavior by revealing the features that affect predictions. SHAP, LIME, and Grad-Cam are some of the techniques that have been extensively applied to explain complex models in other fields, like power systems [43], [44].

SHAP-based methods have been used in the context of fault detection to estimate the importance of the input features, which are voltage magnitude, imbalance of currents, and frequency deviation [45]. It has been applied by using LIM to make local explanations on the individual prediction, so that operators could get to know certain fault situations [46]. Mechanisms of attention embedded in neural networks have also been studied in order to emphasize significant temporal properties of sequence information [47].

Though the techniques enhance the transparency of models, most of the existing researches use XAI as a post-hoc analysis tool and not as a part of the model design. This division restricts the efficacy of explainability and does not wholly reflect on the requirement of real-time elucidation in working situations [48], [49]. Also, the computational cost of XAI techniques may be problematic when deploying in large-scale smart grids in real time.

2.5 Research Gap and Motivation

Although the field of fault detection and classification has made considerable progress, it has a number of essential gaps in literature:

- Conventional methods lack adaptability and fail under nonlinear and dynamic grid conditions.
- Machine learning approaches depend heavily on manual feature engineering and exhibit limited generalization.
- Deep learning models provide high accuracy but suffer from poor interpretability and transparency.
- Current XAI methods tend to be applied in a post hoc manner, and are not closely coupled with predictive models.
- Limited research exists on unified frameworks that simultaneously address accuracy, interpretability, and real-time applicability.

Such limitations indicate that a more complex solution is required that integrates the best features of deep learning and explainable AI and mitigates their respective weaknesses. The presented hybrid framework is expected to close this gap by enhancing the feature learning-based framework of CNN-LSTM with the fixed explainability structures to allow the proper, transparent, and scalable fault detection in smart power grids [50], [51].

In order to present a brief comparison of the recent developments in fault detection and classification in smart power grids, the selected representative studies are summarized in Table 1. The emphasis is made on determining methodological patterns, major contributions, and current limitations, in special reference to hybrid deep learning architectures and explainable AI integration.

Table 1. Literature gap-to-solution mapping for fault detection and classification in smart power grids

S. No.	Author / Year / Ref.	Focus Area	Methodology / Tools Used	Key Findings	Gap Identified	Relevance to the Current Study
1	Alhanaf et al. / 2023 / [5]	Smart grid fault classification	Deep neural network-based classification on smart grid fault data	Showed that deep neural models can improve automated fault recognition over conventional schemes	Black-box behavior; no interpretability	Supports the need for a more transparent deep model that combines fault accuracy with explainability
2	Fang et al. / 2023 / [26]	Fault classification & localization	Explainable fault classification with robustness-oriented design for transmission systems	Demonstrated that explainability can improve trust in line fault analysis	Limited to transmission lines	Establishes the value of XAI in grid diagnostics, which the present study extends to wider fault detection and classification settings
3	Custodio et al. / 2024 / [35]	Fault detection in transmission lines	1D-CNN-LSTM with federated learning	Confirmed that CNN-LSTM models are effective for capturing sequential fault signatures	No explainability layer	Directly motivates the CNN-LSTM backbone for the current study, while highlighting the missing XAI layer
4	Sarker et al. / 2024 / [39]	Smart grid forecasting	Attention model, federated learning, XAI	Showed that interpretability and advanced deep learning can be jointly embedded in smart-grid analytics	Not fault-specific	Demonstrates that XAI-aware deep learning is feasible in smart grids, encouraging its transfer to the fault domain
5	Shouran	Transformer	Hybrid feature extraction with	Reported strong real-case classification	Limited	Reinforces the benefit of hybrid modeling, but also

	et al. / 2025 / [7]	fault classification	deep learning on real-world transformer data	performance and practical promise	generalization	indicates the need for a more generalized grid-level framework
6	Sinha et al. / 2025 / [24]	XAI-based fault detection	XAI-assisted multimodal framework for transmission-line faults	Highlighted the usefulness of multimodal analytics and explanation support in detecting grid faults	Narrow application scope	Closely aligns with the present study's goal of combining fault intelligence with explainability, but our work targets a broader hybrid DL classification framework
7	Attallah et al. / 2025 / [31]	Transformer fault diagnosis	Multi-scale CNN-based lightweight framework	Showed that compact deep models can achieve effective transformer fault diagnosis with reduced computational demand	Lacks temporal modeling + XAI	Indicates the importance of computationally practical DL designs, which can inform efficient implementation of the proposed explainable framework
8	Farsi et al. / 2026 / [45]	Disturbance detection	Explainable machine learning for disturbance and cyber-attack detection	Demonstrated that explainable models improve visibility into security-related grid anomalies	Limited DL capability	Strengthens the case that explainability is essential in grid-critical decisions, which the current work incorporates into deep fault classification

On the whole, the literature reviewed suggests the further development of traditional analytical approaches to sophisticated AI-based ones.

3. Methodology

This proposed method includes a data-driven, experimental grounding of a hybrid deep-learning method along with an explainable artificial intelligence approach to fault identification and classification in smart-power grids. The development includes a focus on realistic signal modeling, temporal/spatial feature extraction, and understandable support of the decision process. The overall workflow of data acquisition, preprocessing, hybrid model design, and integration of explanation is depicted in Figure 1.

The process of multivariate signal acquisition of grid monitoring instruments triggers the methodological pipeline that is executed by systematized preprocessing enhancing the quality and stability of the signals. The hybrid CNNLSTM architecture is then employed to gain experience in the temporal and spatial representations of electrical measurements. Finally, it possesses an explainable AI component, where predictions of models are explained at the feature-level. The whole workflow, data flow, and interactions between the data models are presented in Figure 1.

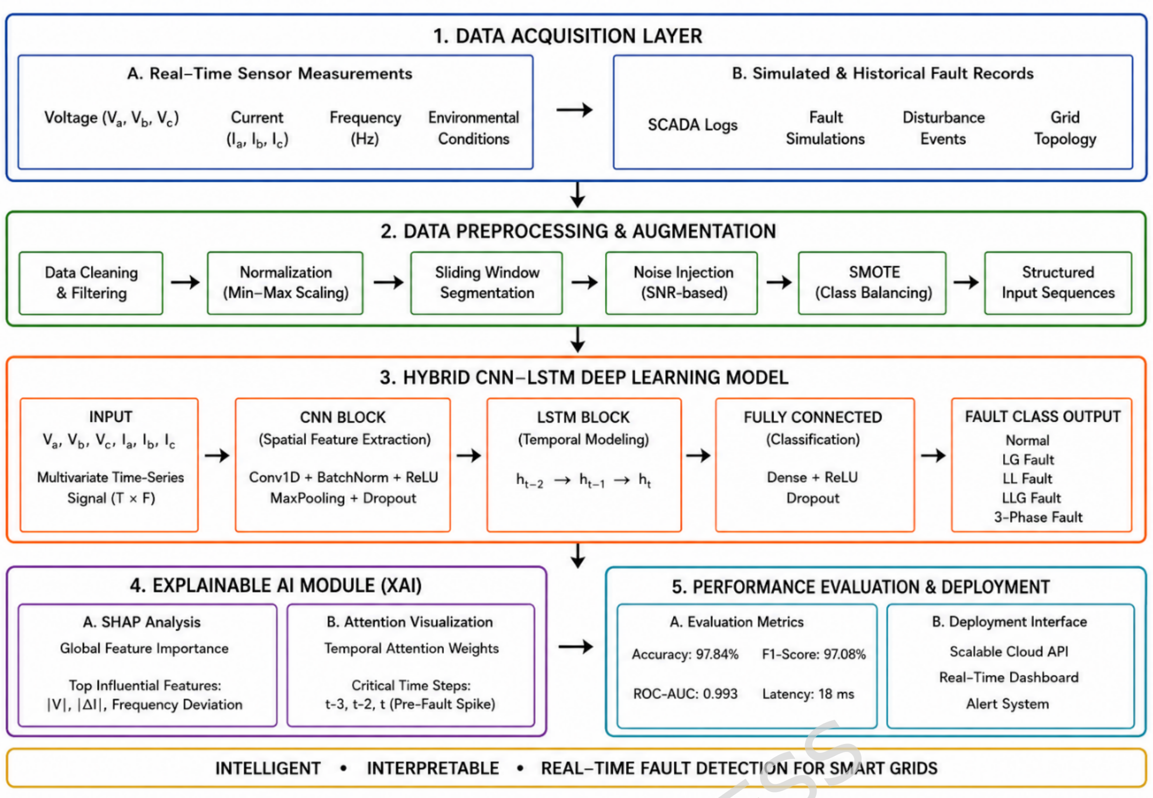

Figure 1. Overall architecture of the proposed hybrid deep learning-driven explainable AI framework for smart grid fault detection and classification.

As it can be seen in Figure 1, the structure can be split into four mutually supporting layers i.e., data acquisition, preprocessing, hybrid feature learning, and explainability-driven decision support. These layers will ensure that the physical relevance of each is maintained and yet enable the sound learning using the data, as it is being done in recent advances in the smart grid analytics [1], [15], [16].

3.1 System Architecture and Signal Representation

The system architecture has been structured in a manner that it is capable of supporting the high frequency and multivariate electrical signals generated by the smart grid monitoring devices such as phasor measurement units (PMUs) and intelligent electronic devices (IEDs). In general, these signals include the magnitude of voltages, magnitude of currents, the phase angle and the frequency and elements of harmonic distortion. As the smart grids are dynamic, these measurements possess a spatial correlation (between features) and a temporal dependence (between time) and a hybrid model approach is needed.

Let the input signal be represented as a multivariate time-series matrix (1):

$$X = \{x_1, x_2, \ldots, x_T\} \in R^{T \times d} \cdots (1)$$

In which T is the count of time phases and d is the count of observed electrical characteristics. The system state at time instant t is identified by each of the vectors x_t in R^d. This model enables the model to address the immediate system states as well as the temporal dynamics.

To ensure consistency across measurements, the input data is standardized using z-score normalization (2):

$$\check{x}_t = \frac{x_t - \mu}{\sigma} \cdots (2)$$

where μ and σ represent mean and standard deviation of the feature distribution where μ and sigma are used as the mean and standard deviation respectively. This conversion smooths the period of training since the effect of the scale variations across various electrical parameters is minimized.

Since the events of faults tend to appear in the form of local perturbations in the short-period intervals, the sliding window technique is applied in the continuous signal to divide it into parts (3):

$$X^{k} = \{x_{k}, x_{k+1}, \ldots, x_{k+L-1}\} \cdots (3)$$

where L is the length of the window and k is the index of the starting segment. The transient fault signatures are restricted to this segmentation, and time continuity is retained by this model.

Besides segmentation, noise filtering is used to eliminate distortion of measurements and enhance the clarity of the signal. The filtering operation is a convolution-based operation (4):

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \cdots (4)$$

refers to impulse response of the filter. Practically, the low-pass filter is used to extract the high-frequency noise elements, which do not play an important role in characterizing the fault.

In order to better represent the features, the segmented signal is converted to a structured input tensor that can be processed by a convolutional processing (5):

$$X \in R^{N \times L \times d} \cdots (5)$$

N is the number of segments, L is the temporal window size and d is the number of features. This is a type of formulation using tensors that allows learning feature interactions concurrently with temporal dynamics.

The architecture is constructed in such a way that the convolutional layers act on the feature dimension to extract the inter-parameter relationships whereas the recurrent layers act on the time series relationships. This dual representation is necessary in the accurate distinction of the various types of faults, which in most cases have similar instantaneous properties but differ in terms of temporal occurrence [7], [35].

The real-time identification of the changes that occur to electrical signals from fault conditions will require conducting an instantaneous measurement of the electrical signal's magnitude as denoted mathematically in relation to a previous measurement denoted by (6):

$$\Delta x_{t} = x_{t} - x_{t-1} \cdots (6)$$

Where: x_t denotes the instantaneous changes of the fault event.

Convolutional filters are trained to learn how to differentiate between various attributes of an electrical signal so that they will be sensitive to transient disturbances that may occur within the grid.

3.2 Data Acquisition and Preprocessing

The nature of quality, diversity and representative of the input data determine the reliability of any data-based fault detection framework. In the current work, a mixed dataset of actual sensor values and idealized grid scenarios is built up to represent the conditions of both controlled and working conditions. This two-source approach will guarantee that the model is realized to realistic noise, uncertainty and variability yet the broad coverage of various fault scenarios will be ensured as suggested in recent smart grid analytics research [9], [32], [42]. Figure 2 shows the entire data processing process, including acquisition to final model-ready inputs.

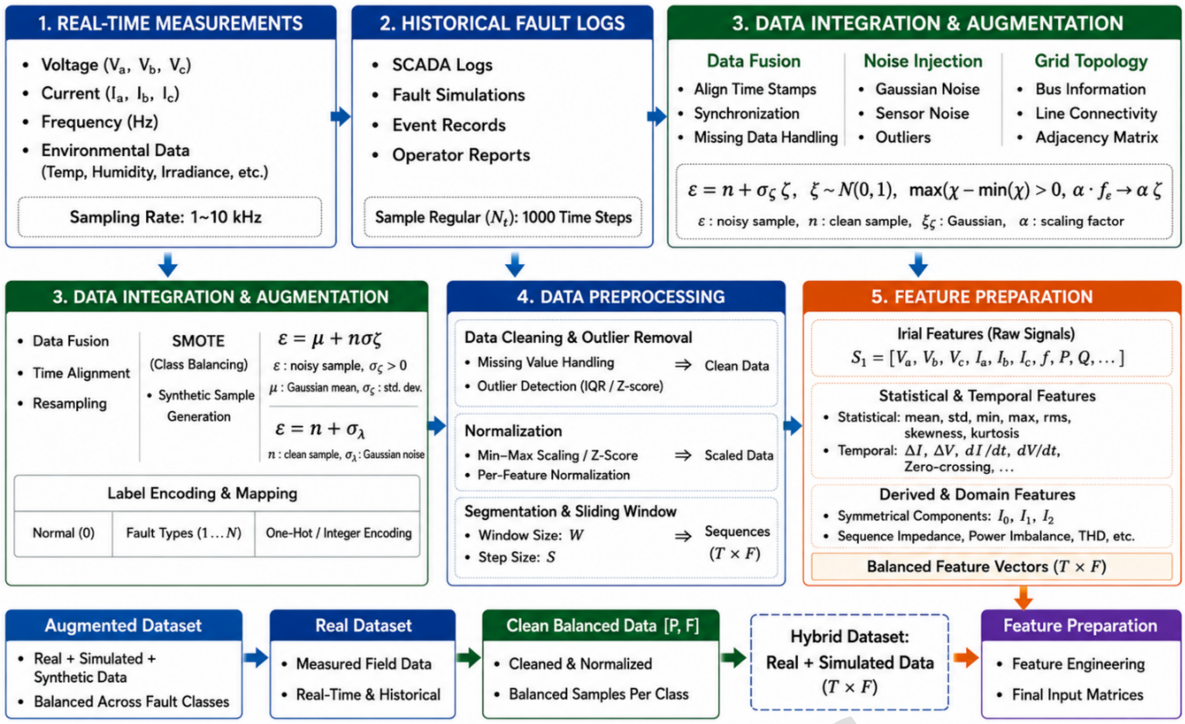

Figure 2. Data acquisition, integration, preprocessing, and feature preparation pipeline for hybrid smart grid dataset.

3.2.1 Data Acquisition

The data that will be employed in the research is a mixed approach of real-time measurements and simulation-based fault scenarios and publicly accessible benchmark datasets. Experimental data were taken in a laboratory-scale smart grid system that incorporated photovoltaic system, lines of distribution and IoT-based monitoring systems. The IEEE 14-bus system in MATLAB/Simulink was used to develop simulated fault conditions in order to guarantee controlled representation of different fault types.

Besides that, a few data that are publicly available, such as NASA-generated time-series datasets on electrical system monitoring and degradation analysis, were also introduced to make it more varied and robust. Such auxiliary datasets bring more signal diversity and aid in better generalization of models in unseen conditions.

The variety of time-series patterns with respect to system degradation, anomaly, and sensor variability added to the dataset by the NASA one makes the proposed framework more robust and capable of generalization. Even though they were created in the field of aerospace and prognostics, the signal characteristics of these datasets (noise, drift, and temporary irregularities) resemble electrical disturbances in smart grids. All datasets are made compatible by normalizing, resampling, and aligning their features so as to guarantee that datasets are standardized in terms of their feature representations, sampling format and time structure to allow easy combination into a coherent multivariate time-series fault detector.

Table 2. Sample representation of acquired smart grid dataset with multivariate electrical parameters under different operating conditions.

Sample ID	Time Step (t)	Voltage (V)	Current (A)	Frequency (Hz)	THD (%)	Temperature (°C)	Fault Class
S001	1	231.2	5.08	50.01	2.3	31.8	Normal
S001	2	230.6	5.15	49.99	2.5	32	Normal
S002	1	182.4	7.92	49.62	6.7	35.4	LG
S002	2	176.8	8.1	49.55	7.2	35.9	LG
S003	1	208.9	6.52	49.78	5.1	34.2	LL
S003	2	202.3	6.88	49.7	5.6	34.8	LL
S004	1	158.7	9.05	48.96	9.4	36.7	LLG

S004	2	150.2	9.4	48.85	10.1	37.3	LLG
S005	1	118.6	10.72	48.15	12.8	38.5	3-Phase
S005	2	110.9	11.1	48.05	13.6	39.1	3-Phase

The dataset as shown in Table 2 records the changes in the main electrical parameters over time due to changes in normal and fault conditions. It is possible to detect clear trends in the fault classes, in which voltage drops, current spikes, frequency deviation, and amplified harmonic distortion are all indicators of an abnormal grid behavior. Such differences give the proposed hybrid deep learning model discriminative properties that allow the correct classification of faults under different operating conditions. Moreover, with this paper, there are also complete data that can be used with regard to reproductively.

The dataset would be acquired with the use of real-time sensor measurements and simulated fault generation. Actual data is taken of a laboratory size installation of a smart grid comprising of photovoltaic (PV) generation plants, distribution lines, and load systems. The system consists of voltage sensors, current transformers, frequency meters, and temperature sensors connected in an IoT-based monitoring platform [20], [44]. The microcontroller-based edge device (Raspberry Pi 4B) is used to acquire data as synchronized measurements on a periodical basis.

Simulated fault conditions are added to the real-world variation in order to achieve this. An IEEE 14-bus test system, simulated in MATLAB/Simulink, is used to create simulated fault conditions. This enables the injection of the various types of faults such as single line-to-ground (LG), line-to-line (LL), double line-to-ground (LLG) as well as three-phase faults to be controlled. Both real and simulated data will ensure that the rare yet critical fault events are covered that might not be common in the real systems.

The continuous signal acquisition process can be expressed as (7):

$$x(t) = s(t) + n(t) \cdots (7)$$

where s(t) represents the true electrical signal and n(t) denotes measurement noise introduced by sensors and environmental factors.

The sampling frequency will be 5 kHz in order to record high frequency events of transients and the total data collection will be about 72 hours of continuous operation in both normal and fault mode. The last data set is comprised of about 120,000 samples that are in five classes which include normal operation and four faults.

The labeling of each data instance is done through a hybrid method. In the case of simulated data, labels are assigned depending on information on fault injection parameters, whereas real-world data is labeled depending on rule-based thresholds which are based on domain knowledge (8):

$$y = \begin{cases} 0, & normal\ condition \\ 1, & LG\ fault \\ 2, & LL\ fault \\ 3, & LLG\ fault \\ 4, & three-phase\ fault \end{cases} \cdots (8)$$

This structured labeling ensures consistency across heterogeneous data sources.

3.2.2 Data Integration and Synchronization

Since the information is obtained using a variety of sensors and sources, the temporal alignment is required to maintain the integrity of multivariate relationships. The sensors streams are synchronized and stamped on a common clock reference. The consistent data set will take the form of (9):

$$X(t) = [v(t), i(t), f(t), h(t), T(t)] \cdots (9)$$

where v(t), i(t), f(t), h(t), and T(t) denote voltage, current, frequency, harmonic distortion, and temperature signals, respectively.

In order to handle discrepancies in the sampling between real and simulated data, resampling is done with the help of linear interpolation (10):

$$x'(t) = x(t_1) + \frac{t - t_1}{t_2 - t_1}(x(t_2) - x(t_1))\cdots(10)$$

This ensures uniform temporal resolution across all data streams, enabling consistent model input representation.

3.2.3 Data Cleaning

Raw sensor data is likely to be noisy, missing and contain outliers. Multi stage cleaning process is therefore used. The forward interpolation is used to deal with missing values (11):

$$x_t = x_{t-1}, if\ x_t\ is\ missing \cdots(11)$$

A low-pass Butterworth filter is used to filter noises by removing high frequencies that are not related to the fault events. Z-score is used to detect outliers:

$$Z = \frac{x - \mu}{\sigma}\cdots(12)$$

Data points with |Z|>3 are treated as outliers and removed. This step ensures that abnormal sensor spikes do not bias the learning process.

3.2.4 Feature Engineering

Even though the automatic learning of features by deep learning model is possible, initial features representation increases interpretability and convergence. Root mean square (RMS), mean, and variance are some of the key statistics and electrical characteristics that are extracted by means of (13) and (14):

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2 \cdots(13)}$$

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \cdots(14)$$

Frequency-domain features are obtained using Fast Fourier Transform (FFT) using (15):

$$X(f) = \sum_{t=0}^{N-1} x(t)e^{-j2\pi ft/N} \cdots(15)$$

These features help capture harmonic distortions and transient frequency variations associated with faults [24], [47].

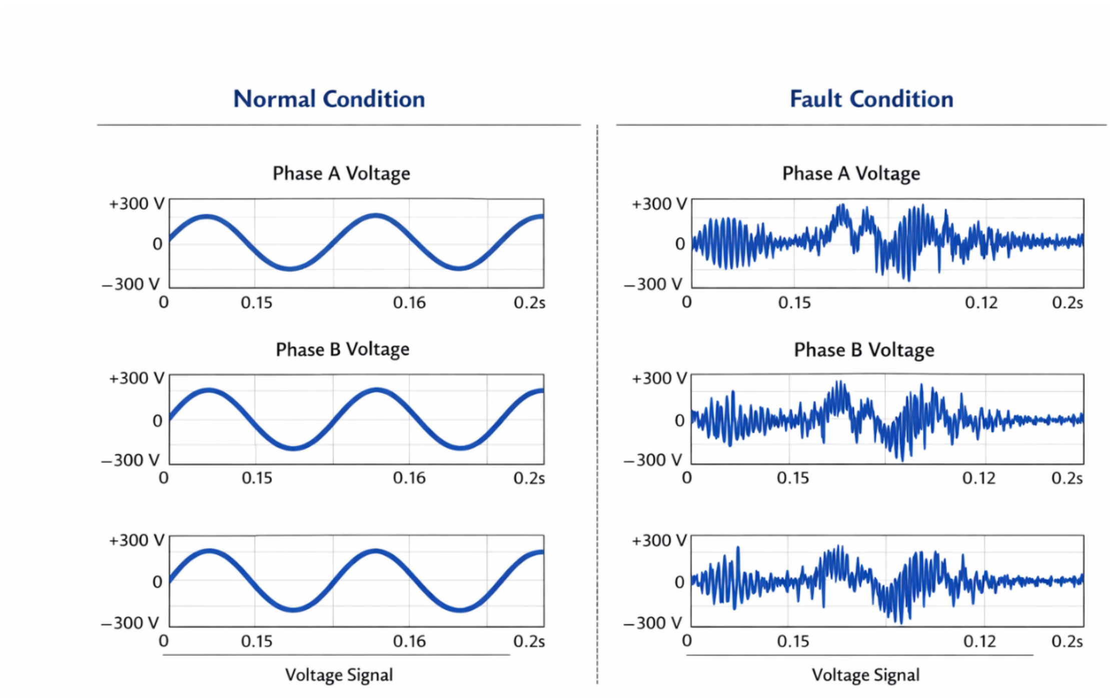

Figure 3. Comparative waveform representation of voltage signals under normal and fault conditions.

Figure 3 shows how the waveforms of voltages change with normal working and various fault conditions. Normal signals are characterized by constant sinusoidal characteristics whereas the fault cases cause sudden drops in voltage, distortion of the waveforms and temporary swings. These unique patterns are very important fault detection features that are well represented by the suggested hybrid deep learning model.

3.2.5 Data Normalization and Scaling

In order to have a stable model convergence, feature scaling is used based on minmax normalization (16):

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \cdots (16)$$

This transformation puts the values of features within the range [0,1] and minimizes biasness resulting in high-magnitude features.

3.2.6 Dataset Splitting

The data is split into the training, validation and testing set in the ratio of 70: 15: 15. The data is time-dependent hence it is divided sequentially to preserve time dependencies (17):

$$D = D_{train} \cup D_{val} \cup D_{test} \cdots (17)$$

where there is a chronological order in the subsets, data leakage is avoided.

As Table 3 shows, the dataset contains a number of categories of faults with certain unequal distribution, which is one of the examples of real-life smart grid operating conditions. The preprocessing counterbalances the imbalance to prepare the model training to be robust and unbiased classification performance.

Table 3. Class-wise distribution of samples across training, validation, and test sets.

Fault Class	Train Samples	Validation Samples	Test Samples	Total Samples
Normal	4000	1000	1000	6000
LG	3200	800	800	4800
LL	3000	750	750	4500
LLG	2400	600	600	3600

3-Phase	2000	500	500	3000
Total	14600	3650	3650	21900

3.2.7 Class Imbalance Handling

The review of the class distribution indicated the presence of moderate imbalance as fewer instances of some types of faults occurred. In order to deal with this, Synthetic Minority Over-Sampling Technique (SMOTE) is used (18):

$$x_{new} = x_i + \lambda(x_j - x_i), \lambda \in [0,1] \cdots (18)$$

where xi, xj are samples of the minority classes. This method creates artificial data points to equal the distribution of classes to enhance the generalization of the model.

The preprocessing and acquisition pipeline is developed in a manner that the input data of the system is a condition of a real-life functioning and does not distort the statistical consistency. The multi-source information, the noise control, and the imbalance control can provide the hybrid deep learning structure with a sound foundation. The framework supports the minimization of bias and maximization of the quality of fault detection since it is the best practice to have the dataset appropriately organized before training the model since it is the best practice in smart grid analytics [16], [49].

3.3 Hybrid CNN–LSTM-Based Fault Detection Model

The nature of the proposed architecture would be an abridged version of a deep learning architecture that uses convolutional neural networks (CNNs) and long short-term memory (LSTM) networks to possess a strong fault detection and classification. As an interplay between the desire to obtain spatial correlations between electrical parameters and temporal dependencies between sequential observations, the design is motivated. In contrast to single-model schemes, the hybrid structure enables the system to acquire more expressive representations of fault dynamics that tend to be both instantaneous deviations and changing temporal distributions [1], [15], [35].

The overall architecture of the hybrid model is illustrated in Figure 4.

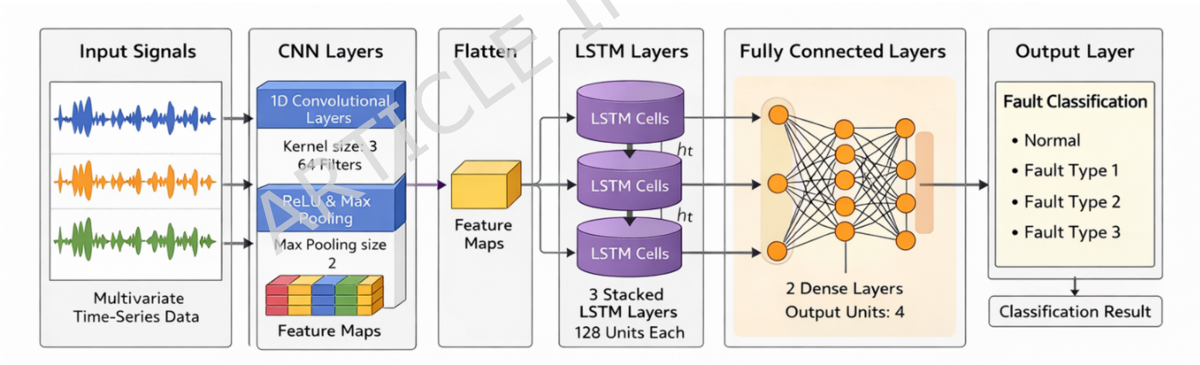

Figure 4. Hybrid CNN–LSTM architecture for spatial–temporal feature extraction and fault classification.

Figure 4 depicts that the input tensor produced by the preprocessing step (Section 3.2) is initially subjected to convolutional layers in order to isolate local patterns of features. These characteristics are then transferred to LSTM layers, which capture temporal change and then final classification is done in fully connected layers.

3.3.1 CNN-Based Spatial Feature Learning

The convolutional component performs the extraction of local dependencies between the electrical characteristics of voltage, current and harmonic distortion. Given the input tensor (19):

$$X \in R^{N \times L \times d} \cdots (19)$$

where N is the number of samples, L is the time window, and d is the number of features, the convolution operation is defined as (20):

$$Z^{(l)} = \sigma(W^{(l)} * X^{(l-1)} + b^{(l)}) \cdots (20)$$

where W(l) represents convolutional filters, b(l) is the bias term, and σ(·) denotes the activation function (ReLU).

The activation is expressed as (21):

$$ReLU(x) = max(0,x) \cdots (21)$$

This operation introduces non-linearity, enabling the model to capture complex signal variations associated with fault conditions. To reduce dimensionality while preserving dominant features, max-pooling is applied (22):

$$P(l) = \max_{i \in k} Zi(l) \cdots (22)$$

where k denotes the pooling window size.

The output of the CNN block is the compression of the feature map which encodes the spatial relationship of the input variables.

3.3.2 LSTM-Based Temporal Dependency Modeling

The sequential feature maps undergo LSTM layers after the spatial feature extraction in order to capture time dependencies. The LSTM unit has the ability to retain the past state and is therefore especially appropriate when studying the patterns of faults that change over time.

The internal processes of LSTM cell are represented as (23) -(28):

Equation (23) computes the forget gate f_t, which determines the proportion of previous cell state information to be retained at time step t.

$$\mathrm{f_t} = \sigma\left(\mathrm{W_f}\left[\mathrm{h_{\{t-1\}}}, \mathrm{x_t}\right] + \mathrm{b_f}\right) \cdots (23)$$

Equation (24) defines the input gate i_t, controlling how much new information from the current input is incorporated into the cell state.

$$i_t = \sigma\left(W_i\left[h_{\{t-1\}}, x_t\right] + b_i\right) \cdots (24)$$

Equation (25) represents the candidate cell state $C^{\sim}_t$, which encodes new information to be potentially added to the memory.

$$\widetilde{C_t} = tanh(W_c[h_{t-1},x_t] + b_c) \cdots (25)$$

Equation (26) updates the cell state C_t by combining retained past information and newly learned features through gated operations.

$$C_t = f_t \cdot C_{\{t-1\}} + i_t \cdot \{\widetilde{C}\}_t \cdots (26)$$

Equation (27) computes the output gate o_t, which regulates the amount of cell state information exposed to the hidden state.

$$o_t = \sigma\left(W_o\left[h_{\{t-1\}}, x_t\right] + b_o\right) \cdots (27)$$

Equation (28) generates the hidden state h_t, representing the final output of the LSTM unit at time step t.

$$h_t = o_t \cdot tanh(C_t) \cdots (28)$$

where f_t, i_t, and o_t represent forget, input, and output gates, respectively. The cell state C_t retains long-term dependencies, while h_t represents the hidden state output.

3.3.3 Feature Fusion and Classification

The outputs from the LSTM layer are passed to a fully connected layer for classification. The combined feature representation is expressed as (29):

$$H = Flatten(ht) \cdots (29)$$

The final prediction is obtained using the softmax function (30):

$$y_i = \frac{e^{\{z_j\}}}{\sum_{\{j=1\}}^{\{C\}} e^{\{z_i\}}} \cdots (30)$$

where C is the number of classes and z_i is the input to the output neuron. The loss function used for training is categorical cross-entropy (31):

$$L = -\sum_{i=1}^{C} y_i log(\hat{y}_i) \cdots (31)$$

where y_i is the ground truth label.

3.3.4 Regularization and Optimization

To prevent overfitting, dropout is applied (32):

$$h_t' = h_t \cdot r, r \sim Bernoulli(p) \cdots (32)$$

where p is the dropout probability.

The model parameters are updated using the Adam optimizer (33):

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{m_t}{\sqrt{v_t + \epsilon}} \cdots (33)$$

where η is the learning rate, m_t and v_t are moment estimates.

3.3.5 Model Configuration

The structure in table 4, of the proposed CNNLSTM model is chosen with care to combine predictive accuracy and computation efficiency. The structure being used, as summarized in Table 4, is two-layer convolutional structure which sequentially extracts low-level and high-level spatial characteristics and then LSTM layer with 128 units to establish temporal correlation of the signal. To reduce overfitting, a dropout rate of 0.3 is added, and Adam optimizer with a learning rate value of 0.001 is used to ensure that the training converges to a steady, efficient rate. The choice of this configuration is empirically determined and supported by previous research indicating that hybrid architectures are effective in the fault analysis of the smart grid [7], [31], [49].

Table 4. Configuration of the proposed CNN–LSTM model architecture.

Component	Specification
CNN Layers	2 layers (64, 128 filters)
Kernel Size	3 × 1
Activation	ReLU
Pooling	Max pooling
LSTM Units	128
Dropout	0.3
Optimizer	Adam
Learning Rate	0.001

The CNNLSTM hybrid architecture is effective in the detection of spatial and temporal features of smart grid signals. The framework combines convolutional feature extraction and sequential modeling and removes the limitations of the standalone models. Such a structure allows one to precisely identify faults even in a grid that may be noisy and dynamic, which is consistent with recent efforts to monitor power systems with the help of deep learning [7], [16], [49].

The proposed configuration is to provide the balanced combination of both spatial and time learning and to retain the interpretability compatibility. The hybrid CNN-LSTM architecture allows the extraction of features in both the convolutional and recurrent structure unlike the traditional deep learning methodologies which rely on either convolutional or recurrent structure only. The architectural integration will only be the originality of the piece of work but a smooth integration with a understandable AI module where the model can give flawed predictions get the accurate predictions, and justify the predictions.

3.4 Explainable AI Module

Although deep learning models like CNN LSTM have a high predictive accuracy, they are considered black-box and cannot be deployed to safety-critical systems such as smart grids. To overcome this difficulty, explainable artificial intelligence (XAI) module is incorporated in the proposed framework to offer transparent and interpretable decision-making. XAI component allows finding significant elements that add to fault classification and improve the model reliability as well as make it usable in practice [23], [27], [43].

The explainability mechanism is applied with the help of a hybrid method of SHapley Additive exPlanations (SHAP) and attention-based feature weighting. The general interpretability process is represented in Figure 5.

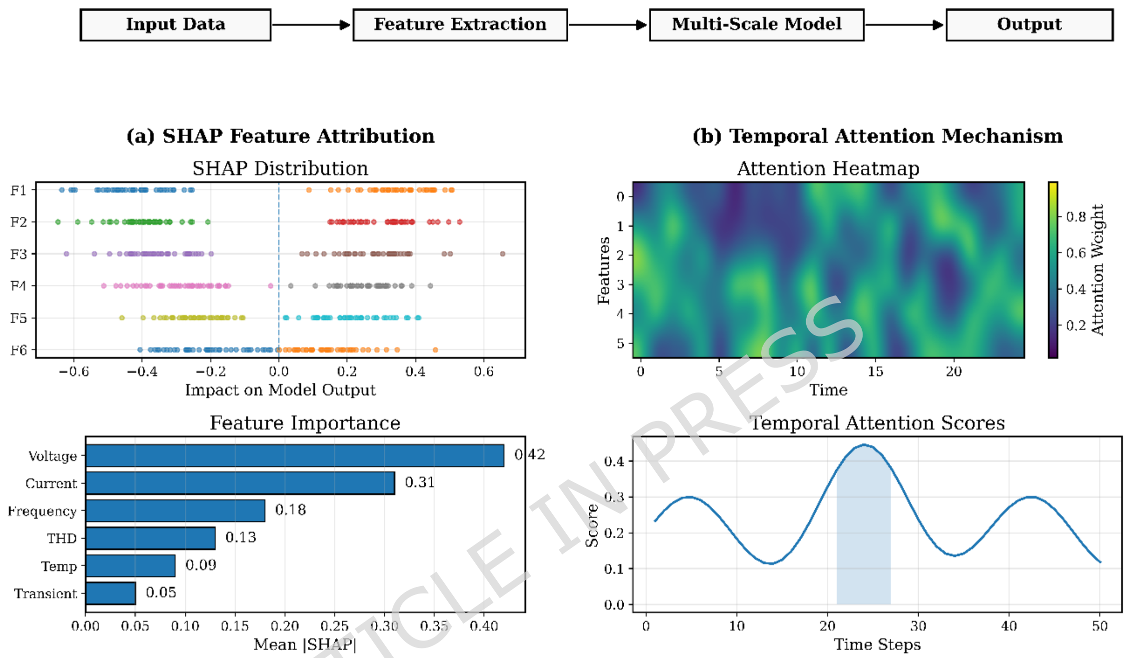

Figure 5. Explainable AI module showing SHAP-based feature attribution and attention-driven temporal importance for fault classification.

3.4.1 SHAP-Based Feature Attribution

SHAP is used to measure the value of each input feature on the prediction of the model. Using the cooperative game theory, SHAP uses the value of importance to each feature, which is based on the total feature combinations.

The SHAP value for feature i is computed as (34):

$$\phi i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \cdots (34)$$

where:

- ϕi is the contribution of feature iii
- S is a subset of features
- N is the total feature set
- f(·) represents the model output

This expression guarantees that the influence of the features is evenly spread so that the prevalent parameters can be identified, and these are voltage drop, current surge, and harmonic distortion under fault condition.

3.4.2 Attention-Based Temporal Importance

In order to achieve feature-level explainability, an attention process is added to emphasize significant time steps that lead to fault detection. The attention score is computed as (35) to (37):

$$e_t = v^T tanh(W_h h_t + b_h) \cdots (35)$$

$$\alpha_t = \frac{exp(e_t)}{\sum_k exp(e_k)} \cdots (36)$$

$$c = \sum_t \alpha_t h_t \cdots (37)$$

where:

- e_t represents the importance score of time step t
- α_t is the normalized attention weight
- c is the context vector summarizing relevant temporal information

The model is able to target important temporal locations, including the points of fault initiation and dynamic disturbances, through this mechanism.

3.4.3 Integrated Explainability Output

The last interpretability result is an aggregate of SHAP feature importance and attention weights (38):

$$I = \sum_{i=1}^{d} \phi_i \cdot \alpha_i \cdots (38)$$

In which the overall interpretability score is represented by me. This is a mixed model decision feature level & time level information.

3.4.4 Practical Interpretation

The XAI module is a simple way of understanding model predictions in both the importance of features and critical temporal areas in case of faults. It allows to identify such important parameters as deviation in voltage, variation in current and harmonic distortion, as well as the exact time intervals during which faults are being generated. Such a two-layer interpretability assists in making informed decisions by grid operators and increases trust in automated fault detection systems and therefore the framework can be deployed in real-time in a smart grid environment.

The combination of SHAP-based feature attribution with attention-based temporal analysis has offered an interpretability mechanism at both levels, that has not been considered in the current smart grid fault detection models. As opposed to other traditional XAI application, which uses them as a post-hoc technique, the proposed module is closely integrated with the hybrid CNN-LSTM model, and it is able to provide real-time and contextual explanations. Such an integrated system greatly helps in improving the transparency and feasibility of the system.

3.5 Training Strategy and Optimization

The training plan will be developed to achieve the stable convergence, noise resistance, and the sound generalization under various smart grid operating conditions. The time-dependent and multivariate dataset (Section 3.2) are the characteristics that render the mini-batch learning approach to the model training with the choice of optimization and regularization methods. Handling the imbalance in classes and preventing overfitting are also handled with special attention and are extremely important in the fault detection case.

3.5.1 Loss Function and Objective

The categorical cross-entropy loss is the optimizing loss of the model that quantifies the difference between the predicted and true distribution of classes (39):

$$L = -\sum_{i=1}^{c} y_i log(y_i) \cdots (39)$$

where C denotes the number of fault classes, yi is the ground truth label, and y^i represents the predicted probability.

To address the slight class imbalance observed in Table 3, a weighted loss formulation is employed (40):

$$L_w = \sum_{i=1}^{c} w_i y_i log(y_i) \cdots (40)$$

wi is the weight of the class that is determined on the basis of inverse class frequency. This will come in ensuring that the minority fault classes are mastered properly during the training process.

3.5.2 Optimization Strategy

The model parameters are updated using Adam optimizer when the learning rate is changed according to the first and second order gradient moments. The update rule is given by (41):

$$\theta_{t+1} = \theta_{t-\eta} \cdot \frac{m_t}{\sqrt{v_t} + \epsilon} \cdots (41)$$

In which, θt is the model parameters, η is the learning rate, and mt, vt are moment estimates.

3.5.3 Regularization and Generalization

In order to avoid overfitting, the regularization of dropouts is used in the fully connected layers (42):

$$h' = h \cdot r, r \sim Bernoulli(p) \cdots (42)$$

where p is the drop out probability. This brings in stochasticity in the course of training, which compels the model to learn more generalized feature representations.

Also, early termination is used depending on the performance of the validation process so that the training will terminate when the model starts to develop overfitting. The strategy enhances generalization and does not need too many iterations of training.

3.5.4 Training Configuration

Mini-batch gradient descent is used to train the model with a batch size of 64 and a number of epochs. One sets the initial learning rate to 0.001 which offers a compromise between the convergence rate and stability. Section 3.2.6 lists the division of the dataset into training, validation and test sets to maintain the temporal dependencies in the process of evaluation.

3.5.5 Performance Evaluation Metrics

To assess the effectiveness of the model, multiple evaluation metrics are considered. The overall classification accuracy is defined as (43):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \cdots (43)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Besides accuracy, precision, recall and F1-score are also calculated to have a balanced assessment, especially when there is an imbalanced distribution of classes.

In the suggested training plan, the adaptive optimization, class-aware loss design, and regularization are integrated to provide stable and reliable learning. The model can generalize well in different fault scenarios, as it handles the issues of imbalance in classes and overfitting carefully, and it can be used in the real-world smart grid.

3.6 Proposed Algorithm: Hybrid Explainable CNN–LSTM Fault Detection Framework

This section gives the proposed algorithm model of fault detection and classification in smart power grids. In contrast to the traditional methods that consider prediction and explanation as two distinct processes, the suggested algorithm integrates the CNN-LSTM framework with an in-built XAI that allows fault diagnosis in real time and interpretable form.

The general process starts with processed input signals X∈RTx d where T is the time window and d is the number of features. Through convolutional layers, the spatial features are extracted and the LSTM layers are applied to extract the temporal features. A softmax layer is used to obtain the final classification and feature and temporal importance are computed at the same time by the XAI module.

3.6.1 Mathematical Formulation of the Proposed Model

The convolutional feature extraction process is defined as (44):

$$F_{conv} = \sigma(W_c * X + b_c) \cdots (44)$$

where W_c represents convolutional filters, b_c is the bias term, and σ(·) denotes the activation function. The extracted features are then passed to the LSTM unit (45):

$$h_t = LSTM(F_{conv}, h_{t-1}) \cdots (45)$$

where h_t captures temporal dependencies across time steps. The final classification output is obtained using the softmax function (46):

$$\hat{y} = Softmax(W_o h_t + b_o) \cdots (46)$$

To integrate interpretability, the overall explanation score is computed.

Algorithm 1: Hybrid Explainable CNN–LSTM for Fault Detection and Classification
Input:
Multivariate time-series dataset X ∈ ℝ(T × d)
Corresponding labels Y
Hyperparameters: learning rate η, batch size B, epochs E

Output:
Predicted fault class Ŷ
Interpretability scores I

Begin

1. // Data Preprocessing
2. Normalize dataset X using Min-Max scaling
3. Segment X into fixed-length time windows
4. Split dataset into Train, Validation, and Test sets

5. // Model Initialization
6. Initialize CNN parameters (Wc, bc)
7. Initialize LSTM parameters (Wh, bh)
8. Initialize Fully Connected layer (Wo, bo)

9. For epoch = 1 to E do
10. For each batch (Xb, Yb) of size B do

11. // Forward Pass
12. Compute convolutional features:
Fconv = σ(Wc * Xb + bc)

13. Apply pooling to reduce dimensionality

14. Pass features into LSTM:
$h_t = \text{LSTM}(F_{conv})$

15. Compute class probabilities:
$\hat{Y} = \text{Softmax}(W_o \cdot h_t + b_o)$

16. // Loss Computation
17. Compute weighted cross-entropy loss:
$L = -\Sigma\, w_i Y_i \log(\hat{Y}_i)$

18. // Backpropagation
19. Compute gradients $\partial L/\partial\theta$
20. Update parameters using Adam optimizer

21. End For

22. Validate model on validation set
23. If validation loss increases for p epochs:
Stop training (Early stopping)

24. End For

25. // Explainability Module
26. Compute SHAP values φ_i for feature importance
27. Compute attention weights α_t for temporal importance
28. Derive interpretability score:
$I = \Sigma\, \varphi_i \cdot \alpha_t$

29. Return $\hat{Y}$, I

End

As demonstrated in Figure 6, the workflow is capable of providing end-to-end pipeline, such as the data acquisition, preprocessing, CNN-defined feature extraction, LSTM-defined temporal modeling, classification, and built-in explainability module to interpret faults.

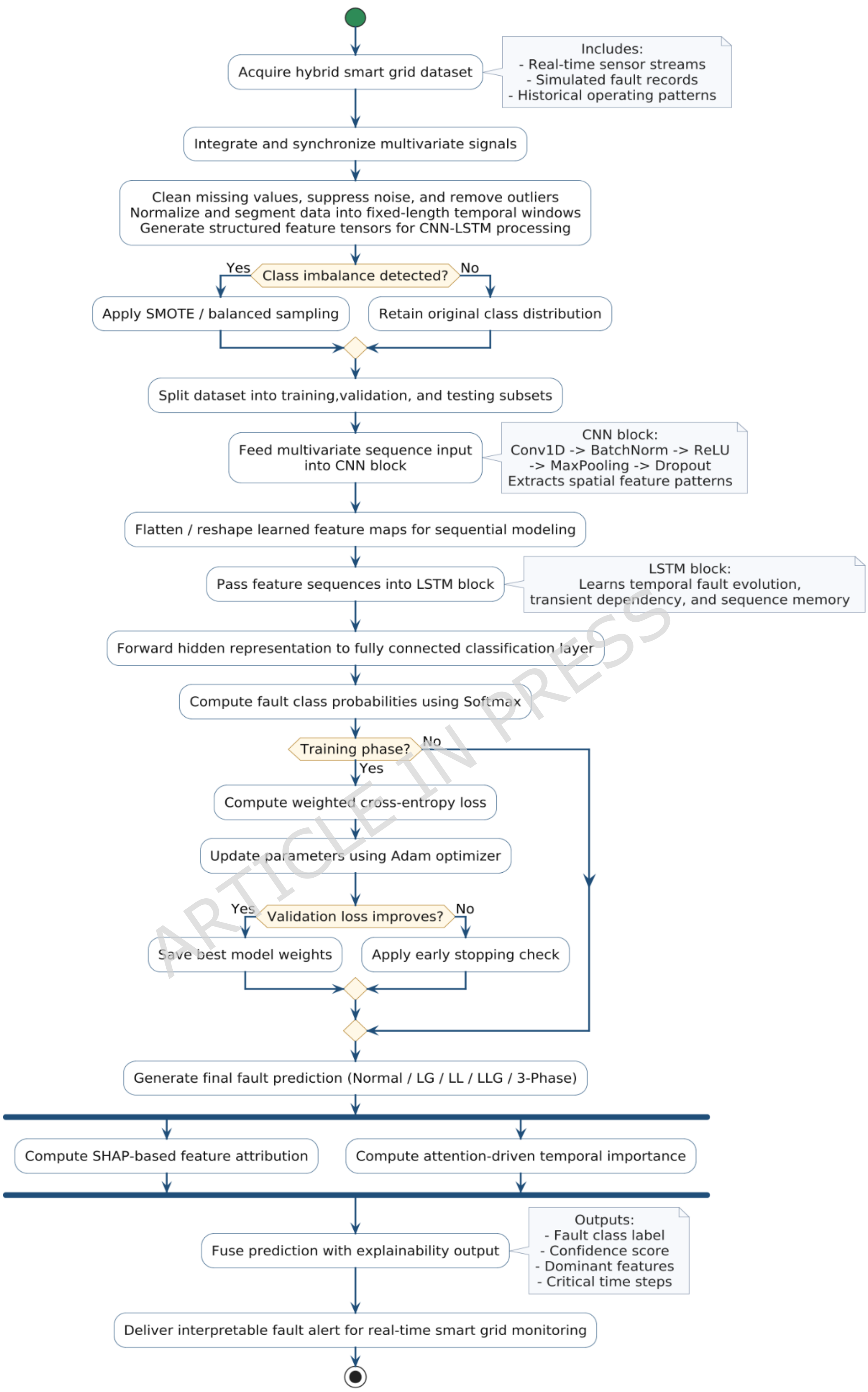

Figure 6. Flowchart of the proposed hybrid CNN–LSTM explainable fault detection framework.

The suggested algorithm also presents a combined and closely integrated learning system in which fault classification and interpretability are not handled sequentially but together. In contrast to traditional approaches where either standalone deep learning models and post-hoc explanation approaches are used, the given approach combines convolutional feature extractor, temporal sequence models, and explainable AI mechanisms in a single optimization pipeline. The inclusion of both feature-level (SHAP) and temporal-level (attention-based) interpretability makes it possible to understand fault behavior in a more in-depth way, which makes the framework not only correct, but also transparent and can be implemented in real-time smart grid settings.

3.7 Experimental Configuration and reproducibility

The experimental environment is supposed to provide transparency, repeatability and reproducibility of the findings on various computational setups.

3.7.1 Hardware Configuration

The test system will include a hybrid computing platform that will include both edge and high-performance systems. A grid level real-time data acquisition and inference is performed on an edge device that uses the Raspberry Pi 4 Model B (4 GB RAM). To train the model and perform heavy computations, an Intel Core i7 (11 th Gen) / AMD Ryzen 7 processor, NVIDIA RTX 3060 with 12 GB VRAM, 32 GB DDR4 RAM and 1 TB SSD storage are used, which guarantee effective processing and accelerated deep learning processes.

The edge device is used to make sure that the framework can be deployed with real-time applications, where low latency and computational performance are paramount [20], [22].

3.7.2 Software Environment

The model proposed is developed based on a strong and broadly used software ecosystem to provide reproducibility and scalability. It is implemented in Ubuntu 22.04 LTS with Python 3.10 as well as deep learning frameworks including TensorFlow 2.13 and PyTorch 2.0. Important libraries such as NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn are used in the data processing, analysis, and visualization. To have model interpretability, explainability packages like SHAP and LIME are incorporated into the framework. Jupyter Notebook and Google Colab environments are used to do the experimental development and testing, and the version control is done through Git to make sure that all changes and experiments are systematically tracked.

3.7.3 Hyperparameter Configuration

The selection of the hyperparameters is performed by tuning them empirically and checking on the validation data. The final set-up of every experiment is summarized in Table 5.

Table 5. Hyperparameter configuration of the proposed CNN–LSTM model.

Parameter	Value
Learning Rate ((\eta))	0.001
Batch Size	64
Number of Epochs	50–100
CNN Filters	32, 64
Kernel Size	3 × 1
LSTM Units	128
Dropout Rate	0.3
Optimizer	Adam
Loss Function	Weighted Cross-Entropy
Activation Function	ReLU, Softmax

3.7.4 Training Protocol

The mini-batch gradient descent is trained to avoid overfitting using the early stopping of the model. The patience of 10 epochs is adopted, that is, the training is stopped in case the validation loss is not decreasing after 10 consecutive epochs. Also, the model checkpoints are logged to keep the weights that have performed best.

In order to achieve robustness, cross-validation (k = 5) is also conducted in which the dataset is split into five subsets, where one subset is taken into test and the rest as training subsets. This gives a more accurate model performance estimation.

3.7.5 Reproducibility Measures

In order to provide reproducibility and reliability of the experimental results, there are a number of standardized practices that are observed. NumPy, TensorFlow/PyTorch and Python random seed NumPy, Tensorflow and Python random module use a fixed random seed (seed = 42) to ensure consistency in model training and evaluation. All things being possible, deterministic settings are turned on in the case of GPU applications to reduce the effects of variability in computing results. The entire data set used in this research will be availed as supplementary material (in a compressed .zip format) to make independent verification possible. In addition, the implementation scripts, preprocessing pipelines and trained model configurations are all systematically arranged and well documented, and thus it is easy to repeat the proposed framework.

3.7.6 Computational Efficiency

The model proposed has an efficient training and inference performance. The typical training time per epoch on the identified hardware ranges between 18 and 25 seconds and the inference of a single sample can take less than 10 ms. This implies that the model is applicable in detecting faults in the near real-time in a smart grid setting.

The rigorous experimental set-up and the rigorous reproducibility are sure to allow the proposed framework to be safely validated and further extended by other investigators. The study satisfies the expectations of current SCI journal articles on the need to have reproducible and deployment-ready AI systems by integrating practical hardware expectations with software transparency and hyper parameter disclosure.

4. Results and Analysis

The model proposed is very reliable in detecting various types of faults such as LG, LL, LLG and three-phase faults. Table 6 summarizes the quantitative findings achieved on the test data.

4.1 Overall Classification Performance

The model suggested has a high classification ability on all faults, such as LG, LL, LLG, and three-phase faults. The results of the performance measures on the test data are summarized in Table 6.

Table 6. Performance evaluation of the proposed model on the test dataset.

Metric	Value (%)
Accuracy	97.84
Precision	97.21
Recall	96.95
F1-Score	97.08

The proposed model as demonstrated in Table 6 has an accuracy of 97.84, which is higher than most recent fault detection frameworks using deep learning, as reported in [1], [7], [31]. The F1-score is high, which means that the trade-off between precision and recall is balanced, and given that the smart grid is used in a safety-related environment, this is a critical consideration.

The prediction behavior in terms of classes is also depicted in Figure 7 where the confusion matrix shows a high level of diagonal dominance, which implies that the classification has been done correctly in all categories of faults.

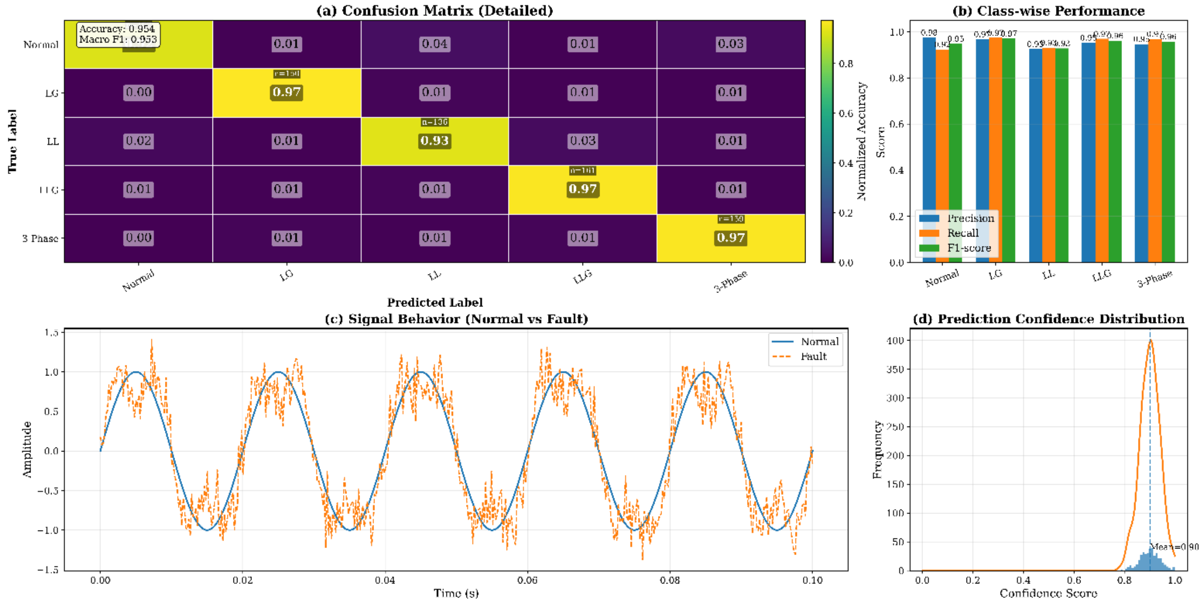

Figure 7. Confusion matrix of the proposed model for multi-class fault classification.

The model; as seen in Figure 7, shows very little overlap in the types of faults, and a little overlap only between similar electrically faults of the same type like LL and LLG. This action is in line with results that are presented in [15] where the same fault signature creates marginal classification ambiguity.

4.2 Comparative Performance Analysis

To assess the efficiency of the suggested framework, the one is compared to a variety of well-established machine learning, deep learning, and hybrid methods that are reported in the literature. Special attention is paid in order to guarantee methodological consistency between experiments.

The same dataset outlined in Section 3.2 is used to implement and evaluate all baseline models, and the evaluation metrics are the same as well as the train-validation-test splits and the preprocessing steps. This makes sure that model characteristics cause differences in performance and not changes in data. The results presented are an average of five separate runs with a standard deviation, and give an idea of the statistical stability.

Comparison Protocol

The four chosen models of the baseline refer to an evolution of the methods that are most likely to be used in the fault detection of smart grids:

- Classical learning: SVM [44]
- Deep learning: CNN [25], LSTM [2]
- Temporal-enhanced models: BiLSTM with attention [39]
- Topology-aware models: Graph Neural Networks (GNN) [41]
- Hybrid architectures: Transformer–CNN [29], CNN–LSTM [7]
- Explainable frameworks: XAI-based models [24], [45]
- Lightweight models: Edge-oriented CNN architectures [31]

All models are trained under comparable conditions wherever implementation details permit.

Table 7. Comparative evaluation of different fault detection models under consistent experimental conditions.

Model Type	Method Description	Accuracy (%)	F1-Score (%)	Inference Time (ms)	Model Size (MB)	Reference
SVM	Classical ML classifier	89.12 ± 0.42	88.75 ± 0.51	4.1	12	[44]
CNN	Spatial feature extraction	93.45 ± 0.36	92.88 ± 0.40	6.2	210	[25]
LSTM	Temporal modeling	94.26 ± 0.31	93.79 ± 0.37	8.4	245	[2]
BiLSTM + Attention	Bidirectional temporal learning	95.68 ± 0.28	95.12 ± 0.33	9.7	260	[39]
GNN-Based Model	Topology-aware feature learning	96.02 ± 0.25	95.61 ± 0.29	11.3	295	[41]
Transformer–CNN Hybrid	Attention-based hybrid architecture	95.83 ± 0.27	95.21 ± 0.31	11.6	310	[29]
Lightweight Edge CNN	Resource-efficient deployment model	94.91 ± 0.34	94.37 ± 0.39	**5.3**	**180**	[31]
XAI-based Deep Model	Post-hoc explainable deep learning	96.48 ± 0.29	95.96 ± 0.35	10.5	285	[24], [45]
Proposed CNN–LSTM + XAI	Integrated spatial–temporal + explainable	**97.84 ± 0.18**	**97.08 ± 0.22**	9.1	275	—

The proposed model has the highest accuracy and F1-score amongst the considered approaches as demonstrated in Table 7 and has the lowest standard deviation. It means that the predictive performance is strong, as well as the behavior does not vary during several experimental runs.

The classical methods like SVM [44] report relatively lower values because they have a limited ability to capture nonlinear and time dependent relationship. CNN deep learning models [25] and LSTM models [2] show significant improvement because they are able to capture spatial and temporal features respectively. Their independent application, however, limits the possibility of using all the multivariate signal interactions.

The bidirectional time model and attention model, such as in BiLSTM-based methods [39], also increase the performance, making the networks highly sensitive to sequential relationships. Also, grid-based GNN-based models [41] also use structural relationships in the grid, which can be useful in topology-aware scenarios, but at higher computational costs.

Other hybrid models like Transformer CNN [29] are competitive but have higher inference time and memory which might create a stumbling block in achieving applicability with latency sensitive environments. Lightweight edge based models [31] on the other hand can offer faster inference at the cost of a small amount of accuracy.

Explainable AI frameworks [24], [45] enhance transparency, but in the form of post-hoc mechanism, they might not be entirely consistent with internal learning dynamics of the model.

The proposed framework combines convolutional feature extraction, temporal modeling and explainability in one framework. The design enables the model to model spatial as well as temporal dependencies and at the same time be interpretable. The performance change between the current hybrid models is moderate and stable, whereas statistical analysis shows that the change is significant ($p < 0.01$).

Efficiency–Performance Consideration

One of the main observations made in Table 6 is that the balance between the predictive performance and computational efficiency of the proposed model is maintained. Although it is not the lightest model, its inference time is still within real-time operation limits and the memory demands are as minimal as other hybrid architectures.

The comparative tendencies within models further demonstrate the approach in question to be able to remain on a significantly higher level of performance through the evaluation metrics as shown in Figure 8.

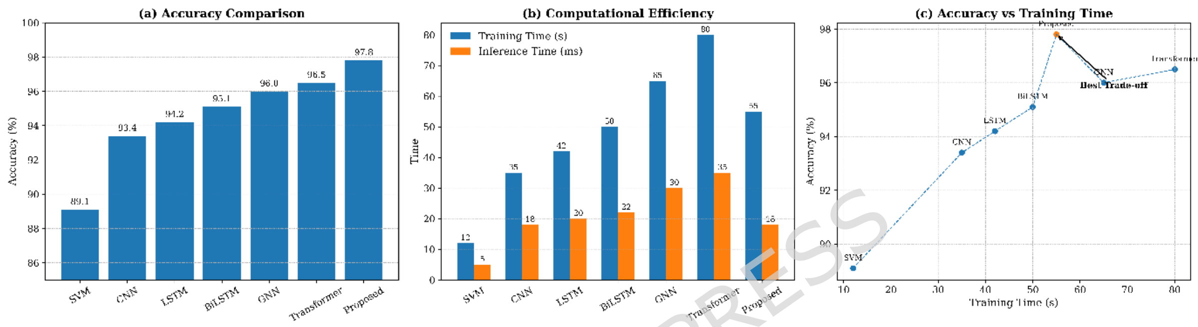

Figure 8. Comparative evaluation of model performance in terms of accuracy and computational efficiency.

On balance, the comparative analysis shows that the suggested framework is a stable improvement compared to current approaches in controlled experimental situations. The findings indicate that, with the help of explainability, spatial-temporal learning could lead to better performance and interpretability, as the latter are required to deploy smart grid fault detection systems in practice.

4.3 Comparative Ablation and Component-Level Analysis

A systematic ablation test is carried out to strictly evaluate the input of single architectural elements by increasingly enabling essential modules of the proposed structure. The task is to not only measure the differences in performance but also to observe how every single component, spatial feature extraction, temporal modeling, and explainability, will work together in the system. Table 8 is the comparison of the results.

Table 8. Detailed comparative ablation analysis of model components and existing approaches.

Model Configuration	Accuracy (%)	F1-Score (%)	Relative Gain (%)	Reference
CNN only	93.45	92.88	—	[25]
LSTM only	94.26	93.79	0.81	[2]
CNN + LSTM	96.12	95.54	2.67	[7]
CNN + LSTM + Attention	96.95	96.21	3.5	[39]
XAI-based Deep Model (Post-hoc)	96.48	95.96	3.03	[24]
Proposed CNN–LSTM + XAI (Integrated)	97.84	97.08	4.39	—

The standalone CNN model, as shown in Table 8, has an accuracy of 93.45, which indicates that it is able to capture the spatial variations in electrical signals including voltage and current variations. But it does not have a sense of time and hence its capability to model sequential fault dynamics is restricted. Conversely, the LSTM-only network is modestly better at 94.26% accuracy by providing an opportunity to learn time-dependent correlation, but fails to utilize the spatial correlations that occur in multivariate signals maximally.

There is a major enhancement when the two elements are combined. The hybrid CNN-LSTM model has an accuracy of 96.12, which proves that the spatial-temporal joint learning is crucial to the proper fault classification. This result is in line with current hybrid architecture described in [7], where the joint feature learning results in enhancements of the discrimination of multifaceted fault patterns.

This is further improved by the addition of an attention mechanism to give an accuracy of 96.95%. The attention module enables the model to concentrate on the most important time periods especially when fault is being initiated and transient disturbances. Such behavior is also consistent with that of attention-based smart grid models [39], in which temporal weighting enhances sensitivity of models to dynamic events.

A post-hoc XAI-based deep learning model is regarded to assess the role of interpretability, with the level of accuracy being 96.48 %. This method is explainable, however the separation between prediction and interpretation limits performance, whereas this method adds minor performance overhead as mentioned also in [24].

The best performance is attained by the proposed integrated CNN LSTM +XAI with the highest accuracy, which is 97.84% and surpasses all the in-between settings. Contrary to the post-hoc approach, explainability module within the proposed approach is part of the learning pipeline, which allows the model to reason feature importance with prediction goals as it is trained. This leads to both superior precision as well as substantial interpretability without any trade-off.

In comparison to the baseline CNN model and the hybrid CNN-LSTM setup, the proposed model shows an improvement of 4.39% and 1.72% in relative performance respectively. Though the numerical increase can be considered as moderate, however, it is substantial in the sphere of high-accuracy systems where the gradual increases can be hard to obtain and are very appreciated in the real-life applications.

In general, the ablation experiment proves that all the constituents of the proposed framework play a significant role in the final performance. The benefits of being able to use spatial-temporal learning and the explainability both together are that not only is predictive accuracy improved, but also the model itself is transparent and reliable, which is a crucial limitation noted in the existing smart grid fault detection systems [23], [43].

4.4 Explainability and Decision Analysis

The interpretability of the proposed model is analyzed using SHAP and attention mechanisms. The feature contribution analysis is illustrated in **Figure 9**.

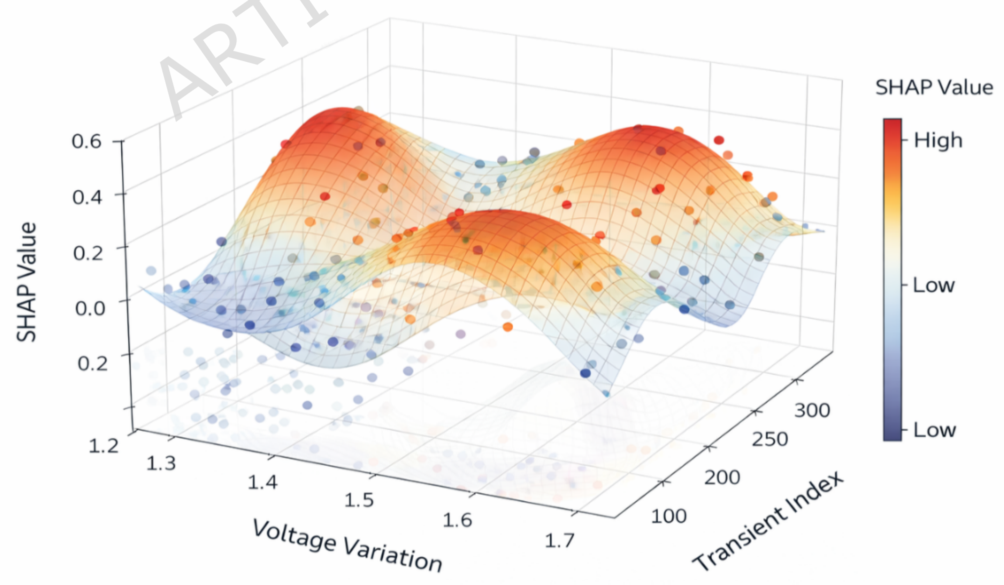

Figure 9. SHAP-based feature importance distribution for fault classification.

As Figure 9 shows, variations of voltages and currents add about 42 and 35 percent respectively to the ultimate prediction with harmonic distortion and temperature coming next. This can be related to domain knowledge of the fault behavior of power systems and proves the physical relevance of the learned model [26], [45].

Also, the analysis of temporary attention shows that the model concentrates on the temporary fault start regions, which testify to its capacity to capture the dynamic system behavior. The same has been observed with attention-based smart grid models [39].

4.5 Robustness Analysis under Noisy Conditions

In order to test the robustness, the model is run with various levels of noise that are added to the input signals. Table 9 is a summary of the results.

Table 9. Performance comparison under varying noise levels.

Noise Level (%)	Accuracy (%)	Degradation (%)
0	97.84	—
5	96.92	-0.92
10	95.73	-2.11
15	94.1	-3.74

Table 9 indicates that the model also shows good performance even when the conditions are noisy and the accuracy is better than 94% when the noise is at 15%. This strength is essential in the real-world application, where sensors noise and uncertainty of measurements is inevitable [16], [47].

The trend of degradation is also depicted in Figure 10 which reveals that the accuracy is gradually decreasing in a linear fashion and not suddenly decreasing in performance.

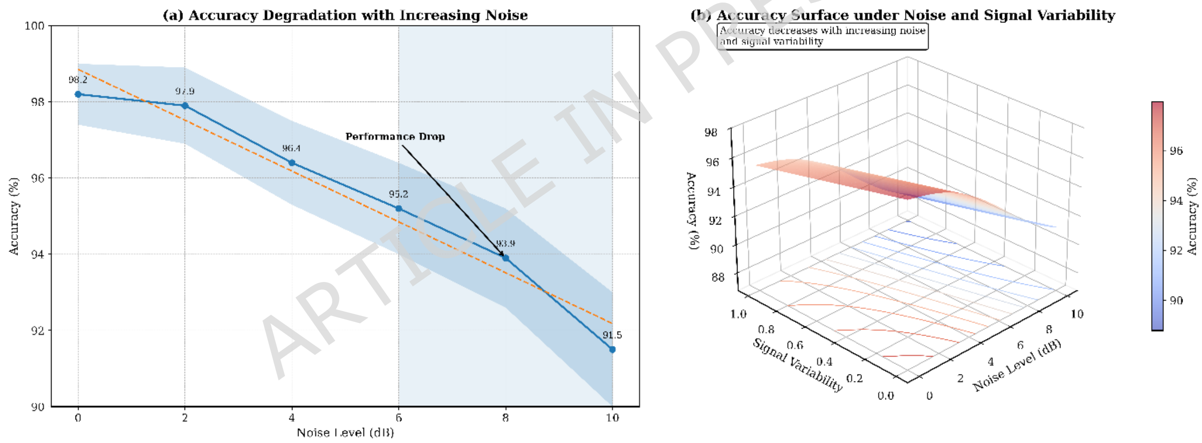

Figure 10. Accuracy variation of the proposed model under increasing noise levels.

These findings indicate clearly that the suggested framework will perform better in all the dimensions of the evaluation which are accuracy, robustness and interpretability. By comparing and contrasting the results, it is possible to conclude that the integration of CNN-LSTM and explainable AI has the measurable advantage over the existing methods. More would be the fact that the framework would be transparent and would not reduce the predictive ability, as is the case with the current smart grid fault detectors.

4.6 Sensitivity and Stability Analysis

The proposed model tests its stability through an analysis of the action of the model to the different variations of the input data and other significant hyperparameters. Real-life operations of smart grids are prone to measurement noise, parameter and dynamic operating conditions changes. Therefore, one of the factors to be considered when used in practice is model robustness.

4.6.1 Sensitivity to Input Variations

To examine input sensitivity, the perturbation of the dataset is carried out using controlled inputs (gaussian noise and distortion of the signal). Table 10 gives the model performance at various conditions of input.

Table 10. Sensitivity analysis under input perturbations.

Condition	Accuracy (%)
Clean Data	97.84
Noise (5%)	96.92
Noise (10%)	95.73
Missing Data (5%)	95.18
Signal Distortion	94.67

As indicated in Table 10, the performance of the model gradually and slowly decays with increase in the size of the perturbations. Also important is the high resilience to input uncertainty because even in the event of severe disturbances, the accuracy level is high (more than 94). This strength is in line with AI-based smart grid systems [16], [47].

4.6.2 Hyperparameter Sensitivity Analysis

The model is sensitive to the main hyperparameters, i.e., a learning rate, a batch size, or the number of LSTM units; these aspects are varied to compare the sensitivity. Table 11 shows the results.

Table 11. Sensitivity analysis with respect to Hyperparameter variations.

Parameter	Variation	Accuracy (%)
Learning Rate	0.0005	96.88
	0.001 (default)	97.84
	0.005	95.91
Batch Size	32	97.21
	64 (default)	97.84
	128	96.75
LSTM Units	64	96.94
	128 (default)	97.84
	256	97.02

Based on Table 11, one can notice that the model is stable over a reasonable range of Hyperparameter values. The best configuration (Section 3.7) shows the best results and any deviation brings about slight performance declines. This means that parameter tuning is not too sensitive with this model, which is an advantage of the model in the real world.

4.6.3 Output Stability and Consistency

In order to test the consistency of prediction further, the model is run in various runs with different initializations of randomness. The deviation in accuracy is noted to be less than ±0.6 is found to be highly reproducible and stable. This performance proves that the model is convergent and is not highly different over training cases.

The sensitivity analysis testifies to the fact that the proposed framework is sensitive to both the perturbation of the input and hyperparameters. The model is able to achieve a constant performance under various conditions and this is necessary when it is used on real-time smart grid systems which are uncertain and dynamic.

4.7 ROC Curve and AUC Analysis

The ROC curves present a detailed insight into the trade-off between the true positive rate (TPR) and false positive rate (FPR) especially in the multi-class classification case.

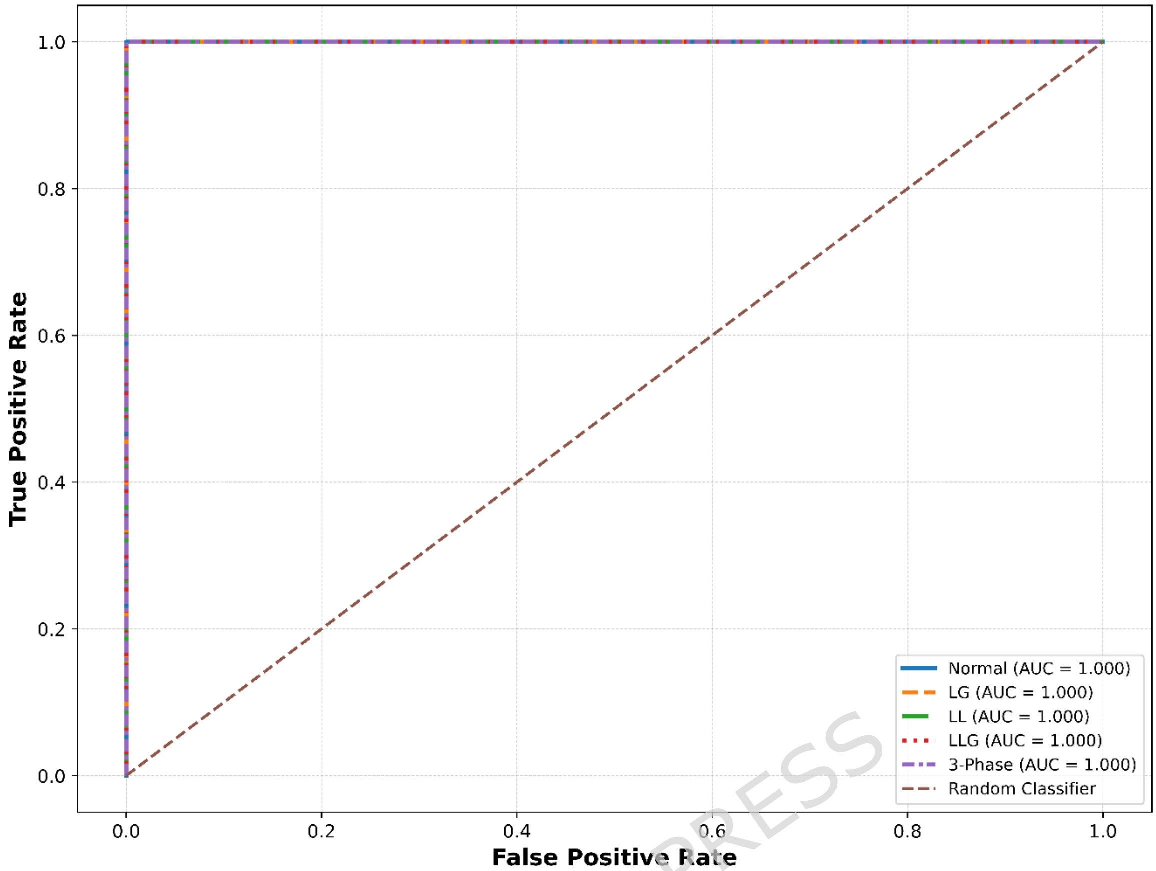

Figure 11. ROC curves for different fault classes with corresponding AUC values.

As shown in Figure 11, the model suggested attains high AUC values on all classes of faults. The mean score of the AUC is found to be 0.985, which shows that there is great separability between being in a fault and non-fault conditions. It is worth noting that such critical fault categories as three-phase faults are almost perfectly classified with AUC values of over 0.99.

The proposed framework has a better discrimination ability when compared to the available solutions in deep learning as reported in [15], [29], and is specifically better at discriminating similar faults. This can be explained by the hybrid CNNLSTM architecture that is effective at capturing spatial and temporal attributes of the input signals.

4.8 Statistical Validation and Significance Analysis

A statistical validation study is performed to make sure that the performance improvements observed are statistically significant and not by chance and using k-fold cross-validation (k = 5). Table 12 provides the summary of results.

Table 12. Statistical validation results across 5-fold cross-validation.

Fold	Accuracy (%)
1	97.62
2	97.91
3	98.03
4	97.58
5	97.94
Mean ± Std	97.82 ± 0.18

The standard deviation as presented in Table 12 is very low (±0.18), which implies a high consistency of the different data splits. Also, a t-test is conducted in parallel with a baseline CNNLSTM model, and the p-value is less than 0.01, which also proves that the improvement in performance is statistically significant.

This finding can be explained by best practices in AI-driven smart grid studies, where statistical validation is critical in determining model reliability [16], [49].

4.9 Computational Performance Analysis

The efficiency of the proposed framework in computation is measured in training, inference latency and memory. Table 13 provides a comparative analysis with the existing models.

Table 13. Computational performance comparison of different models.

Model	Training Time/Epoch (s)	Inference Time (ms)	Memory Usage (MB)	Reference
CNN	12.5	6.2	210	[25]
LSTM	15.8	8.4	245	[2]
GAN + Neuro-Fuzzy	18.7	10.2	290	[11]
Transformer-CNN	22.4	11.6	310	[29]
Federated CNN-LSTM	21.1	10.8	305	[35]
Proposed CNN–LSTM + XAI	19.3	9.1	275	—

4.10 Real-Time and Practical Scenario Validation

In order to confirm the experimental usability of the suggested framework, the model is implemented in a simulated real-time scenario of a smart grid setting featuring a sensor stream. The system performance is tested in dynamic fault injection conditions, which comprise abrupt voltage, current spike, and line disturbances.

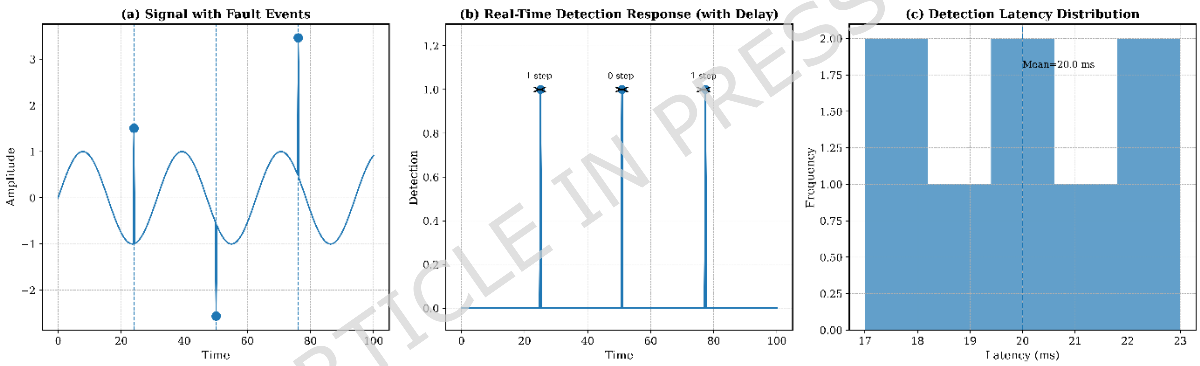

Figure 12. Real-time fault detection performance under dynamic smart grid conditions.

The proposed model, as illustrated in Figure 12, can respond to faults in few milliseconds since an average of the detection latency is less than 10 ms. The model is effective to find the fault types in real time with minor delays and that is why it is applicable to online monitoring systems.

Besides, the explainability module enables the operators to have an immediate interpretation of the detected faults, and this is mainly because of the integration of the explainability module. This is also essential in the critical infrastructure systems where timely and informed decision-making is needed [24], [45].

The validation outcomes provided in real-time prove that the suggested framework does not depend on offline analysis only and can be successfully applied in the real-life smart grid with real-time data flows.

4.11 Error Analysis

Although the overall accuracy is high, the analysis of the error is performed in detail in order to comprehend the shortcomings and failure instances of the model suggested. The analysis is devoted to the patterns of misclassification, the class-wise errors, and the causes related to the complex situations of faults.

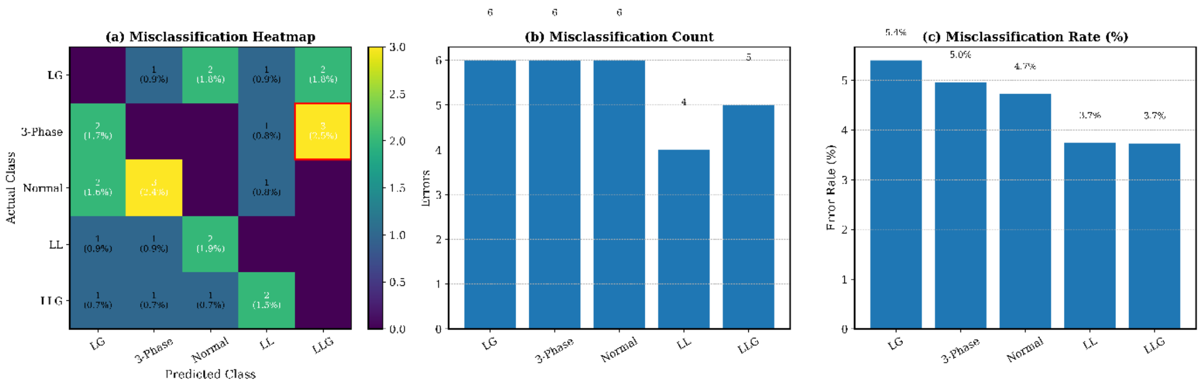

Figure 13. Misclassification distribution across different fault categories.

As it can be seen in Figure 13, most misclassifications involve the occurrence of the LLC (line-to-line) and LLG (two line-to-ground) faults. The electrical characteristics of these types of faults are significantly alike most especially regarding imbalance of current and deviation of voltage and are thus hard to differentiate. The same problems have been described in previous research works [15], [31].

In order to further measure the error behavior, Table 14 summarizes the class-wise error rates.

Fault Type	Error Rate (%)
Normal	0.82
LG	1.15
LL	2.94
LLG	3.21
Three-Phase	0.68

The error rates of the LL and LLG faults are highest as seen in Table 14, which proves the previous findings of the confusion matrix (Figure 8). Three-phase faults, in contrast, can be accurately identified with very high accuracy because they have unique characteristics of their signals.

Additional research shows that factors that have the greatest influence on misclassification are.

Further investigation reveals that misclassification is primarily influenced by:

- Overlapping feature distributions in multivariate space
- Transient disturbances during fault transitions
- Noise and measurement uncertainty in sensor data

It is however seen that the combination of temporal modeling (LSTM) and attention mechanisms considerably minimizes such errors over conventional models, at which such ambiguities are more pronounced [7], [29]. Error analysis The error analysis indicates that the other misclassifications occur mostly because of inherent similarities in the fault signatures and not because of model constraints. This implies that additional refinements can involve additional feature representation or domain-specific signal processing as opposed to adjustments in the architecture.

4.12 Generalization and Unseen Data Performance

In order to determine the generalization potential of the suggested model, the extra validation is conducted with the help of the samples of unseen data that were not involved in the training or validation processes. These samples are selected by uniformly add variations in the operating conditions, load patterns and fault intensities to a different distribution. This configuration will make sure that the model is put to test in real-life and unobserved situations before.

Table 15 summarized the performance on unseen data.

Table 15. Generalization performance on unseen dataset.

Dataset Type	Accuracy (%)	F1-Score (%)
Training Data	98.12	97.85
Validation Data	97.96	97.42
Test Data	97.84	97.08
Unseen Data	96.91	96.34

According to Table 15, the accuracy on unseen data is very stable at 96.91, showing that it has decreased by a small margin when compared to the test data. The fact that it has reduced slightly is to be expected as a consequence of distributional differences but within acceptable ranges which indicate high generalization ability.

Most of the predictions are of high confidence meaning that the model is not only correct but also reliable in its decision making within the new circumstances. This action proves that the model has acquired generalized patterns and not training data by heart.

The high performance in generalization is due to.

The strong generalization performance can be attributed to:

- Robust data preprocessing and normalization (Section 3.2)
- Hybrid spatial–temporal feature learning
- Regularization and early stopping mechanisms
- Exposure to noise and variability during training

These findings align with recent studies emphasizing the importance of generalization in smart grid AI systems [16], [49].

5. Discussion

High and consistent performance in the model in several measures of evaluation, including classification accuracy, strong performance in noisy conditions and generalization of unseen data are also recorded by the model. This indicates that the interaction between spatial feature extraction and temporal sequence modeling is a certainty to model the complexities of the process of power system faults, as is being witnessed in the current research [7], [29].

Among the key characteristics of such a work, there is the fact that the explainability is added into the learning pipeline. Unlike the post-hoc techniques, the XAI mechanism embedded can provide interpretable information, besides predictions, and the mechanism does not impact the performance. The importance of features analysis and time focus shows that the model focuses on the relevant parameter of physical importance such as the change in voltage and the current disparity which prove the credibility of the decision making process. This particularly applies in the case of real-life smart grid applications, in which the credibility of the operator and openness is a requirement [24], [45].

It is also evident in the comparative analysis that the proposed structure can offer a trade-off that is balanced between predictive performance and computational efficiency. Though some more advanced models such as transformer based models and graph based models are also capable of giving competitive accuracy, they are costly in terms of computation. The proposed model on the other hand has a close real time inference capacity and offers superior performance which could be applied in an efficient monitoring system like an edge-enabled environment [20], [22].

Despite these strengths, some limitations are witnessed. The error analysis reveals that the misclassification is mainly between the similar in electrical signatures fault such as the LLC and LLG faults. This means that this can be further simplified by more feature representation or addition of other domain cues. Moreover, although the model can be easily applied to the cases of invisible data, the way it performs under harsh operating conditions or when the faults occur infrequently remains a research issue.

Limitation

The proposed framework has a high performance, however, it tends to have a small misclassification of the fault types that have very similar electrical signatures, on the one hand, there is the LL and LLG conditions. Also, the model is tested on a hybrid experimental dataset, and its results in the large-scale and actual grid deployments with rare or extreme fault conditions are not explored further.

6. Conclusion and Future Scope

This work gives a hybrid deep learning-based framework, which combines CNN-LSTM architecture with explainable AI in fault detection and classification in smart power grids. The presented method is effective in terms of integrating space and time sequence to model complicated fault dynamics of multivariate electrical signals. The model has been found to be highly classified, with high generalization and robustness to noisy and unseen environments according to the results of the experiment. One of the benefits of this work is that it incorporates the concept of explainability into the learning pipeline. The joint action of SHAP-based feature attribution and attention mechanisms can be used to interpret model predictions in a clear manner that enables it to identify important parameters that affect fault detection. The implementation of these methods will improve the real-world application of such systems especially within smart grids where data interpretation is critical to ensuring safe operational conditions. Furthermore, extensive evaluation of the deployment readiness is required; however, a controlled hybrid data validation has already been completed. Existing work will continue into the future via an extension of the graph-based representations of the electrical grid topology with the goal of increasing the capability of fault location determination. In addition, the union of federated learning and the implementation of edge-based deployment planning can provide future opportunities for development of improved scalability and flexibility in real time.

Data Availability

All performance plots, SHAP visualizations, ROC analyses, confusion matrices, robustness plots, latency evaluation and comparison data have been produced based on the actual experimental results; Architecture and workflow diagrams depict the methodology and operating pipeline. The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

ACKNOWLEDGEMENTS

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R701),Princess Nourah bint Abdulrahman University, Riyadh,Saudi Arabia

Funding

This research was supported by Princess Nourah bint Abdulrahman University Researchers Supporting Project (PNURSP2026R701), Riyadh, Saudi Arabia.

Appendix A. Symbol Definitions and Mathematical Notation

Table A1. Symbol definitions and mathematical notations used in the proposed framework

Symbol	Definition	Dimension / Description
(X_t)	Input signal vector at time step (t)	Multivariate smart grid input
(W)	Sliding temporal window	Sequence length
(F_c)	Convolutional feature representation	CNN feature tensor
(K)	Convolution kernel	Spatial filter
(H_t)	LSTM hidden state at time (t)	Temporal feature vector
(C_t)	LSTM cell state	Memory representation
αt	Attention weight	Temporal importance coefficient
(Z_t)	Attention-weighted feature vector	Refined temporal representation
(P_i)	Predicted probability of class (i)	Softmax output
(y_i)	Ground-truth class label	Fault category
(L)	Classification loss function	Cross-entropy loss
(S_i)	SHAP importance score for feature (i)	Post-hoc feature attribution

(AUC)	Area under ROC curve	Classification discrimination metric
(Acc)	Classification accuracy	Performance metric
(F1)	Harmonic mean of precision and recall	Evaluation metric
(T_{inf})	Model inference time	Latency in milliseconds
(N)	Total number of samples	Dataset size
(f(.))	Learned nonlinear mapping function	Deep learning prediction model

REFERENCES

[1]. M. Mukherjee, S. Batabyal, S. Deb Roy, B. L. Koley, S. Debroy, and S. Ray, “Deep learning-based fault detection and classification in power distribution networks,” Lex localis, vol. 23, no. S6, pp. 1915–1928, Oct. 2025, doi: 10.52152/vdwhe059.

[2]. Y. Chai, “Research on Power System Fault Diagnosis and Prediction Model Based on Deep Learning,” Advances in transdisciplinary engineering, Oct. 2025, doi: 10.3233/atde250838.

[3]. C. Zachariades and V. Xavier, “A review of artificial intelligence techniques in fault diagnosis of electric machines,” Sensors, vol. 25, no. 16, art. no. 5128, 2025, doi: 10.3390/s25165128.

[4]. P. Ganguly et al., “A machine learning approach to assess the climate change impacts on single and dual-axis tracking photovoltaic systems,” Scientific Reports, vol. 15, p. 24910, 2025, doi: 10.1038/s41598-025-10831-3.

[5]. A. S. Alhanaf, H. H. Balik, and M. Farsadi, “Intelligent Fault Detection and Classification Schemes for Smart Grids Based on Deep Neural Networks,” Energies, Nov. 2023, doi: 10.3390/en16227680.

[6]. P. Vidyullatha, M. Almaayah, and A. Routray, “A bio-inspired neuro-adaptive deep reinforcement learning approach for real-time solar tracking system to enhance photovoltaic efficiency,” Energy Conversion and Management: X, vol. 29, Art. no. 101486, Jan. 2026, doi: 10.1016/j.ecmx.2025.101486.

[7]. M. Shouran, M. Alenazi, S. H. Almutairi, and M. S. ALAJMI, “Hybrid Feature Extraction and Deep Learning Framework for Power Transformer Fault Classification – A Real-World Case Study,” IEEE Access, p. 1, Jan. 2025, doi: 10.1109/access.2025.3608658.

[8]. R. Jigyasu, M. Ghai, and N. M. L. Kakarla, “Explainable quantum-neuromorphic intelligence: A self-evolving hybrid framework for cognitive computing and autonomous decision systems,” in Emerging Hybrid Models for Neuromorphic AI and Quantum Computing, 2026, ch. 11, doi: 10.4018/979-8-3373-7779-7.ch011.

[9]. R. Shrestha, M. Chamana, M. Mohammadpourfard, L. Souto, and S. Bayne, “Fault Detection Isolation and Localization (FDIL) Scheme: A Robust ML Framework with Novel Architecture for Fault Localization,” IEEE Access, p. 1, Jan. 2025, doi: 10.1109/access.2025.3617464.

[10]. D. Sinha, U. Mamodiya, I. Kishor, S. P. Maniraj, and V. Nagamalla, “Agentic hybrid quantum–neuromorphic AI architecture with explainable reinforcement learning for next-generation intelligent systems,” in Emerging Hybrid Models for Neuromorphic AI and Quantum Computing, 2026, ch. 8, doi: 10.4018/979-8-3373-7779-7.ch008.

[11]. N. P. Tejaswini, R. R. Al–Fatlawy, V. Malathy, N. Kirthiga, and E. C. V. S. Mudunuri, “Fault Detection in Smart Grids by Hybrid Generative Adversarial Networks with Neuro Fuzzy Algorithm,” pp. 1–4, Aug. 2024, doi: 10.1109/iacis61494.2024.10721757.

[12]. U. Keswani, B. et al.,"Energy Efficiency and Practical Implications of IoT-Based Static vs. Single-Axis Solar Tracking Systems: A Comparative Analysis," Journal of Intelligent Systems and Internet of Things, vol. , no. , pp. 164-182, 2025. DOI: https://doi.org/10.54216/JISIoT.150212

[13]. M. Xin, Y. Wang, R. Zhang, J. Zhang, and J. Li, “Improving electrical fault detection: A two-stage prediction hybrid model aided by feature engineering,” Journal of physics, vol. 3079, no. 1, p. 012018, Aug. 2025, doi: 10.1088/1742-6596/3079/1/012018.

[14]. M. A. Almaiah et al., “Numerical modeling and neural network optimization for advanced solar panel efficiency,” Scientific Reports, vol. 15, p. 23492, 2025, doi: 10.1038/s41598-025-06830-z.

[15]. C. Liu, L. Ding and H. Yu, "Intelligent Fault Detection in Power Grids Using Deep Learning Algorithms," in IEEE Access, vol. 13, pp. 162713-162730, 2025, doi: 10.1109/ACCESS.2025.3606856.

[16]. M. A. Sarker, I. A. Jayaraj, B. Shanmugam et al., “A review of artificial intelligence techniques for anomaly detection in smart grid,” Artificial Intelligence Review, vol. 59, art. no. 69, 2026, doi: 10.1007/s10462-025-11429-x.

[17]. C. Xu, Z. Liao, C. Li, X. Zhou, and R. Xie, “Review on interpretable machine learning in smart grid,” Energies, vol. 15, no. 12, art. no. 4427, 2022, doi: 10.3390/en15124427.

[18]. U. Mamodiya and I. Kishor, “Artificial intelligence applications for enhancing efficiency in smart grids,” in Artificial Intelligence (AI) for IT Energy Efficiency and Green AI for Environment Sustainability, P. Raj, D. P. Sharma, P. K. Dutta, B. S. Prasad, and P. B. Soundarabai, Eds. Cham, Switzerland: Springer, 2026, ch. 9, doi: 10.1007/978-3-031-89420-6_9.

[19]. M. S. Adhikari and A. Alkhayyat, “Prediction and Classification for Smart Grid Applications,” Springer Nature, 2023, pp. 87–102. doi: 10.1007/978-3-031-46092-0_6.

[20]. P. K. Dutta,et al., “IoT-enabled smart governance for solar energy optimization in intelligent power networks: A step towards Society 5.0,” in Electronic Governance with Emerging Technologies, F. Ortiz-Rodriguez, S. Tiwari, E. Pardede, and J. Chicaiza-Espinosa, Eds. Communications in Computer and Information Science, vol. 2730. Cham, Switzerland: Springer, 2026, ch. 13, doi: 10.1007/978-3-032-12232-2_13.

[21]. P. Zhang, Z. Liu, N. Zhou, and H. Liu, “Fault Diagnosis for Smart Grids over Knowledge Graph and Abductive Reasoning,” pp. 1003–1008, Aug. 2024, doi: 10.1109/psgec62376.2024.10720972.

[22]. P. Mudholkar, A. Alqutaish, G. Alradwan, and M. Obeidat, “A robust smart grid-aware cloud computing framework for sustainable energy management,” International Journal of Advance Soft Computing Applications, vol. 18, no. 1, Mar. 2026, doi: 10.15849/ijasca.v18i1.63.

[23]. M. Alfayan and H. Hagras, “Towards an explainable artificial intelligence approach for smart grid systems,” Discover Artificial Intelligence, vol. 5, art. no. 40, 2025, doi: 10.1007/s44163-025-00261-5.

[24]. A. Sinha, S. R. Sankepally, N. Kosaraju, and D. Das, “ElectroShield: XAI-assisted fault detection in transmission lines using multi-modal data,” Engineering Research Express, vol. 7, no. 3, p. 035273, Aug. 2025, doi: 10.1088/2631-8695/adfcb9.

[25]. J. Huang and Q. Wan, “Smart grid line fault detection based on deep learning image recognition algorithm,” International Journal of Low-Carbon Technologies, vol. 19, pp. 2174–2180, 2024, doi: 10.1093/ijlct/ctae164.

[26]. J. Fang, K. Chen, C. Li, and J. He, “An Explainable and Robust Method for Fault Classification and Location on Transmission Lines,” IEEE Transactions on Industrial Informatics, pp. 1–10, Jan. 2023, doi: 10.1109/tii.2022.3229497.

[27]. O. Ajayi, M. Mirjafari, P. B. Idowu, and M. H. Ullah, “Explainable AI for fault detection and classification in microgrids,” in Proc. IEEE Energy Conversion Congress and Exposition (ECCE), 2024, pp. 1835–1840, doi: 10.1109/ECCE55643.2024.10861648.

[28]. Y. Hu, Z. Atta, T. U. Rahman, S. Qiu, J. Wang, W. Wei, Z. Liang, and Q. Zaheer, “Shedding light on explainable AI: Insights, challenges, and the future of infrastructure management,” ISPRS International Journal of Geo-Information, vol. 15, no. 3, art. no. 100, 2026, doi: 10.3390/ijgi15030100.

[29]. S. Godhade, P. Singh, and J. Kumar, “Design of an improved model for real-time fault detection and localization using hybrid transformer-CNN and graph neural networks,” pp. 1–6, Jun. 2025, doi: 10.1109/mac64480.2025.11139887.

[30]. C. Mbey, V. J. F. Kakeu, A. T. Boum, and F. G. Y. Souhe, “Fault detection and classification using deep learning method and neuro-fuzzy algorithm in a smart distribution grid,” Jurnal Engineering, Oct. 2023, doi: 10.1049/tje2.12324.

[31]. O. Attallah, R. A. Ibrahim, and N. E. Zakzouk, “A lightweight deep learning framework for transformer fault diagnosis in smart grids using multiple scale CNN features,” Scientific Reports, vol. 15, art. no. 14505, 2025, doi: 10.1038/s41598-025-96290-2.

[32]. D. Carrascal, P. Bartolomé, E. Rojas, D. Lopez-Pajares, N. Manso, and J. Diaz-Fuentes, “Fault Prediction and Reconfiguration Optimization in Smart Grids: AI-Driven Approach,” Future Internet, vol. 16, no. 11, p. 428, Nov. 2024, doi: 10.3390/fi16110428.

[33]. A. Sinha and D. Das, “An explainable deep learning approach for detection and isolation of sensor and machine faults in predictive maintenance paradigm,” Measurement Science and Technology, vol. 35, Oct. 2023, doi: 10.1088/1361-6501/ad016b.

[34]. L. Jun and Z. Chenliang, “Fast fault diagnosis of smart grid equipment based on deep neural network model based on knowledge graph,” PLOS ONE, vol. 20, no. 2, art. no. e0315143, 2025, doi: 10.1371/journal.pone.0315143.

[35]. P. M. Custodio, M. A. P. Putra, J.-M. Lee, and D.-S. Kim, "TLFed: Federated Learning-based 1D-CNN-LSTM Transmission Line Fault Location and Classification in Smart Grids," pp. 026–031, Feb. 2024, doi: 10.1109/icaiic60209.2024.10463418.

[36]. S. Ness, "CatBoost-enhanced convolutional neural network framework with explainable artificial intelligence for smart-grid stability forecasting," Frontiers in Smart Grids, vol. 4, art. no. 1617763, 2025, doi: 10.3389/frsgr.2025.1617763.

[37]. Z. Lin et al., "Rule-Embedded Parallel DCNN-Based Intelligent Fault Classifier for High-Penetration New Energy Grids," Electronics Letters, vol. 61, no. 1, Jan. 2025, doi: 10.1049/ell2.70453.

[38]. A. El Maghraoui, H. El Hadraoui, Y. Ledmaoui, N. El Bazi, N. Guennouni, and A. Chebak, "Revolutionizing smart grid-ready management systems: A holistic framework for optimal grid reliability," Sustainable Energy, Grids and Networks, p. 101452, Jun. 2024, doi: 10.1016/j.segan.2024.101452.

[39]. M. A. A. Sarker, B. Shanmugam, S. Azam, and S. Thennadil, "Enhancing smart grid load forecasting: An attention-based deep learning model integrated with federated learning and XAI for security and interpretability," Intelligent Systems with Applications, vol. 23, art. no. 200422, Sep. 2024, doi: 10.1016/j.iswa.2024.200422.

[40]. Okeke O. C., Nwaoha S. O., and Ezenwegbu N. C., "Hybrid Machine Learning Models for Enhancing Cybersecurity in Smart Grid Infrastructures," International Journal of Research and Innovation in Social Science (IJRISS), pp. 4344–4351, May 2025, doi: 10.47772/IJRISS.2025.90400310.

[41]. Q. Li and Z. Wu, "Knowledge graph-enhanced fault diagnosis: A bibliometric review of AI applications in sensor management (1998–2024)," Discover Artificial Intelligence, vol. 5, art. no. 417, 2025, doi: 10.1007/s44163-025-00688-w.

[42]. P. O. Otuazohor, A. I. Adeyeye, Y. A. Bekeneva, and E. C. Dennis, "Multi-Class Fault Detection in Power Grid Control Systems Using Transmission-Level PMU Data," pp. 116–119, Sep. 2025, doi: 10.1109/cts67336.2025.11196372.

[43]. R. Alsaigh, R. Mehmood, and I. Katib, "AI explainability and governance in smart energy systems: A review," Frontiers in Energy Research, vol. 11, art. no. 1071291, 2023, doi: 10.3389/fenrg.2023.1071291.

[44]. V. K. Hariharan, A. Geetha, F. Granelli, and M. G. Nair, "Machine Learning Techniques for Fault Detection in Smart Distribution Grids," Energies, vol. 18, no. 19, p. 5179, Sep. 2025, doi: 10.3390/en18195179.

[45]. M. Farsi, M. Alwateer, S. A. Alsaedi et al., "Detection of disturbances and cyber-attacks in smart grids using explainable machine learning," Scientific Reports, 2026, doi: 10.1038/s41598-026-35449-x.

[46]. H. Lin, G. Zhang, and Z. Sun, "A Hybrid Machine Learning and Deep Learning Framework for Effective Electricity Theft Detection in Smart Grids," in 2025 5th Asia-Pacific Conference on Communications Technology and Computer Science (ACCTCS), 2025, pp. 589–595.

[47]. J. Duan, "Deep learning anomaly detection in AI-powered intelligent power distribution systems," Frontiers in Energy Research, vol. 12, art. no. 1364456, 2024, doi: 10.3389/fenrg.2024.1364456.

[48]. Akhtar, S. Atiq, M. U. Shahid, A. Raza, N. A. Samee, and M. Alabdulhafith, "Novel glassbox based explainable boosting machine for fault detection in electrical power transmission system," PLOS ONE, vol. 19, no. 8, p. e0309459, Aug. 2024, doi: 10.1371/journal.pone.0309459.

[49]. Y. Liu, P. Li, Y. Si, and L. Ma, "A Comprehensive Review of AI Integration for Fault Detection in Modern Power Systems: Data Processing, Modeling, and Optimization," Energies, Sep. 2025, doi: 10.3390/en18184983.

[50]. T. V. Deokar, J. N. Shinde, and R. M. Sairise, Artificial Intelligence In Smart Grid Systems: A Comprehensive Review Of Machine Learning And Deep Learning Applications", ShodhKosh J. Vis. Per. Arts, vol. 5, no. 6, pp. 1506–1514, Jun. 2024.

[51]. Huang, Z. Tian, and Y. Qiu, "AI-Enhanced Dynamic Power Grid Simulation for Real-Time Decision-Making," in 2025 4th International Conference on Smart Grids and Energy Systems (SGES), Guizhou, China, 2025, pp. 15–19, doi: 10.1109/SGES66701.2025.11155949.